Localised Entry, Global Frontier: How Generative-AI Subscriptions Are Priced Across Countries

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October 2026

Abstract

Generative-AI assistants are sold through one subscription menu in every country while national incomes differ a hundredfold. Two audits of tax-inclusive prices in every national Apple App Store storefront, the second covering ChatGPT, Claude, Gemini, six other AI assistants, 21 general subscription apps and four apps with embedded AI features, show which tiers are priced locally. Within the same firm and storefront, where currency, tax and the platform's conversion cancel, the relative prices of the $20 standard and $100–500 high-usage AI tiers vary little with income (absolute slopes at most 0.022), whereas the same firms' YouTube and X Premium subscriptions are 50–73% cheaper in the median low- or lower-middle-income storefront (Microsoft 365 Personal 15%). Two capped AI entry tiers are about 40% cheaper (ChatGPT Go, Character.AI Lite); every localised AI tier is a capped entry tier, but not every capped tier is localised. Relative to the platform's automatically generated price, estimated from 30 small-developer apps, the frontier tiers' deviations are centred on zero; Google Play's lowest ChatGPT price and Microsoft's web prices closely match the App Store. In a four-option versioning model with an income-invariant inference cost, an unconstrained vendor would pass an entry-tier discount on to the frontier tier; instead, the frontier-to-entry dollar gap widens where incomes are low, which rejects that benchmark but cannot separate a held frontier price from frontier valuations that do not scale with income. A $20 plan costs 0.65% of per-capita income in the median high-income economy and 29% in the median low-income one.

Keywords: generative AI; international price discrimination; versioning; platform pricing; affordability; digital divide.
JEL codes: L11; L15; L81; L86; D42; O33.

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E-mail: noah.taehwan@gmail.com. Data and code: https://doi.org/10.5281/zenodo.23006365 (replication package, all versions; described in Appendix B). An earlier version circulated as SSRN working paper 7535900 under the title *Affording Generative AI Across Countries: Uniform Prices, Unequal Burdens*. This version: 3.1, October 2026.

1 Introduction

Some digital goods are priced to market: online retailers charge less where incomes are lower, with cross-country price-income elasticities of about 0.14–0.18 for tradable goods (Simonovska, 2015), streaming services set country-specific prices (Colbjørnsen et al., 2022), and Spotify's country prices raise its revenue over a single world price while transferring surplus to lower-income countries (Waldfogel, 2020). Many digital subscriptions are not: in the App Store benchmark of this paper only 8 of 21 general subscription apps localise any plan (Appendix Table E.8). The informative comparison is therefore within the firm: generative-AI firms localise some of their products and not others. OpenAI, Anthropic and Google sell the same $20 standard tier and the same $100–500 high-usage tiers in every country, and the only AI products whose prices are lower where incomes are lower are capability-capped entry tiers. Which tiers of the menu do vendors price for local markets, and what keeps the others at the U.S.-anchored price? This paper documents the pattern with two audits of every national Apple App Store storefront, compares each firm's AI tiers with the other subscriptions it sells through the same storefronts, estimates how much of the remaining variation is the platform's own tax-and-currency conversion rather than vendor pricing, and asks whether a simple versioning model with a positive, dollar-denominated marginal cost of inference and cross-border arbitrage can account for where localisation appears and where it does not.

The question matters for two reasons. For industrial organization, AI subscriptions are versioned products with a positive marginal cost of serving each subscriber: unlike a copy of a song or a film, every conversation with a frontier model consumes computation that is priced in dollars worldwide. Versioning with positive marginal or usage costs has been studied in a single market (Jones and Mendelson, 2011; Chellappa and Mehra, 2018); what the audit adds is the multi-market margin, where the same dollar cost meets a hundredfold spread of incomes, a platform that converts prices automatically, and the possibility of buying across borders. For policy, the pattern determines who can afford frontier capability. A uniform dollar price for the standard tier costs 0.65% of per-capita income a year in the median high-income economy and 29% in the median low-income one, and the entry tiers that vendors do localise withhold the frontier model.

The evidence comes from four sources collected for this paper. The first is a two-wave audit of the tax-inclusive local prices that Apple's App Store listed on 27 September and 16 October 2026 for every plan of ChatGPT, Claude and Gemini in every national storefront (218 codes, of which 170–173 return local price lists). The second wave extends the audit to six other AI assistants (Perplexity, Grok, Mistral, Poe, Character.AI and Microsoft 365 Copilot), 21 general subscription apps (among them YouTube, YouTube Music, Google One, Duolingo, Telegram, Discord, Snapchat, LinkedIn, Dropbox, X and NordVPN) and four apps that sell AI features inside a general product (Canva, CapCut, Notion, Grammarly): 34 apps, 7,412 app-storefront queries and 30,817 item-storefront prices, with the fingerprint of every page recorded in the package and the pages themselves deposited with the replication package. The third is a panel of 30 small-developer subscription apps in the same storefronts, used to estimate the price that Apple generates automatically from a U.S. base price; the estimate is exact only at price points that small developers use, such as $19.99, and is extrapolated with a storefront multiplier elsewhere. The fourth is a cross-channel check of Google Play's lowest in-app price for twelve apps in 30 countries and of Microsoft's web store in 57 countries.

Three findings follow. First, within the same firm and the same storefront, where currency, taxes

and the platform's conversion cancel, the relative prices of the $20 and $100–500 AI tiers vary little with income (absolute slopes at most 0.022, against 0.15–0.34 for YouTube and X Premium), whereas the same firms' subscriptions without an AI allowance rise with income 0.20 log points per log point faster than their frontier AI tiers in a stacked regression with firm-by-storefront and item fixed effects (randomisation $p = 0.0005$), and those with a small allowance 0.12 faster; among AI products only capped entry tiers are localised (ChatGPT Go, Character.AI Lite), and not all of them (Google AI Plus, SuperGrok Lite). Second, relative to the estimated platform schedule, the deviations of the $20 and $100–500 tiers of six vendors have slopes between -0.016 and 0.004 and medians of zero in every income group below high income, while ChatGPT Go's deviation has a slope of 0.110 (s.e. 0.006) and a median 38% below the schedule in low- and lower-middle-income storefronts; Google Play's lowest ChatGPT item is within 5% of the App Store Go price in 22 of 25 same-currency countries, and Microsoft's web prices are within 5% of the App Store prices for both of its plans in all 49 same-currency countries (identical to the cent in 37). Third, within the same storefront ChatGPT Plus costs 2.5 times Go in the United States and 4.0 times in the median low-income storefront, and the dollar gap between the two plans, relative to the gap between their schedule prices, is 25% wider in the median poorer storefront (income slope -0.047, s.e. 0.005; in dollars relative to the U.S. gap, -0.010, s.e. 0.007); an unconstrained vendor would narrow, not widen, that gap where incomes are low.

Section 3 develops a model under which a profit-maximising vendor would price this way, and states which of its implications the cross-section can and cannot test. A multi-market monopolist sells a free tier, a capped entry tier and a frontier tier to consumers whose valuations scale with local income. Serving a frontier subscriber costs c_H a month in inference, a cost that is denominated in dollars and does not fall with local income; serving a capped entry subscriber costs much less. With lognormal valuations, the profit-maximising price of each tier's increment over the tier below has an income elasticity that falls with the marginal-cost share of that increment, from one when the marginal cost is zero towards zero when the margin is thin (Proposition A.2). Because the frontier price is the entry price plus that increment, an unconstrained vendor would pass any entry-tier discount on to the frontier tier, so a frontier price that stays put while the entry tier is localised is the signature of a binding constraint within the nested menu (Proposition A.3); frontier valuations that do not scale with income would produce the same pattern (Remark 1). The platform's default schedule is the zero-effort price, and deviating from it carries a small fixed cost, so vendors deviate only where the gain is large: for the entry tier in poorer markets, and not for the frontier tier, where inference cost leaves little profit at stake in re-pricing. Cross-border arbitrage caps the dollar gap that any tier can open between storefronts; a gap that is a large discount on an $8 capped tier is a small one on a $200 tier, so arbitrage can help hold the high-usage tiers at the default but not, on its own, the $20 tiers (Proposition A.4). The model yields five predictions, P1–P5, of which Section 5 tests P1–P3 against listed prices and treats P4 as a consistency check, with P5 discussed in Section 7; its demand side implies that the margin on which affordability policy would bite is paid access to frontier capability.

The paper contributes to four literatures. It adds a dated, archived, multi-vendor record of local AI subscription prices to the literature on international price discrimination and pricing to market (Krugman, 1986; Goldberg and Knetter, 1997; Atkeson and Burstein, 2008; Cavallo et al., 2014), on uniform pricing by multi-market and multi-product firms (DellaVigna and Gentzkow, 2019; Shiller and Waldfogel, 2011) and on the decomposition of cross-border price differences into taxes, costs and markups (Goldberg and Verboven, 2001; Gopinath et al., 2011); the within-firm, within-storefront design and the

estimated platform schedule address a problem specific to platform-distributed goods, where listed prices embed the platform's own tax and currency rules, which Enache et al. (2023) exploit as a source of price variation over time and this paper uses cross-sectionally as the benchmark that separates platform conversion from vendor localisation. It brings versioning theory (Mussa and Rosen, 1978; Deneckere and McAfee, 1996; Bhargava and Choudhary, 2001, 2008; Sundararajan, 2004), and its extensions to goods with positive marginal or usage costs (Jones and Mendelson, 2011; Chellappa and Mehra, 2018) and to token-priced language models (Bergemann et al., 2025), to a multi-market setting in which the marginal cost is denominated in dollars and does not fall with local income, and shows that a high inference-cost share of the frontier increment, rather than degradation as such, can keep the frontier tier at the platform default by leaving little profit at stake in re-pricing it; listed prices cannot separate this from frontier valuations that do not scale with local income (Appendix A.5). It extends the cross-country evidence on streaming-service pricing (Waldfogel, 2020), on pharmaceutical pricing under arbitrage (Danzon and Chao, 2000) and on textbook pricing (Cabolis et al., 2007), and the theory of parallel trade as a limit on international price discrimination (Malueg and Schwartz, 1994; Valletti, 2006), to AI, and it treats the platform's default price as the zero-effort option that a menu cost of overriding it preserves, in the sense of the price-adjustment costs measured by Levy et al. (1997) and Zbaracki et al. (2004) and of the customer reactions to price changes that Anderson and Simester (2010) document. And it supplies the consumer-price side of the economics of language models, which has so far priced tokens in the business market (Demirer et al., 2026, 2025) and compared usage with income but not with the prices that users face (Liu et al., 2025; Misra et al., 2025; Chatterji et al., 2025; Appel et al., 2025; Daepp and Slaughter, 2026; Iscenko et al., 2026); Section 6 relates the public usage indicators to the menu descriptively, with the measurement caveats that those indicators require. The uneven access to frontier models that such pricing produces has been described as a compute divide (Lehdonvirta et al., 2024), as disparities in the control of and access to large language models (Sathish et al., 2024) and, within a country, as a premium digital divide between free and paid tiers (Kanzamanova and Myeong, 2026); the price record measures the margin that vendors' own pricing sets.

Section 2 describes the menus and the App Store's pricing mechanics. Section 3 presents the model and its predictions. Section 4 documents the two audit waves, the default panel, the cross-channel data and the archive. Section 5 reports the within-firm relative prices, the income slopes of listed prices, the decomposition against the estimated schedule and its validation, the within-storefront ratio and increment, and affordability. Section 6 summarises what the public usage indicators show. Section 7 discusses interpretation, policy and limitations, and Section 8 concludes. The formal model, data construction, the cross-channel comparison, the usage-indicator analysis, additional tables, a descriptive account of the ChatGPT Go rollout and a measurement protocol are in the appendices.

2 Institutional background

2.1 The price menu

Table 1 summarises the consumer plans of the three leading assistants as documented on vendors' official pages on 27 September 2026, together with the U.S. App Store price of the same plans. Three features of the menu stand out.

Convergence. The three menus have converged on nearly the same rungs: a free tier, an entry tier at

Table 1: The consumer price ladder of three leading AI assistants, September 2026

Tier	OpenAI (ChatGPT)	Anthropic (Claude)	Google (Gemini)
Free	Free	Free	Free
Entry	Go: $8 [$8.00]; launched in India Aug. 2025, then in stages elsewhere, U.S. Jan. 2026; ads tested	—	AI Plus: [$4.99]
Standard	Plus: $20 [$19.99]	Pro: $20 monthly, $17 a month billed annually [$20.00]	AI Pro: [$19.99]
High usage	Pro $100: 5× Plus usage [$100.00]	Max 5×: "from $100" [$124.99]	AI Ultra $100: 5× Pro usage [not listed in the App Store]
Highest usage	Pro $200: 20× Plus usage [$200.00]; new sign-ups paused 10–29 Sep. 2026; Pro $500 added 29 Sep. 2026 [$500.00 in Oct.]	Max 20×: $200 [$249.99]	AI Ultra $200 (was $250): 20× Pro usage [$199.99]

Notes: Official U.S. web prices per month as documented on 27 September 2026, with U.S. App Store prices in brackets (same day, before sales tax; the Pro 500 price from the 16 October 2026 audit). Web prices and allowance multiples supply context; the cross-country regressions use only the App Store records. OpenAI: "Both Pro tiers include the same core capabilities. The main difference is usage allowance" (OpenAI, 2026a); Plus from OpenAI (2026e), Go from OpenAI (2026d), Pro 500 and the reopening of Pro 200 from OpenAI (2026c,b). Anthropic: Anthropic (2026b) (prices exclude tax). Google: the $100 Ultra tier and the $200 Ultra (down from $250) announced 19 May 2026 (Google, 2026); the App Store lists one Ultra plan, at $199.99, in 94–98 storefronts (its absence elsewhere is list truncation, Section 4); Google's plan page shows region-dependent prices, so App Store prices are given for AI Plus and AI Pro.

$5–8, a $20 standard tier, and tiers at about $100 and $200. Perplexity Pro, SuperGrok and Microsoft 365 Premium with Copilot sit on the same $20–30 rung, and SuperGrok Heavy on a $300 rung.

Quantity, not intelligence, above the standard tier. The upper rungs are differentiated primarily by usage allowances ("5×" and "20×" the standard tier) and, secondarily, by access to the most capable model variants and features. The free and entry tiers, by contrast, are degraded versions of the standard tier: ChatGPT Go offers an "Instant" model with lower caps, while Plus offers the "Thinking" model with higher limits (OpenAI, 2026d); Google AI Plus and Character.AI Lite are likewise capped versions of the products above them. A higher price therefore buys more frontier capability per month, though not a capability comparable across vendors, languages or tasks. The menu is a textbook instance of second-degree price discrimination by quantity and quality (Mussa and Rosen, 1978; Varian, 1985), and, as in versioning models with positive marginal or usage costs (Jones and Mendelson, 2011; Chellappa and Mehra, 2018) and in the pricing of access services with allowances (Essegaier et al., 2002), every conversation with a frontier model consumes computation whose cost is incurred in dollars, so the marginal cost of serving a subscriber is positive and does not fall with the subscriber's income. The question this paper asks is what that dollar cost does to the menu when it is sold across a hundredfold range of incomes.

Localisation at the bottom of the ladder. The entry tiers are the vendors' instrument for reaching lower-income markets: ChatGPT Go launched in India in August 2025 and was rolled out in stages before its U.S. launch at $8 in January 2026 (OpenAI, 2026d). Section 5 shows that it is one of two AI tiers in the audit whose local price departs from the platform's schedule with income. Promotional give-aways, such as bundles of premium plans with mobile subscriptions in India, are temporary and country-specific and are not captured by a listing audit (Appendix F). A tariff is a bundle of fee, allowances, eligible models, overage terms and reset rules, as the tariff change of 29 September 2026, which added

a $500 Pro tier and reopened Pro 200 with a lower allowance (OpenAI, 2026c,b), illustrates; the audit observes the fee (Appendix G).

2.2 How App Store prices are set

The prices this paper audits are listed by Apple's App Store, and four platform rules shape them. First, a developer chooses a base storefront and a price there on a grid of price points; App Store Connect then "provides comparable prices for all 175 App Store countries and regions, taking into account taxes and foreign exchange rates" and rounding to local price conventions, and a developer may override the generated price in any storefront (Apple, 2026b,a). The listed prices include value-added or goods-and-services taxes in the storefronts where Apple collects them (Section 4). The automatically generated price is the *default schedule* referred to below; the AI vendors' U.S. prices sit on the price-point grid ($19.99, $199.99), consistent with a U.S. base storefront. Second, for auto-renewable subscriptions Apple does not refresh the generated prices after later tax or exchange-rate changes: each of its periodic price updates since February 2025 states that the update "excludes auto-renewable subscriptions" (Apple, 2026c). A subscription's default is therefore the comparable price of the date on which its price was set or last reset, which makes the schedule estimated in Section 5.3 a mixture of vintages; Section 5.3 tests the implications. Third, Apple decides which storefronts exist and in which currency each is denominated: 45 of the 218 two-letter codes audited have no storefront of their own and redirect to the U.S. store, and 102 of the 164 storefronts in the first-wave price regressions, including every storefront in a low-income economy, are denominated in U.S. dollars; in 16 of the U.S.-dollar storefronts the generated price nonetheless embeds a tax of more than 5%. Fourth, Apple retains an ad valorem commission on subscription revenue, 15 or 30% depending on the developer's size and the subscription's tenure but, under Apple's standard terms, the same share in every storefront for a given plan, so the commission scales all of a plan's prices alike (Wang and Wright, 2017); with a positive marginal cost it raises the cost share of the price, which the model of Section 3 absorbs into the cost. Because the default schedule embeds taxes, which rise with income across storefronts, a plan left at the default will show a small positive income slope in dollars even though the vendor has done nothing; Section 5.3 measures that slope from the small-developer panel and treats a plan's deviation from the default as the vendor's active pricing decision, and Section 5.1 shows what can be learned without the schedule at all.

3 A model of multi-market versioning with inference costs

This section sets out a compact model of a vendor that sells a menu of tiers in many markets through a platform, and derives five predictions that the audit can confront. The formal statements and proofs are in Appendix A.

3.1 Demand

In market c a share κ_c of the population is connected and chooses among four options: non-use, a free tier F, a capped entry tier L and a frontier (standard or high-usage) tier H, with service levels $0 < q_F < q_L < q_H$ and monthly prices $0 = p_F < p_{L,c} < p_{H,c}$. An individual with valuation v per unit of service obtains $u_k = vq_k - \tau - p_{k,c}$ from tier k and zero from non-use, where τ is the monetary equivalent of participation costs common to all tiers. Valuations are lognormal with location $\mu_c = \mu_0 + \ln(y_c/y_0)$

and dispersion s, so that a doubling of income doubles every valuation. When the total outlay $\tau + p_{k,c}$ is convex in the service level, the three thresholds are nested,

$$\frac{\tau}{q_F} < \frac{p_{L,c}}{q_L - q_F} < \frac{p_{H,c} - p_{L,c}}{q_H - q_L}, \tag{1}$$

and the shares of the connected population using any tier, a paid tier and the frontier tier are $\Phi(z_{D,c})$, $\Phi(z_{L,c})$ and $\Phi(z_{H,c})$, where $T_{D,c}$, $T_{L,c}$ and $T_{H,c}$ are the three terms of (1) and $z_{k,c} = (\mu_c - \ln T_{k,c})/s$ is the standardised distance by which the mean log valuation exceeds the log threshold (Appendix A). In this regime the price of a paid tier does not affect whether people use AI at all, which is governed by τ/q_F; it affects only which tier they use.

3.2 Supply: inference costs, the platform default and arbitrage

Three features of the supply side distinguish AI subscriptions from the information goods of the versioning literature.

Positive, income-invariant marginal cost. Serving a subscriber on tier k for a month costs $c_k \geq 0$ in computation, with $c_H > c_L \geq c_F$ and a cost per unit of service that rises from the entry to the frontier increment, $(c_L - c_F)/(q_L - q_F) < (c_H - c_L)/(q_H - q_L)$, which keeps the vendor's optimal menu nested. The cost is incurred in dollars at the vendor's data centres, so it does not fall with the subscriber's income, and for the frontier tier it is a substantial share of the price: heavy users of a $20 plan can consume more computation than the plan collects, which is why vendors cap usage and sell higher allowances at higher prices.

The platform default. For each tier the vendor sets a U.S. base price P_k. Unless it overrides it, the platform lists $P^D_{k,c} = t_c e_c P_k$ in storefront c, where e_c converts dollars into the storefront currency and $t_c \geq 1$ adds the taxes the platform collects there, so that, net of tax and currency, the default is a uniform dollar price; overriding costs a fixed amount $K > 0$ per plan and storefront, the administrative and monitoring cost of maintaining a bespoke price point, a menu cost of the kind measured by Levy et al. (1997) and Zbaracki et al. (2004). The vendor therefore deviates from the default only in storefronts where the profit gain from the market-specific price exceeds K; that gain is zero at the U.S. income level and scales with the number of potential subscribers, so deviations appear in storefronts well below U.S. income and, other things equal, with large markets; the income implication is tested below, the market-size implication is not.

Cross-border arbitrage. A consumer in a richer market c' can buy in storefront c at a friction cost χ (a foreign payment method, address or identity check) that does not depend on the plan she buys. A subscriber who buys her own tier abroad obtains an identical product, so the vendor cannot sustain $p_{k,c} < p_{k,c'} - \chi$ on either paid tier without losing customers to its own cheaper storefront. A frontier subscriber gains no more from a cheap foreign entry tier than the entry tier's own price gap, because the capped product is worth $v(q_H - q_L)$ less to her than the frontier tier she buys at home. Arbitrage therefore caps the dollar gap between two storefronts by the same amount for every tier, and a gap that is a large proportional discount on a cheap capped tier is a small one on the frontier tiers.

3.3 Predictions

Let $\Delta p_{k,c}$ denote the price increment of tier k over the option below it, $p_{L,c}$ for the entry tier and $p_{H,c} - p_{L,c}$ for the frontier tier, Δc_k the corresponding increment in marginal cost, and $\ell_k = \Delta c_k / \Delta p_{k,c}$ the cost share of the increment. In the nested regime the vendor's profit is the sum of its profits on the two increments, so each is priced on its own. Appendix A shows that when log valuations have an increasing hazard rate, as lognormal valuations do, the income elasticity of the optimal increment, absent the platform default and arbitrage, is

$$\beta_k \equiv \frac{\partial \ln \Delta p^*_{k,c}}{\partial \ln y_c} = \frac{\xi_k}{\xi_k + \ell_k} = 1 - \ell_k \rho_k, \qquad \xi_k > 0, \tag{2}$$

where $\rho_k = 1/(\xi_k + \ell_k)$ is the rate at which the optimal increment passes through its marginal cost and ξ_k is the derivative, with respect to the log of the income-normalised price, of the cost share that the Lerner rule associates with that price (Proposition A.2). For lognormal valuations β_k falls monotonically with the cost share: an increment with zero marginal cost is priced in proportion to income ($\beta = 1$), and one whose price is mostly marginal cost is priced almost uniformly ($\beta \to 0$).

The frontier price is the sum of the two increments, $p^*_{H,c} = p^*_{L,c} + \Delta p^*_{H,c}$. At the vendor's unconstrained optimum a dollar off the entry price is therefore a dollar off the frontier price, and the frontier price inherits part of the entry tier's localisation even when its own increment is nearly uniform (Proposition A.3). A frontier price that stays at the platform default while the entry tier is localised is not the unconstrained optimum; it needs a constraint that holds it up. Arbitrage can play that role for the high-usage tiers but not for the $20 tiers: it caps dollar gaps equally on all tiers, and ChatGPT Go is listed $3–4 cheaper in the poorest storefronts than in the United States without, in the model, triggering arbitrage, a gap that would leave room for a 15–20% discount on a $20 tier but only 0.6–4% on a $100–500 tier. The override cost plays that role for the $20 tiers precisely when inference cost is a large share of the frontier increment, because the profit at stake in re-pricing the frontier tier is then small (Proposition A.4). The model then predicts:

(P1) *Frontier tiers at the default.* When inference cost is a large share of the frontier increment, the gain from re-pricing the frontier tier, including passing on the entry tier's discount, falls short of the override cost K; on the high-usage tiers, a friction of the size of the entry tier's discount would in addition limit any discount to a few per cent. The listed price then equals the default: the slope of $\ln(p_{H,c}/P^D_{H,c})$ on $\ln y_c$ is zero, and listed prices bunch at the default.

(P2) *Entry tiers localised, but less than proportionally.* For the capped entry tier L, a low cost share leaves a margin to cut, the dollar gaps at stake are no larger than the arbitrage friction, and the gain from deviating from the default is zero at U.S. income, positive below it and proportional to the number of potential subscribers. Where that gain clears K, the deviation $\ln(p_{L,c}/P^D_{L,c})$ rises with income, with a slope strictly between zero and one, deviations are concentrated in poorer storefronts, and holding the frontier price at the default restrains rather than deepens them. The prediction is one-sided: an AI tier whose price is lower where incomes are lower is a capped tier, but a capped tier whose margin or market is too small to clear K stays at the default.

(P3) *A dollar gap and a price ratio that rise as income falls.* Because $p_{H,c}$ is held at the default and $p_{L,c}$ is below it in poorer markets, the dollar increment $p_{H,c} - p_{L,c}$, net of the platform's conversion,

and the ratio $p_{H,c}/p_{L,c}$, both observed in the same storefront and free of currency and tax effects, rise as income falls. The ratio would also rise as income falls at the unconstrained optimum, but the dollar increment would fall, so the increment's sign separates a held frontier price from an unconstrained one.

(P4) *Within-firm contrasts.* Holding the dispersion of valuations and their scaling with income fixed, products of the same firm with near-zero marginal cost or with costs that scale with local revenue (streaming video or music, social-network features) are localised more than its AI tiers; among a firm's AI products, localisation is confined to capped tiers. Firm-level pricing capacity cannot then explain the AI pattern, although product-level differences in demand may contribute to it.

(P5) *Cheap access through less service, not a lower price.* A tier that is cheap where incomes are low must offer less service than the frontier tier: the frontier product at a lower local price would be bought by arbitrageurs and could fail to cover its inference cost. Service is one-dimensional in the model, so a strict usage cap on the frontier model and a weaker model are the same instrument here; which of the two vendors choose lies outside the model.

Predictions P1–P3 are tested against listed prices in Sections 5.3 and 5.4; P4 and P5 are consistency checks against the within-firm comparisons of Section 5.1 and the composition of the menus in Section 7, and Appendix A.5 sets out which statements listed prices can and cannot test. Appendix A also derives the model's demand-side implications for who uses AI and who pays (Proposition A.1), which Section 6 relates descriptively to the public usage indicators, since those do not observe paid adoption.

3.4 The affordability burden

The paper's affordability measure is the normalised price

$$AAB_c(p_c) = \frac{12\,p_c}{y_c} \times 100, \tag{3}$$

the annual price of a plan as a percentage of GNI per capita (Atlas method, current dollars). It follows the convention of the ITU's broadband affordability statistics and their 2% benchmark (International Telecommunication Union, 2025a), and it is reported at quintile means, relative to household consumption and at purchasing-power parity as sensitivities. Under uniform pricing the burden is a monotone transformation of income, so its cross-country gradient is arithmetic; the paper reports it because the price pattern of Section 5 is what makes it so. Dividing the dollar price by PPP income equals the Atlas-income burden under a constructed price $p_c = p \cdot PLR_c$, with $PLR_c = y_c/y_c^{PPP}$ a proxy for the local price level, that is, the burden if the vendor indexed its price to local price levels (Remark 2 in Appendix A).

4 Data

Table 2 lists sources and coverage, and Appendix B describes how the variables are constructed. The App Store has 218 two-letter storefront codes, the 217 World Bank economies plus Taiwan; 170 of them returned a local price list for at least one of the three assistants in the first wave and 173 for at least one app in the second; 162–165 of the first-wave storefronts (by app, including the United States) and up to 170 of the second-wave storefronts have income data, which the price regressions

require, and the regressions against the platform schedule also require an estimated storefront multiplier (161 storefronts).[1] The affordability calculations use the 201 economies with GNI per capita, and the availability table (Appendix Table E.4) counts all 217 classified economies.

Income. The main denominator is GNI per capita (Atlas method) for 2025, available for 182 economies. For the remaining 19 I use the latest value from 2022–2024. Nigeria's 2022 GNI per capita, for instance, has been revised from $2,140 to $2,940 in the current WDI vintage, while its 2025 value is $1,360 after the naira's depreciation. I use the official FY2027 groups throughout. For Argentina ($14,650) and Türkiye ($16,300) the 2025 values in the July 2026 WDI update exceed the high-income threshold, although both are classified as upper-middle income; the World Bank keeps them there because the Atlas method does not fully smooth their persistent high inflation (World Bank, 2026a).

Local prices, wave 1. For each of the 218 storefront codes and each of the three assistants, I retrieved the App Store product page on 27 September 2026 and extracted the storefront currency and the list of in-app purchases with their displayed prices. Plans are identified by name. Where a plan appears with monthly and annual variants, the monthly price is the modal price after discarding variants priced at five or more times the lowest one. Prices are converted to dollars at market rates of the audit day (open.er-api.com; an independent source agrees to within 1.6%). Requests redirected to the U.S. storefront are coded as no local listing observed (45 codes). Pages returning "not found" are coded as unavailable in the audit (ChatGPT: Belarus, China, Hong Kong SAR, Macao SAR, Russia and Venezuela; Claude: the same plus Afghanistan, Myanmar and Yemen; Gemini: Afghanistan, Belarus, China and Russia). App Store prices include value-added or goods-and-services taxes in almost all storefronts and exclude sales taxes in the United States and Canada, and they can exceed web prices because of platform commissions (Claude Max 5× costs $124.99 in the U.S. App Store against "from $100" on the web). The first wave stored parsed listing records rather than the original pages.

Local prices, wave 2, and the benchmark apps. On 16 October 2026 the same procedure was applied to 34 apps: ChatGPT, Claude and Gemini; Perplexity, Grok, Mistral, Poe, Character.AI and Microsoft 365 Copilot; 21 general subscription apps (YouTube, YouTube Music, Google One, Duolingo, Telegram, Discord, Snapchat, LinkedIn, Dropbox, Headspace, Calm, Strava, Tinder, Bumble, Truecaller, Babbel, X, Lightroom, NordVPN, Picsart, Evernote); and four apps that sell AI features inside a general product (Canva, CapCut, Notion, Grammarly). Of 7,412 app-storefront queries, 7,218 returned a page (5,688 local, 1,530 redirected to the U.S. store) and 194 returned "not found"; 173 storefronts returned at least one local price list, 122 local pages carried no purchase list, and the parsed lists contain 30,817 item-storefront prices. Apple's product page lists at most ten in-app purchases (4,632 of the 5,566 non-empty lists have exactly ten), so an item missing from a full list is censored rather than unavailable, and a plan's monthly variant can be pushed off the list while its annual variant remains. Two rules address this (Appendix B): a price recorded as monthly that exceeds three times the platform's reference for the item's U.S. monthly price is dropped, which flags 151 of the 21,938 item-storefront cells with a U.S.

[1] Storefront arithmetic: 218 codes minus 45 that redirect to the U.S. store leaves 173 active storefronts; Belarus, China and Russia returned "not found" for all three assistants in the first wave (170). The default panel was retrieved in the 167 storefronts in which ChatGPT returned a local page (all active storefronts except Belarus, China, Hong Kong SAR, Macao SAR, Russia and Venezuela); 164 of them have a multiplier (all but Nigeria and Türkiye), 161 of those have income data (Taiwan, the British Virgin Islands and Yemen do not), and 108 of the 111 storefronts with a clean multiplier have income data.

price and income data, concentrated in five items, and changes the Character.AI+ slope from -0.092 to 0.034 (Appendix Table E.11); and items that the same firm lists under two apps or two spellings are counted once, which removes five of the 152 items listed with a U.S. price in at least 60 storefronts from the item-level statistics. Apple changed the product page's format between the two waves, and the parser handles both; OpenAI renamed Pro $5\times$ and $20\times$ to Pro 100 and Pro 200 and added Pro 500. The second wave recorded the SHA-256 fingerprint of every page, so the extraction can be replayed against the page archive, which is deposited with the replication package (Appendix B). Items are classified as subscriptions or consumables (credits, gems, boosts) by name. All 1,586 plan-storefront prices of the ten AI plans observed in both waves were unchanged, in local currency, 19 days later (Appendix Table E.12); the window is too short to speak to medium-run repricing or exchange-rate pass-through.

The platform schedule. Apple's own equalisation table is available only through a developer account, so the price that the platform generates automatically is estimated from a panel of 30 small-developer subscription apps (scanner, weather, habit, fitness and similar utilities), chosen on their U.S. listings from 85 candidates returned by App Store searches for such categories (candidates, search terms and U.S. purchase lists are in the package) and retrieved on the same day in the 167 storefronts in which ChatGPT returned a local page (5,007 of 5,010 requests returned a page; two of the 30 apps list no priced subscription item). Such developers rarely set storefront-specific prices, so for each storefront and U.S. price point the modal local price across the panel's items, when it is unique and accounts for at least half of the items, is taken as the schedule price; a storefront multiplier is the median of the schedule price divided by the U.S. price over price points of $2.99 and above. The procedure yields 1,551 schedule prices and a multiplier for 164 storefronts (all but Nigeria and Türkiye); the multiplier divided by the market exchange rate is the tax factor the platform applies. The panel's items are priced between $0.99 and $69.99 a month, with a few at $74.99–169.99, so the schedule price is observed directly only for AI plans priced at $4.99, $9.99 or $19.99, and for the $19.99 plans only in about three quarters of the storefronts; for every other plan, including ChatGPT Go at $8.00 and the $100–500 tiers, the reference is the storefront multiplier times the U.S. price. Section 5.3 describes the schedule, its validation and its vintage.

Cross-channel prices. To check that the pattern is not specific to Apple's store, the same day's Google Play product pages were retrieved for ChatGPT, Claude, Gemini, Perplexity, Grok, YouTube, X, Snapchat, Duolingo, Spotify, Telegram and Microsoft's Copilot listing in 30 countries (the `gl` parameter, not the requesting address, sets the country, and the currency of every page matched its country), and Microsoft's web store pages for Microsoft 365 Personal, Family and Premium in 63 locales covering 57 countries. Google Play shows only the lowest and highest in-app price per app, so the lowest ChatGPT item ($8.00 in the United States, the Go price) and the lowest Claude item ($20.00, Pro) are the comparable quantities; Gemini shows no price range because its plans are sold through Google One; the Copilot listing is the former Office hub app, whose range mixes products; Microsoft's Gulf prices are in dollars; and no page states whether its prices include tax. OpenAI's and Google's own web pages localise by the requesting address and could not be collected for many countries at zero cost. Appendix C reports the comparison.

Diffusion. The Microsoft AI User Share is the estimated share of people aged 15–64 who used a generative-AI product during the period, derived from anonymised Windows telemetry on visits to 19 AI services and scaled for device, mobile and internet shares (Misra et al., 2025); 36 of its 147 economies, including 20 of the 23 low-income economies in the data, are pooled into regional aggregates with identical series. The Anthropic AI Usage Index for country c is the country's share of Claude.ai conversations divided by its share of the world's working-age population, from privacy-preserving samples of conversations (Appendix Table B.1) (Handa et al., 2025; Appel et al., 2025, 2026; Massenkoff et al., 2026a,b; Anthropic, 2026a). The consumer series pools free and paid accounts, excludes API and enterprise traffic, and is published only for countries with at least 200 sampled conversations. I recompute the index from the sampled counts for the 2025 and February 2026 windows for all geolocated economies (165–169) and apply the publication floor in the cross-sectional regressions (113–120 economies), as Appendix B describes. Google's ATLAS reports a per-capita adoption quintile for Gemini products in April 2026 (Iscenko et al., 2026).

Computational workflow. Generative AI tools (ChatGPT and Codex by OpenAI, Claude by Anthropic) assisted with code development, extraction review and drafting; an AI browser agent executed the second-wave, default-panel and cross-channel collection scripts under the author's direction; and AI subagents performed numerical, mathematical and bibliographic audits of an earlier version. The parsers and classification rules were written with these tools and checked by hand on samples of pages, and the list-truncation guard and the de-duplication rule were added after an audit of the extraction found two errors. The author verified the sources, analysis and text. The replication package (Appendix B) contains the frozen inputs, scripts and outputs; re-running it reproduces every number in the paper, and independent reimplementations of the key regressions match the reported estimates.

4.1 Estimands and inference

Four estimands recur. For items A and B sold by the same firm in storefront c, the *within-firm relative price* is $d_{AB,c} = \ln(p_{A,c}/p_{B,c}) - \ln(P_A^{US}/P_B^{US})$: both prices are listed in the same storefront under the same currency, tax rule and platform conversion, so all three cancel, and $d_{AB,c}$ measures how much more the firm has localised A than B relative to their U.S. prices. Its slope on $\ln y_c$ is the paper's main estimate of where localisation appears; it needs no estimate of the platform's schedule. The *U.S.-dollar price regression* $\ln p_{k,c} = \alpha_k + \beta_k \ln y_c + u_{k,c}$ for plan k is the cross-country association of listed consumer prices with income; because taxes and the platform's currency conversion enter the listing, it is not the vendor's localisation rule, and the estimated schedule has a slope of 0.033 against which it should be read. The *deviation regression* $\ln(p_{k,c}/P_{k,c}^D) = a_k + b_k \ln y_c + e_{k,c}$ removes the estimated schedule and is the paper's estimate of active localisation of plan k; b_k corresponds to the slope that predictions P1 and P2 concern. Shares of storefronts within $\pm 3\%$ of the schedule and more than 10% below it, and median deviations by income group, describe the same object without a functional form. The *within-storefront ratio* $p_{H,c}/p_{L,c}$ and the dollar increment $p_{H,c} - p_{L,c}$ use two listings observed in the same storefront and cancel currency and any proportional tax. All regressions are ordinary least squares with equal storefront weights and HC1 standard errors (HC3 errors, used in the first-wave tables, differ from the HC1 errors of the second-wave plans by less than 0.0002); because storefronts that share a currency share the platform's conversion and 102–104 of them are priced in U.S. dollars, the headline estimates are

also reported with standard errors clustered by currency (about 40 clusters, one of them the dollar bloc), with randomisation inference, and separately for dollar and local-currency storefronts, and the deviation slopes are reported weighted by working-age population. Because $\ln AAB_{k,c} = \ln(1200) + \ln p_{k,c} - \ln y_c$, the burden slope on the same sample is $\beta_k - 1$; a constant reference price imposes a slope of -1 by construction, which is why the usage indicators of Section 6 are related to income rather than to burdens.

5 Results

5.1 Relative prices within firm and storefront

The cleanest evidence needs no estimate of the platform's schedule. When one firm lists two products in the same storefront, both prices carry the same currency, the same tax and the same platform conversion, so the log ratio of the two local prices, net of the ratio of their U.S. prices, measures how much more the firm has localised one product than the other. Table 3 reports, for 21 within-firm pairs,[2] the slope of that relative price on log income, its median in low- and lower-middle-income storefronts, the share of storefronts in which the first product is more than 10% cheaper relative to the second than in the United States, and the share in which the two relative prices coincide. Figure 1 plots the slopes.

Three results stand out. First, the relative prices of the AI tiers with large allowances vary little with income. Measured as the higher tier relative to the firm's standard tier, the nine frontier pairs have slopes between -0.022 and 0.011: ChatGPT Pro 100, Pro 200 and Pro 500 relative to Plus -0.021, -0.007 and -0.022 and SuperGrok Plus relative to SuperGrok -0.008 (randomisation $p < 0.0005$, 0.0035, < 0.0005 and 0.11), so these higher tiers are relatively dearer, not cheaper, in poorer storefronts; Google AI Ultra relative to AI Pro and SuperGrok Heavy relative to SuperGrok 0.000; Claude Max 5× and 20× relative to Pro 0.009 and 0.007 and X Premium Plus relative to SuperGrok 0.011, slightly cheaper (p 0.0015, 0.0275 and 0.013). The two products are listed at the same relative price as in the United States in 45–95% of the storefronts, depending on the pair. The cross-vendor version of the same comparison, which measures every plan relative to ChatGPT Plus or to the median of the three $19.99 frontier plans in the same storefront, gives slopes between -0.022 and 0.003 for the $20–500 tiers of all six vendors and a median relative price of exactly one in poorer storefronts (Appendix Table E.10). Second, the same firms localise their other subscriptions heavily. Relative to Google AI Pro, YouTube Premium rises with income with a slope of 0.20 (currency-clustered s.e. 0.09), YouTube Premium Lite 0.34 (0.08) and YouTube Music 0.33 (0.07), and all three are 57–73% cheaper relative to AI Pro in the median poorer storefront than in the United States; relative to X Premium Plus or SuperGrok, X Premium has a slope of 0.15–0.16 (0.02) and is 55% cheaper in the median poorer storefront, and X Premium Basic relative to SuperGrok Lite 0.16 (0.04) and 50% cheaper. Microsoft's contrast is moderate: Microsoft

[2] Inclusion rule. Table 4 stacks every named monthly consumer plan of the five firms that sell both AI tiers and other subscriptions through the App Store (OpenAI, Google, X Corp/xAI, Microsoft, Character.AI) with a U.S. price of $4 or more and listed in at least 60 storefronts, excluding annual and family variants, storage plans (Google One 100 GB, Microsoft 365 Basic), the Google One app's second listing of Google AI Plus (2 TB) and X's creator and post-promotion products; Table 3 measures each of those plans, and each of Anthropic's, relative to its firm's standard frontier tier (SuperGrok, $30, for X Corp/xAI; Character.AI+, $9.99, for Character.AI), whether the plan is a lower or a higher tier, and adds two X Corp pairs that hold the app fixed (X Premium against X Premium Plus) or compare the two entry products (X Premium Basic against SuperGrok Lite). The excluded same-firm items and their slopes are in Appendix Table E.9: the storage plans Google One 100 GB (0.033, s.e. 0.007) and Microsoft 365 Basic (0.029, 0.011) and X's creator subscription (0.032, 0.009) are not localised either, and the family variants YouTube Premium Family (0.36, 0.04), YouTube Music Family (0.38, 0.04) and Microsoft 365 Family (0.056, 0.010) move with the plans they extend.

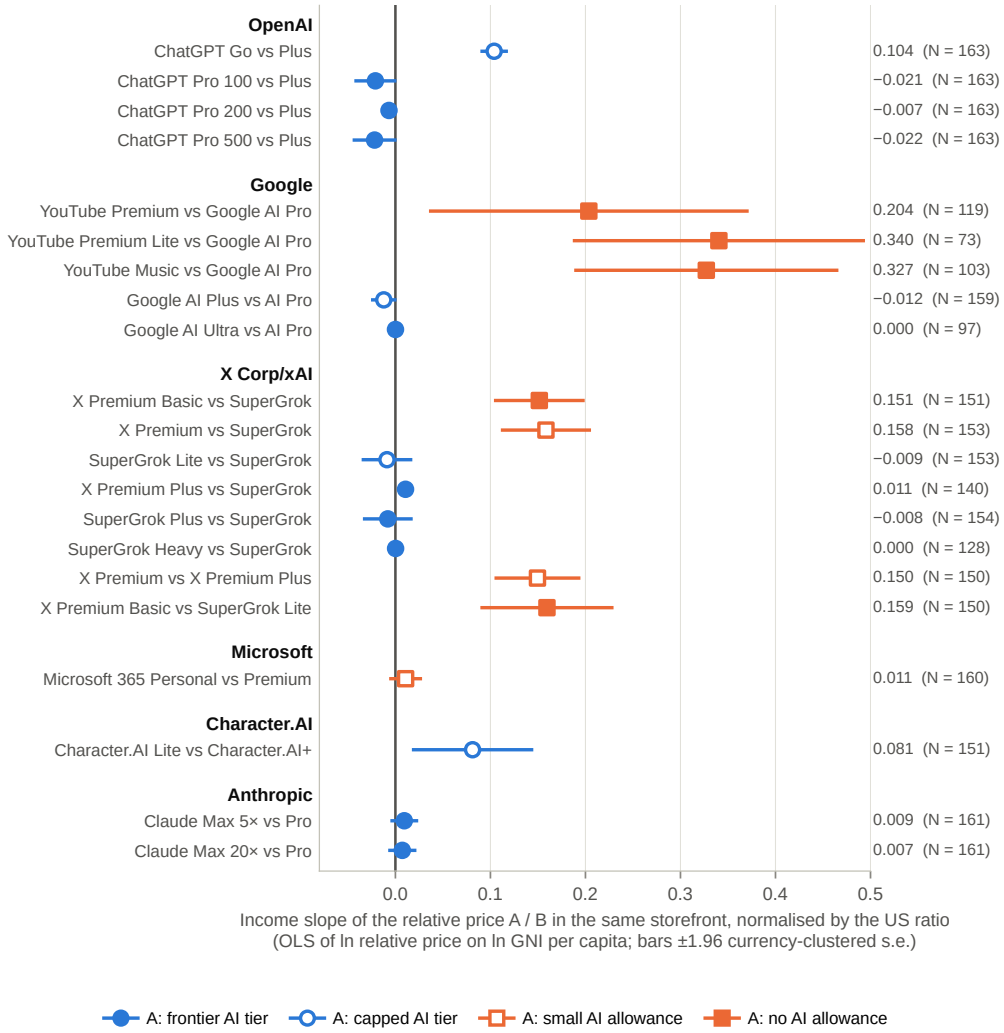


Figure 1: Income gradients of relative prices within firm and storefront. Slopes of Table 3 with ±1.96 currency-clustered standard errors; a positive value means that the first product of the pair is localised more than the second.

365 Personal relative to Premium has a slope of 0.011 (0.009), and Personal is 15% cheaper in the median poorer storefront but also cheaper in many rich ones, a level difference as much as an income gradient. The same firms' low-priced storage plans, Google One 100 GB and Microsoft 365 Basic, are not localised either (U.S.-dollar slopes 0.033 and 0.029; Appendix Table E.9); like AI tiers they carry a dollar-denominated cost of serving, so the within-firm evidence is consistent with a contrast between products that are costly to serve in dollars and products whose costs are near zero or scale with local revenue, rather than between AI and non-AI products as such; at $1.99 the storage plans' small stakes could also keep them at the default. Third, localisation among AI products is confined to capped tiers, but not every capped tier is localised. ChatGPT Go relative to Plus has a slope of 0.104 (0.007), 38% cheaper in the median poorer storefront and more than 10% cheaper in 80% of storefronts; Character.AI Lite relative to Character.AI+ has a slope of 0.08 (0.03) and is 40% cheaper. Google AI Plus relative to AI Pro and SuperGrok Lite relative to SuperGrok, by contrast, have slopes of −0.012 and −0.009 and a median relative price of one: Google and xAI keep even their capped AI tiers at the U.S.-anchored ratio.

Table 4 pools the comparisons. In a regression of the relative price on income with firm-by-storefront and item fixed effects, in which the omitted group is each firm's AI tiers with large allowances, the prod-

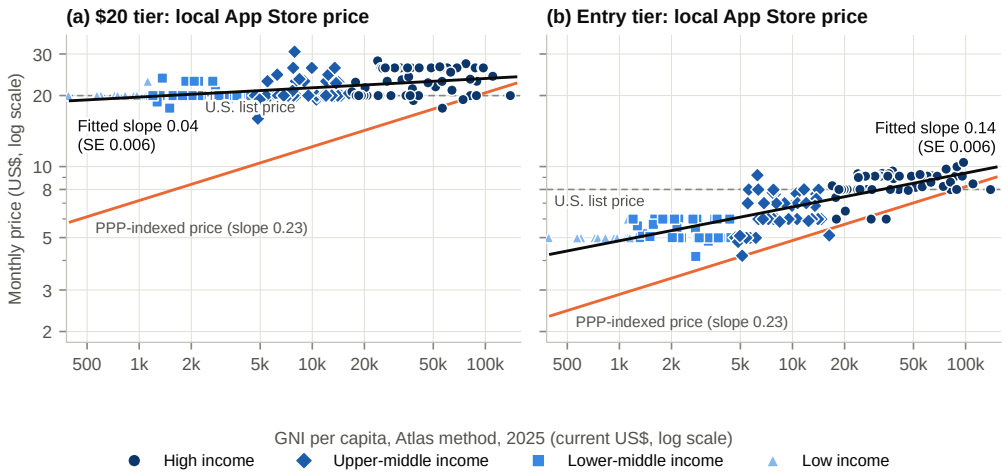


Figure 2: Local App Store prices of ChatGPT Plus and ChatGPT Go against income. Tax-inclusive monthly prices in each national App Store storefront, converted to dollars at market exchange rates on 27 September 2026. Black lines: fitted log-log regressions. Orange lines: prices indexed to local price levels, anchored at the U.S. price, with the slope on income (0.23) of the Atlas-to-PPP income ratio, a price-level proxy. Grey lines: U.S. list prices.

ucts with no AI allowance rise with income 0.20 log points per log point faster than those tiers (two-way clustered s.e. 0.044), the products with a small AI allowance 0.12 faster (0.051), and the capped AI tiers 0.04 faster (0.031). Because the comparison rests on five firms, inference is also by exact randomisation: permuting the group labels among the items within each firm in every one of the 252,000 possible ways, a gap as large as the observed one arises for the no-allowance group in 0.05% of relabellings and for the small-allowance group in 0.006%, but for the capped group in 41%. The gradients are present within the dollar storefronts for the products without an AI allowance (0.10, s.e. 0.03) and larger within the local-currency storefronts, so they are not an artefact of the platform's currency rules. The within-firm comparisons show that firm-level pricing capacity or attention does not explain the uniform AI tiers: the firms that keep Google AI Pro, SuperGrok and Microsoft 365 Premium at the U.S.-anchored price set income-graded prices for YouTube, X Premium and Microsoft 365 Personal through the same storefronts in the same month. They do not rule out product-level differences that coincide with the AI allowance, such as content royalties, the use to which a product is put, brand position or product age (Section 7). Products with large AI allowances are left at the U.S.-anchored price, products with no or small allowances are often localised, and AI tiers as a class are not unusual among digital subscriptions: at the app level, two of eight AI apps and eight of 21 non-AI apps localise at least one subscription (Fisher exact $p = 0.67$; Appendix Table E.8), so the identifying comparison is within firm rather than across classes of app.

5.2 Income slopes of listed prices

Figure 2 plots the local App Store price of ChatGPT Plus and ChatGPT Go against per-capita income in the first wave, and Table 5 reports, for the seventeen AI and Microsoft plans of the second wave, the slope of the log dollar price on log income and the deviation from the platform's schedule described in Section 5.3. The first-wave estimates for nine of the ten plans of the three assistants, with medians by income group and currency-clustered intervals, are in Appendix Tables E.1 and E.2 (Claude Max 20×, whose second-wave slope is 0.033, s.e. 0.006, is omitted from the first-wave table).

The standard and high-usage tiers of every vendor have nearly identical dollar prices across countries. In the first wave the income slope is 0.039 (s.e. 0.006) for ChatGPT Plus, 0.026 for Claude Pro and 0.037 for Google AI Pro, and the slopes of the high-usage tiers of the first wave ($100–250) lie between 0.019 and 0.035; the second wave gives 0.016–0.039 for the $20 and $100–500 tiers of the six vendors (Table 5), against 0.033 for the platform's schedule. For reference, the price-level proxy (the Atlas-to-PPP income ratio) has a slope of 0.228 across 197 economies and 0.243 on the 164 economies of the first-wave price sample, and proportional pricing would imply a slope of one. The median local price of ChatGPT Plus is $19.99 in upper-middle-, lower-middle- and low-income storefronts alike and $22.99 in the median high-income storefront, a difference that the next section attributes to taxes. The dollar slopes above 0.10 belong to two capped AI tiers: ChatGPT Go has a slope of 0.143 in both waves, with first-wave medians of $9.05, $5.99, $5.34 and $4.99 across the four income groups and lows of $4.16 in India and $4.19 in Indonesia, and Character.AI Lite 0.12 (Appendix Table E.9). Microsoft 365 Personal, which bundles a small Copilot allowance into an office suite, has a slope of 0.047; Google's entry tier, AI Plus, is flat at $4.99 in each of the three non-high-income groups, and SuperGrok Lite has a slope of 0.018.

5.3 The platform's schedule: decomposition, validation and vintage

The small positive slopes of the frontier tiers could be vendor pricing or the platform's own conversion. The small-developer panel separates the two. The estimated schedule has an income slope of 0.033 (s.e. 0.006; 161 storefronts): because the platform adds value-added taxes that are levied mostly by higher-income storefronts, a plan left entirely at the schedule shows a slope of about 0.03 in dollars, so the frontier tiers' slopes of 0.026–0.039 are the platform's conversion rather than vendor pricing. Figure 3 and the deviation columns of Table 5 measure each plan's departure from the schedule price for its U.S. price point in each storefront.

Frontier tiers. Prediction P1 holds in the sense that the frontier tiers show no income-graded localisation: the deviation slopes of the $20 and $100–500 tiers of the six vendors lie between -0.016 (ChatGPT Pro 500) and 0.004 (ChatGPT Plus), every 95% interval lies within ± 0.025, a quarter of ChatGPT Go's slope, and the median deviation is zero in low-, lower-middle- and upper-middle-income storefronts (in high-income storefronts it is zero for seven of the 13 tiers and within ± 0.04 for the rest). It does not hold in the sense of every listing sitting on the schedule. ChatGPT Plus is within 3% of the reference in 87% of storefronts, 93% where the reference is an observed modal price and 68% where it is extrapolated, and on the 108 storefronts whose multiplier is clean (its spread across price points below 20%) in 95%; the other frontier tiers are within 3% in 55–73% of storefronts. Listings depart from the reference in both directions. Claude Pro, Claude Max, Perplexity Pro and SuperGrok are listed more than 3% above the extrapolated reference in 25–37% of storefronts and more than 10% above in 4–15%; because different vendors list identical round prices in the same storefronts (the two $20.00 plans, Claude Pro and Perplexity Pro, both at $25.00 in all 16 taxed dollar storefronts, and both at $22.00 in the same eight dollar storefronts whose panel tax factor is one), much of this is likely the platform's own conversion, either its rounding of round-number price points, which the panel's items, almost all at $X.99 price points, cannot reveal, or tax updates that post-date the panel's prices, rather than vendor pricing. Google AI Pro, Google AI Ultra, Microsoft 365 Premium and ChatGPT Pro 500 are more than 10% below the

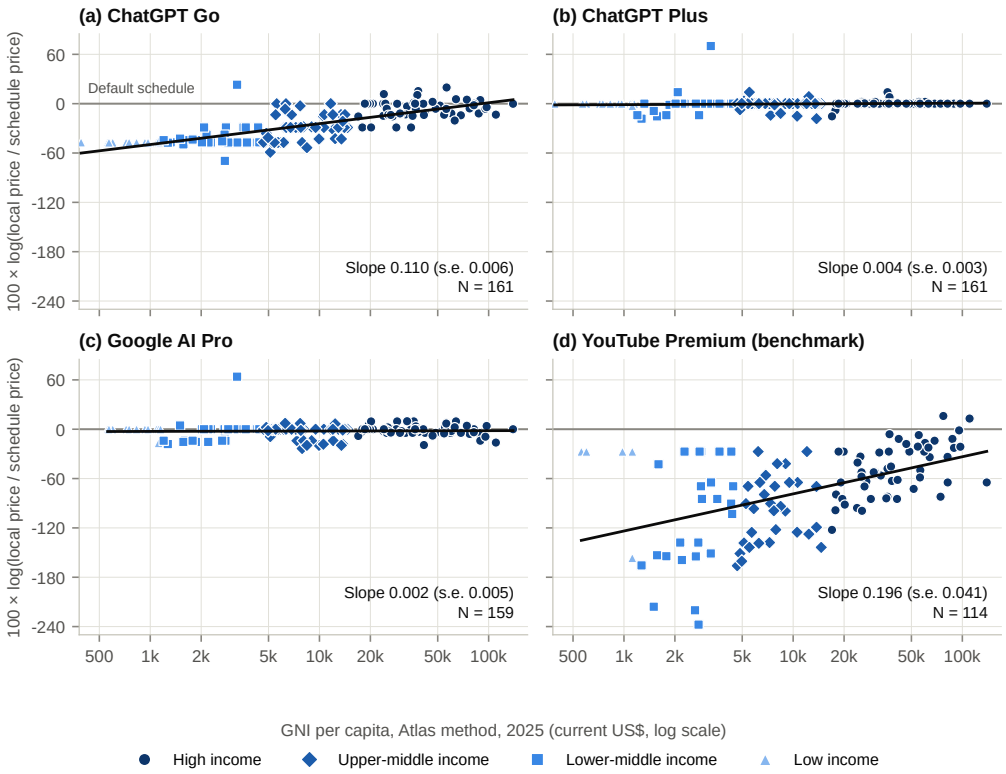


Figure 3: Deviation of listed prices from the platform's schedule, 16 October 2026. Each dot is a storefront; the vertical axis is $100 \times \ln$(listed price / reference price), where the reference is the panel's modal price for the plan's U.S. price point when one is observed (ChatGPT Plus and Google AI Pro, in about three quarters of storefronts) and the storefront multiplier times the U.S. price otherwise (every storefront for ChatGPT Go and YouTube Premium); the horizontal axis is GNI per capita on a log scale; lines are OLS fits with HC1 standard errors. Panels (a)–(c): AI plans; panel (d): YouTube Premium, a subscription without an AI allowance sold by the vendor of Google AI Pro through the same storefronts.

reference in 11–16% of storefronts. Part of this is the vintage of the schedule, examined below; part is deliberate. The slopes are robust to the exchange-rate source, to the treatment of ties in the modal rule, to restricting the sample to the 60 storefronts not priced in dollars, to population weights and to HC3 errors (Appendix E). In the 16 dollar-priced storefronts whose schedule embeds a tax of more than 5%, Google lists AI Pro at exactly $19.99 in 15 and AI Ultra at $199.99 in all 8 in which it is offered, Microsoft lists 365 Premium at $19.99 in 10 and OpenAI lists Plus at the tax-inclusive reference in 11 (Appendix Table E.13): some vendors override the platform's tax-inclusive price to keep a round dollar price. Ten of these 16 storefronts are low- or lower-middle-income economies; Google lists AI Pro at $19.99 in all ten and OpenAI lists Plus at $19.99 in five. A vendor that overrides the default there and still chooses the U.S. dollar price is not held by the cost of overriding; these listings fit a deliberate uniform dollar price, as in the segmented-demand case of Remark 1, better than the override-cost mechanism. The robust fact is narrower than "frontier tiers sit on the schedule", and it is the fact that matters for the model: frontier prices are not lower where incomes are lower relative to the platform's conversion, even though vendors actively set storefront-specific prices for them.

Entry tiers. Prediction P2 holds for ChatGPT Go. Its deviation has a slope of 0.110 (s.e. 0.006): only 14% of storefronts list it within 3% of the reference, 75% list it more than 10% below, and in low- and lower-middle-income storefronts the median listed price is 38% below the reference (98% of the 50 such

storefronts are more than 10% below). The deviation is smaller but present in high-income storefronts, where the median is 6% below, and the plan is within 0.5% of the reference in 18 storefronts, 17 of them priced in U.S. dollars and most of them small high-income economies in the Gulf, the Caribbean and the Pacific. Microsoft 365 Personal shows the same pattern in weaker form, a slope of 0.021 (s.e. 0.007) and a median 15% below in poorer storefronts, and Character.AI Lite a slope of 0.091 (0.012) and a median 40% below. Google AI Plus and SuperGrok Lite, the two other capped AI tiers, have deviation slopes of -0.003 and -0.016 and a median deviation of zero: as Section 5.1 found, localisation among AI products is confined to capped tiers, but two of the four capped tiers are not localised.

Validation of the schedule. Because the schedule is an estimate from small developers' prices, Table 6 tests it in five ways. Rebuilding it from the panel's items reproduces the shipped multipliers exactly. Its income slope is 0.0327 on the main rule, 0.0272 on the strict-majority rule and 0.0331 on the panel's subscription items alone; leaving out each app in turn moves the slope within 0.029–0.035, and across 300 random splits of the apps into two halves the slopes of the half-samples lie between 0.021 and 0.041 (5th–95th percentile). The schedule also coincides with what the AI vendors list: where the panel observes a modal price at a frontier plan's price point, ChatGPT Plus is listed at that price to the cent in 117 of 126 storefronts, Google AI Pro in 73 of 124 and Microsoft 365 Premium in 67 of 125, and in the 33 storefronts whose schedule departs from exchange-rate parity by more than 20% a frontier plan sits at the panel's price in 24 of 69 cases. The schedule is weak exactly where it matters most for the poorer markets: in India, Brazil, Egypt, Uganda, Nigeria and Türkiye the panel's items spread across prices (median modal share 0.33–0.50), no price point of a frontier plan has a unique mode, and Nigeria and Türkiye have no multiplier (Appendix Table E.14); for ChatGPT Go, whose $8.00 price point no panel item uses, and for the $100–500 tiers, the reference is extrapolated with the multiplier everywhere. These are the reasons the paper leads with the within-firm comparisons.

Vintage. Apple does not refresh the generated price of an auto-renewable subscription after later tax or exchange-rate changes (Section 2.2), so the panel's prices and the AI plans' prices are each the platform's conversion of the date on which the price was set, and a deviation can arise from a difference in vintages rather than from a decision. If so, deviations of the frontier tiers should be centred on zero, should be larger in storefronts whose currency has moved more since the plans were priced, and should be rare in dollar storefronts, where only tax changes can create a vintage difference. The first two hold and the third largely (Appendix Table E.15). Pooling the ten frontier plans (1,489 plan-storefront rows), the deviation has an income slope of -0.0014 (s.e. 0.0028, clustered by storefront) and a median of zero; its absolute value rises with the range of the currency's log exchange rate since March 2024, with a coefficient of 0.54 (0.23), and 81% of rows are within 3% of the reference in dollar storefronts, where most of the rest are the deliberate overrides of Appendix Table E.13 or tax updates that post-date the panel's prices, against 41% in local-currency storefronts whose currency moved by less than 10% and 22% where it moved by more than 30%. Controlling for the currency's movement leaves the frontier income slope at -0.007 (0.007) and ChatGPT Go's at 0.108 (0.010): the entry tier's localisation is not a vintage effect, and the frontier tiers' scatter largely is.

Other channels. The App Store is one channel, and its rules (dollar denomination of small storefronts, a 15–30% commission) could in principle produce uniform frontier prices that vendors do not practise

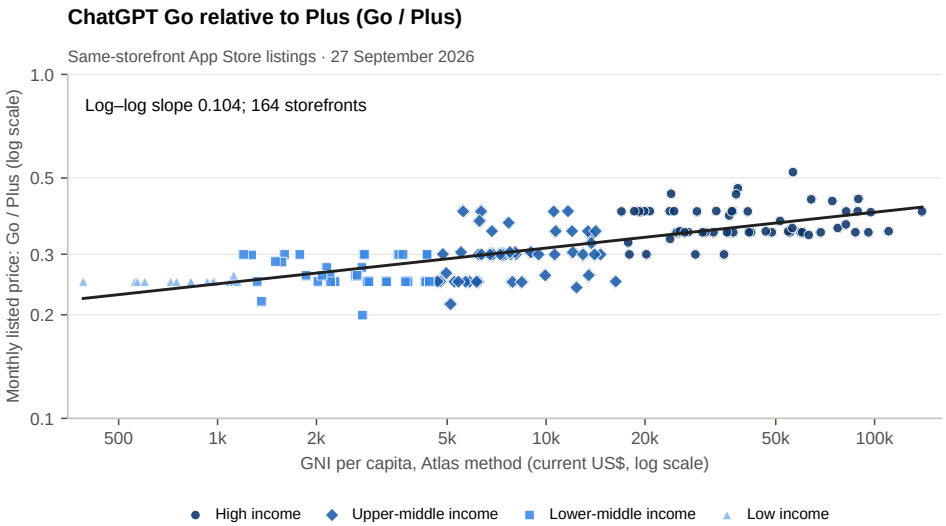


Figure 4: The ChatGPT Go-to-Plus price ratio within storefronts. Both plans as listed in the same storefront on 27 September 2026 (unchanged on 16 October); each dot is a storefront. The line is the log–log fit.

elsewhere. Appendix C compares the same day's Google Play and Microsoft web-store prices. Google Play's lowest ChatGPT item, the Go price, is within 5% of the App Store Go price in 22 of the 25 countries where the two stores price in the same currency (identical in 20; median log difference zero) and is 37% below the U.S. price in the median lower-income country, while Claude's lowest item, Pro, is within 5% of the App Store price in only 12 of 25 and at 1.05 times the U.S. price in the median lower-income country; Microsoft's web prices for 365 Personal and Premium are each within 5% of the App Store prices in all 49 same-currency countries (identical to the cent in 37 of 49, median difference zero), and on the web the Personal-to-Premium ratio rises with income with a slope of 0.082 (0.018), with India 31% below the U.S. ratio on both channels. The localised entry tier and Microsoft's frontier price are therefore vendor policies rather than artefacts of Apple's store; because Google Play shows only the lowest item, the check is silent on the frontier tiers of OpenAI and Google, and for Claude it is weak (Appendix C).

5.4 The within-storefront ratio and the dollar increment

Because both plans are listed in the same storefront under the same currency and tax rule, the ratio of the standard price to the entry price is free of the conversion that affects dollar prices. Prediction P3 holds for the ratio: ChatGPT Plus costs 2.5 times ChatGPT Go in the United States, 2.9 times in the median high-income storefront, 3.3 in the median upper-middle-, 3.8 in the median lower-middle- and 4.0 in the median low-income storefront, identically in both waves (Figure 4; 167 storefronts in the second wave). The log Go-to-Plus ratio has an income slope of 0.104 (currency-clustered s.e. 0.007; 164 economies; equivalently -0.104 for the Plus-to-Go ratio), which on a common sample is exactly the difference between the Go and Plus slopes; it is 0.113 within the 102 USD-priced storefronts, 0.112 without them and 0.125 without USD and EUR storefronts (Appendix Table E.3), so the entry-tier gradient is not an artefact of currency conversion.

The ratio rises as income falls under every regime of the model, so the discriminating quantity is the dollar increment between the two plans. Table 7 measures it two ways. Relative to the increment between the two plans' schedule prices, the Plus-to-Go dollar gap is 4% wider than the schedule gap

in the median high-income storefront, 17% wider in the median upper-middle-income storefront and 25% wider in the median lower-middle- and low-income storefronts, with an income slope of −0.047 (s.e. 0.005; 161 storefronts); in dollars relative to the U.S. increment of $11.99, the slope is −0.010 (0.007; 163 storefronts), the difference between the two normalisations being the schedule's own slope. Character.AI's increment behaves the same way (−0.059, s.e. 0.012), whereas the increment of Pro 200 over Plus, two frontier tiers, and that of AI Pro over AI Plus, neither of which is localised, are flat relative to the schedule; the increment of 365 Premium over Personal, whose lower plan is moderately localised, is 15% wider in the median poorer storefront (slope −0.014, s.e. 0.008), the same pattern in weaker form. A wider gap where incomes are lower is what a held frontier price produces, whereas at the unconstrained optimum the frontier price follows the entry price down and the increment narrows (Proposition A.3). The data reject the unconstrained optimum of the nested menu. They are consistent with a frontier price held near the default and equally with the segmented-demand case of Remark 1, which listed prices cannot separate.

5.5 Affordability

With frontier prices at the U.S.-anchored level everywhere, burdens are set by income. Table 8 consolidates the affordability measures by income group, and Figure 5 shows the $20 plan across the income distribution. At a uniform $20 the median burden is 0.65% of per-capita income in high-income economies, 3.09% in upper-middle-, 8.73% in lower-middle- and 28.9% in low-income economies, a 45-fold difference between the first and last groups; at the tax-inclusive local price of ChatGPT Plus in the economies with a listing the medians are 0.75%, 3.27%, 8.89% and 28.9%. The $20 tier meets the 2% benchmark in every high-income economy, in 22% of upper-middle-income economies and in no lower-middle- or low-income economy, and the 113 of 201 economies above it are home to 60% of the covered working-age population. Indexed to local price levels the burdens would fall to 0.40%, 1.20%, 3.38% and 10.7%, still five times the benchmark in the median low-income economy. For the poorest fifth of households a $20 plan costs a median 1.6% of mean income in high-income economies but 33% in lower-middle- and 73% in low-income economies, and it fails the benchmark even for the richest fifth in every low-income economy with survey data and in 98% of lower-middle-income economies. Only the entry tiers fall below 3% in the median lower-middle-income economy: ChatGPT Go, the localised tier, at 2.6%, and Google AI Plus, which is not localised but has a $4.99 U.S. price, at 2.1% (Appendix Table E.4); in the median low-income economy both cost 7.2%. Bringing the $20 tier to the benchmark under uniform pricing would require a subsidy of 35%, 77% and 93% of the price in the three poorer groups ($84, $185 and $223 per subscriber a year), or 0.08%, 0.43% and 1.55% of GNI if one tenth of the working-age population were covered; applying the entry tier's U.S.-dollar income slope (0.143, which includes the schedule's 0.033) to the $20 tier would lower the median low-income burden only to 14.8%, and applying only its deviation from the schedule (0.110) to 17.3% (Appendix Table E.7). These are accounting magnitudes, not estimates of behaviour, and the 2% benchmark is a broadband convention rather than a threshold derived for AI subscriptions.

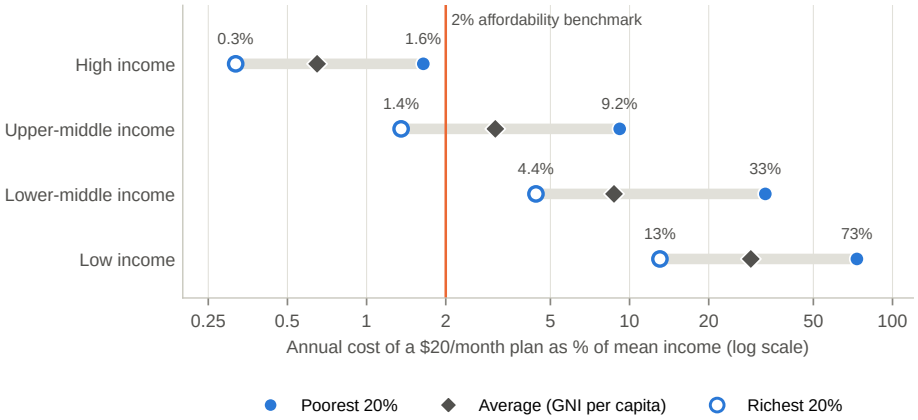


Figure 5: Burden of a $20 plan across the income distribution. Medians across economies by income group of the annual cost of a $20/month plan as a share of the mean income of the poorest quintile, of the population (GNI per capita), and of the richest quintile. Quintile incomes are approximated as GNI per capita times the survey income share relative to 20%. The vertical line marks the 2% broadband affordability benchmark.

6 The public usage indicators: what they can and cannot show

The model's demand side implies that, with a free tier in every market and paid tiers priced above the free hurdle, whether people use AI at all depends on connectivity, skills and language rather than on the price of paid tiers, whereas paying for more capability depends on the ratio of price to income (Proposition A.1). No public indicator observes paid adoption, so the two usage indicators cannot test that implication; Appendix D reports what they show. Any use is moderately income-graded: across the economies with country-specific Microsoft estimates, the elasticity of the AI user share with respect to GNI per capita is 0.370 (s.e. 0.021), unchanged by controls for internet use, education, urbanisation, age structure, digital payments and region, and median user shares in the second quarter of 2026 were 32.3%, 18.8%, 12.0% and 7.6% from the high- to the low-income group. Per-capita activity on one frontier assistant is twice as income-graded: on the 96 economies covered by both indicators, the elasticity of the Anthropic AI Usage Index is 0.711 (s.e. 0.036) against 0.371 (s.e. 0.026) for the Microsoft user share, a difference of 0.340 (clustered s.e. 0.034) that survives the robustness checks of Appendix Table D.4. What the contrast identifies is limited by the measurement of the two indicators. The index pools free and paid accounts: its gradient is the sum of the gradients of the share using any generative-AI product (which the Microsoft indicator measures), of Claude's share among AI users and of conversations per Claude user, and the last of these is an attenuated version of the paid-margin gradient that Proposition A.1 concerns (Appendix A); Anthropic's publication floor censors small low- and lower-middle-income economies, so the common sample contains one low-income economy; Microsoft's estimator shrinks noisy countries toward the global average; and Claude's share among AI users varies with language support, brand reach and the occupational composition of work (Daepp and Slaughter, 2026; Gmyrek et al., 2025). The contrast is therefore consistent with the model's ordering of margins, which the $20 tier's slope satisfies unless more than about nine in ten users pay (Appendix A), and equally with these measurement and composition explanations; testing paid upgrading would require country-level user counts by subscription status and effective prices. The staggered rollout of ChatGPT Go offers price variation unrelated to income, but its comparison group was partly treated within days, the treated economies lie in one region and telecom bundles in India confound the largest case, so Appendix F

reports it as a descriptive episode without causal interpretation.

7 Discussion

7.1 Interpretation

The price evidence is consistent with the model's predictions: frontier tiers on or near the platform schedule, with deviations centred on zero (P1); localised AI tiers that are all capped entry tiers, although not every capped tier is localised (P2); a within-storefront dollar gap between the frontier and the entry tier that widens as income falls (P3), the one implication that separates a held frontier price from an unconstrained one; and within-firm contrasts in which the same firm localises its products without an AI allowance and not its frontier AI tiers (P4). What the increment test establishes is narrower than the model's mechanism: it rejects the unconstrained optimum of the nested menu, and it does not attribute the hold to inference cost. The model's account is that an unconstrained vendor would pass the entry tier's discount on to the frontier tier, that with a high inference-cost share little profit is at stake in doing so, and that, where vendors have not overridden it, the platform's default holds the frontier price, which is why the dollar gap between Plus and Go widens in poorer storefronts (Proposition A.3). Where vendors have overridden the default (Appendix Table E.13), they chose the U.S. dollar price, which points to segmented frontier demand or a deliberate uniform dollar price rather than to the override cost. Four explanations therefore remain side by side, and the evidence speaks to each differently. A high inference-cost share together with an override cost fits the held frontier price and the localised entry tiers, but not the overrides to $19.99 in taxed dollar storefronts. Cross-border arbitrage caps the dollar gap that any tier can open; because ChatGPT Go is sustained $3–4 below its U.S. price, a friction common to all plans could help hold the $100–500 tiers within a few per cent of the default but not the $20 tiers (Proposition A.4), and because tax-inclusive gaps of tens of dollars persist between rich storefronts (Appendix A.4, footnote), frictions are probably large enough that arbitrage binds on no tier. A uniform-dollar pricing policy, or anchoring on the U.S. price point, fits every frontier listing, including the overrides, but says nothing about why the entry tiers are localised. Segmented demand among frontier buyers, whose valuations do not scale with local income (Remark 1), produces the same signs as the held frontier price, including the widening dollar gap, and fits the overrides as well. The cross-section cannot separate these. Other explanations cover parts of the pattern. Inattention to storefront-specific pricing for new products does not: the same vendors localised YouTube, X Premium and, moderately, Microsoft 365 Personal through the same storefronts, and OpenAI localised ChatGPT Go within weeks of its launch. Market-entry sequencing does not either: Go's deviation from the schedule rises smoothly with income across the 161 storefronts rather than marking the launch markets in India and Asia. Product-level demand may matter: YouTube and X Premium face local competitors and local content, and a productivity subscription may have a clientele whose valuations scale less with local income than a consumer subscription's; the firm-by-storefront fixed effects absorb a firm's pricing capacity, not these differences, and the comparison across firms does not settle them (X Premium pays no content royalties and is localised as heavily as YouTube; Microsoft 365 Personal, a productivity product, only moderately). The closest contrast in the data is within Google, between YouTube Premium, at $20.99 in the United States, and Google AI Pro, at $19.99, sold by the same firm through the same storefronts under the same App Store account rules: the first is 57% cheaper than the second in the median

lower-income storefront (Table 3), so neither the firm, the price level nor the platform's rules account for the difference, and what differs is the product. Measuring the arbitrage friction beyond the lower bound that the entry tier reveals would require observing a change in frictions, such as the introduction of local identity or payment checks, with prices held fixed.

Prediction P5 concerns how vendors make frontier capability cheaper where they do. The localised AI tiers in the audit are all capped or degraded versions: ChatGPT Go withholds the frontier reasoning model and Character.AI Lite caps usage and features; Google AI Plus and SuperGrok Lite are capped tiers that are not localised, and Microsoft 365 Personal carries a small Copilot allowance. No vendor sells the frontier model at a localised price with a usage cap, although the allowance structure of the upper tiers shows that caps are available to them. In the model, service is one-dimensional, so a strict cap on the frontier model and a weaker model are the same instrument: either lowers the service and the inference cost of the cheap tier and its value to arbitrageurs who would otherwise buy the frontier tier. Why vendors weaken the model rather than cap it lies outside the model; one candidate is that a capped frontier tier would be an identical product for light users with high-value tasks, who could trade down to it at home or buy it abroad, whereas a weaker model is not. The model gives a reason why the frontier tier stays where it is, and what relaxing that would require: either a cap tight enough to make the local price cover inference cost, or a third party that pays the difference.

7.2 Policy

Three implications follow for the debate on access to generative AI in lower-income countries. First, the margin that pricing policy can move is paid access to frontier capability. Cheap tiers already exist in most poorer markets at \$5–8 a month, and the burden of ChatGPT Go is below 3% of per-capita income in the median lower-middle-income economy, as is that of Google AI Plus, whose cheapness there comes from a low U.S. price point rather than from localisation; the \$20 frontier tier costs 29% of per-capita income in the median low-income economy and more than 13% even at the synthetic mean resources of its richest fifth. Second, vendors' own localisation is unlikely to close that gap: applying OpenAI's entry-tier localisation to the \$20 tier would lower the median low-income burden to 14.8% (17.3% if only the vendor's deviation from the schedule is applied), still seven to nine times the benchmark, and the model gives a reason why vendors may not go further. Institutional purchase and subsidies, which pay the inference cost that a local price cannot cover, are the instruments the magnitudes point to: bringing the \$20 tier to the benchmark costs \$84–223 per subscriber a year in the three poorer income groups, or 0.08–1.55% of GNI for a tenth of the working-age population. Schools, universities, public agencies and employers can decouple access from household income, as a supervised tutoring programme in Nigeria did (De Simone et al., 2025), and most evidence on AI's productivity effects comes from such employer-provided tools (Brynjolfsson et al., 2025; Cui et al., 2024). Third, quantity rationing could reconcile frontier access with the cost constraint: a frontier model at a localised price with a strict usage cap bounds the vendor's cost per subscriber and deters arbitrage by heavy users, whereas a degraded tier bounds cost by withholding the model. Third-degree price discrimination tends to raise welfare when it opens markets that uniform pricing would leave unserved (Varian, 1985; Aguirre et al., 2010), the relevant case here. Connectivity remains a precondition for any use, and, as an accounting matter, about half of the use gradient coincides with the connectivity gradient (Appendix D) (World Bank, 2026c; International Telecommunication Union, 2025a; Hjort and Poulsen, 2019).

7.3 Limitations

The audit observes listed prices in one distribution channel on two days 19 days apart, with a cross-channel check of Google Play's lowest in-app price and of Microsoft's web store in a subset of countries; web checkout prices of the other vendors, carrier billing and promotional bundles are not observed, and the first wave stored parsed listings rather than pages. The estimated platform schedule comes from small-developer apps whose own pricing may deviate in some storefronts; it is exact only at the price points those apps use, is extrapolated with a storefront multiplier at the AI vendors' \$8, \$20 and \$100–500 price points, and is weak in six lower-income storefronts, which the validation bounds but does not eliminate and the within-firm comparisons avoid. The model is a single-vendor versioning model with one-dimensional service that omits competition, employer access and dynamic pricing; the cross-section bounds the role of arbitrage but does not separate the cost share of the frontier tier from the income scaling of its buyers' valuations. The usage indicators are modelled or sampled estimates that censor small low-income economies and pool free and paid accounts, their income gradients are descriptive, and burdens are normalised prices rather than expenditures.

8 Conclusion

Generative-AI assistants are priced as a global product with a local entry ramp. Two audits of every national App Store storefront show that the \$20 standard tiers and the \$100–500 high-usage tiers of OpenAI, Anthropic, Google, Perplexity, xAI and Microsoft sit on or near the price that the platform generates automatically from the U.S. price, with deviations centred on zero and no income-graded localisation (slopes between -0.016 and 0.004), and that within the same firm and storefront their prices relative to one another vary little with income. The same firms localise their video, music, social-network and office subscriptions through the same storefronts, YouTube and X heavily and Microsoft 365 moderately, which is consistent with uniformity being a property of products that are costly to serve, among them the frontier AI tiers, rather than of the firms. The only AI tiers whose prices are lower where incomes are lower are capped entry tiers, and not all capped tiers are localised: ChatGPT Go is 38% below the schedule in the median poorer storefront, so that ChatGPT Plus costs two and a half times Go in the United States and four times in the median low-income storefront, while Google AI Plus and SuperGrok Lite stay at the schedule.

A versioning model in which inference cost leaves little profit at stake in re-pricing the frontier tier, cross-border arbitrage caps the dollar gap any tier can open, and a capped tier is cheap to serve and worth localising, is consistent with where localisation appears and where it does not, and its one implication that an unconstrained optimum reverses, a frontier-to-entry dollar gap that widens where incomes are low, holds in the data; the same sign also follows from frontier valuations that do not scale with local income, which listed prices cannot rule out. Its implication for policy is that the affordability gap in frontier capability, 0.65% of per-capita income in the median high-income economy against 29% in the median low-income one, is unlikely to be closed by vendors' own localisation if the frontier price is kept uniform by a cost that does not fall with local income, as the model suggests; it can be narrowed by institutions that pay the inference cost on users' behalf, or by frontier tiers whose usage caps let a local price cover that cost. Measuring which of these works requires what no vendor yet publishes: paid subscriptions, effective prices and entitlements by country.

Declarations

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. During the preparation of this work the author used ChatGPT and Codex (OpenAI), Claude (Anthropic) and Gemini (Google) to assist with literature search, coding and data processing, drafting and translation, and to check the methods and the replication package. An AI browser agent collected the second-wave, default-panel and cross-channel prices by executing the collection scripts under the author's direction; AI subagents performed numerical, mathematical and bibliographic audits of earlier versions; and feedback from an AI review tool (Google Deep Review, Gemini) on two earlier versions was consulted. After using these tools, the author reviewed and edited the content as needed, verified all results and takes full responsibility for the content of the publication.

Data availability. All data are publicly available. The replication package (raw extracts, page archive fingerprints, code and outputs) is archived at Zenodo, https://doi.org/10.5281/zenodo.23006365 (all versions; version 3.1, https://doi.org/10.5281/zenodo.23428432, accompanies this manuscript together with the two page archives; versions 2.1, 2.0 and 1.8 of the record belong to earlier versions of the paper). Appendix B describes its contents.

Funding and competing interests. The author received no specific funding for this work and declares no competing interests.

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A A model of multi-market versioning with inference costs

Conventions. Storefronts (markets) are indexed by c and tiers by $k \in \{F, L, H\}$. Prices $p_{k,c}$ are monthly and in dollars net of the taxes that the platform adds to the listing, so that the listed price is $t_c e_c p_{k,c}$ and the platform default $P^D_{k,c} = t_c e_c P_k$ is the uniform price $p_{k,c} = P_k$ in every storefront.[3] For a valuation threshold T, $z = (\mu_c - \ln T)/s$ is the standardised distance by which the mean log valuation exceeds the log threshold; the share of the eligible with $v \geq T$ is $\Phi(z)$, and $h(z) = \phi(z)/\Phi(z)$, where ϕ and Φ are the standard normal density and distribution function, is s times the hazard rate of $\ln v$ at $\ln T$. This convention is used throughout the appendix. The population of market c is M_c, of which a share κ_c is eligible (connected); χ is the cross-border arbitrage friction, K the cost of overriding the platform default for one plan in one storefront, Π_c the vendor's profit in storefront c and π_c the paid share of users.

A.1 Demand: four options and nested thresholds

Consider a connected individual in market c choosing among non-use, a free tier F, a capped entry tier L and a frontier tier H. Let $v > 0$ denote the individual's monthly monetary valuation of one unit of effective service, let $0 < q_F < q_L < q_H$ describe the tiers' service levels, $0 = p_F < p_{L,c} < p_{H,c}$ their

[3] If demand responds to the tax-inclusive price, y_c below should be read as y_c/t_c; because t_c is one in the median storefront and lies between 0.98 and 1.29 in nine tenths of the storefronts with a multiplier (the extremes, 0.46 in Egypt and 1.57 in Colombia, reflect the vintage of the schedule after large currency movements) and the default schedule has an income slope of 0.033 (Section 5.3), none of the results depends on the distinction. The platform's commission is the same share of the price in every storefront; it can be absorbed into the costs, which are then the inference costs divided by one minus the commission rate.

monthly prices, and $\tau > 0$ a monthly monetary equivalent of the time, learning and other participation costs common to all tiers. Utilities are

$$u_0 = 0, \qquad u_k = vq_k - \tau - p_{k,c}, \quad k \in \{F, L, H\}. \tag{A.1}$$

The service levels are theoretical objects that need not correspond to particular models or measured capability. Pairwise comparison of adjacent options gives three thresholds: the free tier beats non-use when $v \geq \tau/q_F$, the entry tier beats the free tier when $v \geq p_{L,c}/(q_L - q_F)$, and the frontier tier beats the entry tier when $v \geq (p_{H,c} - p_{L,c})/(q_H - q_L)$. When the total outlay $\tau + p_{k,c}$ is convex in the service level, the origin included, so that

$$T_{D,c} \equiv \frac{\tau}{q_F} < T_{L,c} \equiv \frac{p_{L,c}}{q_L - q_F} < T_{H,c} \equiv \frac{p_{H,c} - p_{L,c}}{q_H - q_L}, \tag{A.2}$$

the thresholds are nested and each option is chosen on an interval of valuations: non-use below $T_{D,c}$, the free tier on $[T_{D,c}, T_{L,c})$, the entry tier on $[T_{L,c}, T_{H,c})$ and the frontier tier above $T_{H,c}$. (Utilities are linear in v with slopes that increase with q_k, so the choice follows the upper envelope of four lines; non-adjacent crossings, such as that of non-use and the entry tier at $(\tau + p_{L,c})/q_L$, lie between adjacent thresholds and do not alter it.) If (A.2) fails for the entry tier, that tier is never chosen and the menu collapses to the three-option case; if it fails for the free tier, a paid tier draws individuals directly from non-use and the paid price affects any use as well. Within the nested regime a marginal change in $p_{H,c}$ or $p_{L,c}$ leaves the any-use threshold unchanged.

Let κ_c be the population share with the connectivity and complementary resources needed to participate, and conditional on eligibility assume

$$\ln v \sim N(\mu_c, s^2), \qquad \mu_c = \mu_0 + \ln(y_c/y_0), \qquad s > 0, \tag{A.3}$$

where y_c is a country income index and y_0 a reference value; equivalently $v = y_c \tilde{v}$, where $\ln \tilde{v} \sim N(m, s^2)$ with $m = \mu_0 - \ln y_0$ in every market. This location-shift assumption is a stylisation, not an identity linking national GNI to household incomes. Holding qualities, participation costs and dispersion common across markets, the population shares using any tier, a paid tier and the frontier tier are

$$D_c = \kappa_c \Phi(z_{D,c}), \qquad P_c = \kappa_c \Phi(z_{L,c}), \qquad H_c = \kappa_c \Phi(z_{H,c}), \qquad z_{k,c} = \frac{\mu_c - \ln T_{k,c}}{s}, \tag{A.4}$$

because $\Pr(v \geq T_{k,c}) = 1 - \Phi(-z_{k,c}) = \Phi(z_{k,c})$. If participation costs vary as $\tau_c \propto y_c^{\lambda}$, the any-use slope below becomes $\varepsilon_\kappa + (1 - \lambda) h(z_{D,c})/s$ even at a fixed free service level; more generally, if the mean log valuation of the connected rises with income at rate $\sigma_\mu \leq 1$ (positive selection into connectivity where penetration is low) and the service levels q_F, $q_L - q_F$ and $q_H - q_L$ rise with income at rates $\gamma_F, \gamma_L, \gamma_H \geq 0$ (language support, task relevance), the three gradients below acquire the factors $(\sigma_\mu - \lambda + \gamma_F)$, $(\sigma_\mu - \beta_L + \gamma_L)$ and $(\sigma_\mu - \beta_\Delta + \gamma_H)$ in place of 1, $(1 - \beta_L)$ and $(1 - \beta_\Delta)$. These possibilities are substantively relevant and are not ruled out by an income regression.

Proposition A.1 (Local income gradients)**.** *Within the strict nested-threshold regime, define* $h(z) = \phi(z)/\Phi(z)$ *and, for the price schedules that the vendor lists,* $\beta_L = \partial \ln p_{L,c}/\partial \ln y_c$, $\beta_H = \partial \ln p_{H,c}/\partial \ln y_c$,

$\beta_\Delta = \partial \ln(p_{H,c} - p_{L,c})/\partial \ln y_c$ *and* $\varepsilon_\kappa = \partial \ln \kappa_c / \partial \ln y_c$. *Then*

$$\varepsilon_D = \varepsilon_\kappa + \frac{h(z_{D,c})}{s}, \qquad \varepsilon_P = \varepsilon_\kappa + (1-\beta_L)\frac{h(z_{L,c})}{s}, \qquad \varepsilon_H = \varepsilon_\kappa + (1-\beta_\Delta)\frac{h(z_{H,c})}{s}. \tag{A.5}$$

Uniform pricing of every tier ($\beta_L = \beta_H = 0$) gives $\varepsilon_H > \varepsilon_P > \varepsilon_D$. Comparing schedules with different elasticities at a common anchor market, where the thresholds, and hence the $z_{k,c}$, take the same values, $\varepsilon_P > \varepsilon_D$ persists if $\beta_L < \beta^* \equiv 1 - h(z_{D,c})/h(z_{L,c})$, where $0 < \beta^* < 1$; and if the frontier price is uniform while the entry price is localised, $\beta_\Delta = -p_{L,c}\beta_L/(p_{H,c} - p_{L,c}) < 0$ and ε_H exceeds its uniform-pricing value.

Proof. Differentiating the logarithms of equation (A.4) with respect to $\ln y_c$, using $d\ln\Phi(z)/dz = h(z)$, $\partial \mu_c / \partial \ln y_c = 1$, $\partial \ln T_{D,c}/\partial \ln y_c = 0$, $\partial \ln T_{L,c}/\partial \ln y_c = \beta_L$ and $\partial \ln T_{H,c}/\partial \ln y_c = \beta_\Delta$, gives (A.5). The strict ordering (A.2) implies $z_{H,c} < z_{L,c} < z_{D,c}$, and h is strictly decreasing, because $h'(z) = -h(z)[z + h(z)]$ and $z + h(z) > 0$ for every z; so $h(z_{H,c}) > h(z_{L,c}) > h(z_{D,c})$, which with $\beta_L = \beta_\Delta = 0$ orders the three gradients. Subtracting ε_D from ε_P and solving for equality gives β^*, which lies in $(0,1)$ because $0 < h(z_{D,c}) < h(z_{L,c})$. Finally, $\beta_\Delta = (p_{H,c}\beta_H - p_{L,c}\beta_L)/(p_{H,c} - p_{L,c})$, which equals $-p_{L,c}\beta_L/(p_{H,c} - p_{L,c}) < 0$ when $\beta_H = 0 < \beta_L$. □

The threshold β^* depends on how far the two margins sit in the tail of the valuation distribution. Writing $D_c/\kappa_c = \Phi(z_{D,c})$ for the share of eligible individuals who use any tier and π_c for the share of users who pay, so that $\Phi(z_{L,c}) = \pi_c \Phi(z_{D,c})$, gives $\beta^* = 0.44$ when 23% of the eligible use AI and 10% of users pay, 0.34 when a quarter pay, and 0.58 when 1% pay; β^* falls below the observed slope of the \$20 tier (0.026 for Claude Pro, 0.04 for ChatGPT Plus), which is the cheapest paid tier of a menu without an entry tier such as Claude's, only if more than about nine in ten users pay. The comparison across β is local: it compares schedules $p_\beta(y) = p_0(y/y_0)^\beta$ at their common anchor, and it establishes neither a global effect of a localisation policy nor welfare effects.

Bridge to the activity index. Section 6 relates these margins to Anthropic's activity index, which pools free and paid conversations. The index of country c is proportional to conversations per working-age person, the product of the share using any generative-AI product, D_c^*, the vendor's share among AI users, ω_c, and conversations per user of the vendor, m_c; hence $\varepsilon_{AUI} = \varepsilon_{D^*} + \varepsilon_\omega + \varepsilon_m$ exactly, and the vendor's own any-use share is $D_c = D_c^* \omega_c$, so that $\varepsilon_D = \varepsilon_{D^*} + \varepsilon_\omega$. With $m_c = (1-\pi_c)m_F + \pi_c m_P$, $\pi_c = P_c/D_c$ the paid share of users and $m_F < m_P$ the within-tier intensities held fixed across countries, differentiating $\ln m_c$ with respect to $\ln y_c$ gives

$$\varepsilon_m = \theta_c(\varepsilon_P - \varepsilon_D), \qquad \theta_c \equiv \frac{\pi_c(m_P - m_F)}{(1-\pi_c)m_F + \pi_c m_P} \in (0,1), \tag{A.6}$$

the share of incremental paid conversations in all conversations. The measured gap $\varepsilon_{AUI} - \varepsilon_{D^*} = \varepsilon_\omega + \theta_c(\varepsilon_P - \varepsilon_D)$ therefore attenuates the paid-margin gradient by θ_c and adds a vendor-share term, so it bounds neither $\varepsilon_P - \varepsilon_D$ from above nor from below without knowledge of θ_c and ε_ω.

A.2 Supply: the income elasticity of an optimal price increment

In the nested regime the vendor's profit is the sum of its profits on two price increments, the entry price over the free tier and the frontier price over the entry price (Appendix A.3). Consider the increment of

tier k over the option below it: price increment Δp_k ($\Delta p_L = p_{L,c}$, $\Delta p_H = p_{H,c} - p_{L,c}$), service increment $\Delta q_k > 0$ ($q_L - q_F$, $q_H - q_L$) and marginal-cost increment $\Delta c_k \geq 0$ ($c_L - c_F$, $c_H - c_L$). The increment is bought by consumers with $v \geq \Delta p_k / \Delta q_k$, and a consumer who moves down to the option below forgoes it, so that the vendor loses its margin $\Delta p_k - \Delta c_k$ and keeps the margin on the option below. The problem $\max_{\Delta p}(\Delta p - \Delta c)[1 - G(x)]$ below is therefore exact rather than an approximation. Dropping the tier subscript, $\tilde{v} = v/y_c$ has density g and distribution function G, independent of income; $x = \Delta p/(\Delta q\, y_c)$ is the income-normalised increment and $\hat{c} = \Delta c/(\Delta q\, y_c)$ the income-normalised cost. Define the demand elasticity $\eta(x) = xg(x)/[1 - G(x)]$ and $\Lambda(x) = 1 - 1/\eta(x)$. At an interior optimum the Lerner condition $(\Delta p - \Delta c)/\Delta p = 1/\eta$ holds, so $\Lambda(x^*)$ equals the cost share of the increment, $\ell \equiv \Delta c/\Delta p^*$: $\Lambda(x)$ is the cost share that the Lerner rule associates with the normalised price x (one minus the Lerner index), and it increases exactly when η does. Because $\eta(x)$ is the hazard rate of $\ln \tilde{v}$ evaluated at $\ln x$, the natural regularity condition concerns the distribution of log valuations rather than that of valuations (Bagnoli and Bergstrom, 2005).

Proposition A.2 (Income elasticity of an optimal price increment)**.** *Suppose that* $\ln \tilde{v}$ *has a differentiable, strictly increasing hazard rate, so that* $\eta' > 0$*, with* $\eta(x) \to 0$ *as* $x \to 0$ *and* $\eta(x) > 1$ *for large* x*. This holds whenever* $\ln \tilde{v}$ *has a strictly log-concave density, in particular for lognormal valuations, although the lognormal density* g *of* $\tilde{v}$ *is itself not log-concave.*

(i) The profit $(\Delta p - \Delta c)[1 - G(x)]$ *is strictly quasi-concave in* Δp *on* $\Delta p > \Delta c$*, its maximiser is the unique solution of* $\Lambda(x^*) = \ell$*, and the income elasticity of the optimal increment is*

$$\beta \equiv \frac{\partial \ln \Delta p^*}{\partial \ln y_c} = \frac{\xi}{\xi + \ell} = 1 - \ell\rho, \qquad \xi \equiv \left.\frac{d\Lambda}{d\ln x}\right|_{x^*} > 0, \qquad \rho \equiv \frac{\partial \Delta p^*}{\partial \Delta c} = \frac{1}{\xi + \ell}, \tag{A.7}$$

where ρ *is the rate at which the optimal increment passes through its marginal cost (Weyl and Fabinger, 2013). Thus* $\beta = 1$ *when* $\Delta c = 0$ *and* $0 < \beta < 1$ *when* $\Delta c > 0$*.*

(ii) If, in addition, Λ *is concave in* $\ln x$*, then* β *is strictly decreasing in the normalised cost* $\hat{c}$*, and hence in the cost share* ℓ*, which rises with* $\hat{c}$*: the optimal increment is less income-elastic in poorer markets and for increments with a larger cost share.*

(iii) If tier-k valuations scale as $v = y_c^{\sigma_k}\tilde{v}$ *with* $0 \leq \sigma_k \leq 1$*, the elasticity is* $\sigma_k \xi/(\xi + \ell)$*. If the marginal cost is proportional to local income, or is a share of local revenue (as a royalty is),* $\beta = 1$*.*

(iv) For lognormal valuations, $\ln \tilde{v} \sim N(m, s^2)$*, the conditions of (i) and (ii) hold, and with* z^* *the solution of* $h(z^*) = s/(1 - \ell)$*,*

$$\beta = \frac{z^* + h(z^*)}{z^* + 2h(z^*) - s}, \tag{A.8}$$

which equals one at $\ell = 0$*, decreases strictly in* ℓ *and tends to zero as* $\ell \to 1$*.*

Proof. (i) Write $\Delta p = \Delta q\, y_c x$ and the profit as $\Delta q\, y_c\, \varpi(x)$ with $\varpi(x) = (x - \hat{c})[1 - G(x)]$. For $x > \hat{c}$,

$$\frac{d \ln \varpi(x)}{dx} = \frac{1 - (1 - \hat{c}/x)\,\eta(x)}{x - \hat{c}}.$$

The numerator is strictly decreasing in x, because $1 - \hat{c}/x$ is positive and non-decreasing and η is positive and strictly increasing; it is positive near $x = \hat{c}$ (near $x = 0$ when $\hat{c} = 0$) and negative for large x. So ϖ is strictly quasi-concave with a unique maximiser, which solves $(1 - \hat{c}/x)\eta(x) = 1$, that is $(\Delta p -$

$\Delta c)/\Delta p = 1/\eta$, or $\Lambda(x^*) = \hat{c}/x^* = \ell$. Writing the condition as $x\Lambda(x) = \hat{c}$ and differentiating, $[\Lambda(x^*) + x^*\Lambda'(x^*)]\,dx^* = d\hat{c}$, so that $dx^*/d\hat{c} = 1/(\ell+\xi)$, where $\xi = x\Lambda'(x) = x\eta'(x)/\eta(x)^2 > 0$. Because $\Delta p^* = \Delta q\, y_c x^*$ and $\hat{c} = \Delta c/(\Delta q\, y_c)$, $dx^*/d\hat{c}$ is also $\rho = \partial\Delta p^*/\partial\Delta c$; and because $\partial \ln \hat{c}/\partial \ln y_c = -1$,

$$\beta = 1 + \frac{\partial \ln x^*}{\partial \ln y_c} = 1 - \frac{\hat{c}}{x^*}\frac{dx^*}{d\hat{c}} = 1 - \ell\rho = \frac{\xi}{\xi+\ell}.$$

(ii) Along the optimum, $d\ln\ell/d\ln\hat{c} = 1-\ell\rho = \beta > 0$, so ℓ rises with $\hat{c}$, that is, as income falls or cost rises; and x^* rises with $\hat{c}$. If Λ is concave in $\ln x$, ξ is non-increasing in x and hence in $\hat{c}$, so ξ/ℓ falls strictly and $\beta = 1/(1+\ell/\xi)$ falls strictly as $\hat{c}$ rises.

(iii) With $v = y_c^{\sigma_k}\tilde{v}$ the normalised increment is $x = \Delta p/(\Delta q\, y_c^{\sigma_k})$; repeating (i) with $y_c^{\sigma_k}$ in place of y_c gives $\partial \ln x^*/\partial \ln y_c = -\sigma_k \ell/(\xi+\ell)$ and $\beta = \sigma_k + \partial \ln x^*/\partial \ln y_c = \sigma_k \xi/(\xi+\ell)$. If $\Delta c = c_0 y_c$, then $\hat{c} = c_0/\Delta q$ is the same in every market, so is x^*, and $\beta = 1$; if the cost is a share r of revenue, profit is $(1-r)\Delta p[1-G(x)]$, whose maximiser is that of the zero-cost problem, and $\beta = 1$.

(iv) For $\ln\tilde{v} \sim N(m, s^2)$, $1-G(x) = \Phi(z)$ with $z = (m-\ln x)/s$ and $g(x) = \phi(z)/(sx)$, so $\eta(x) = h(z)/s$. Because $h'(z) = -h(z)[z+h(z)]$, $z+h(z) > 0$ for every z, and $0 < h(z)[z+h(z)] < 1$ (Sampford, 1953), h is strictly decreasing; since z falls as x rises, η is strictly increasing, with $\eta \to 0$ as $x \to 0$ and $\eta \to \infty$ as $x \to \infty$. Then $\Lambda = 1 - s/h(z)$ and

$$\xi = \frac{d\Lambda}{d\ln x} = \frac{s\,h'(z)}{h(z)^2}\frac{dz}{d\ln x} = \frac{z+h(z)}{h(z)} = 1 + \frac{z}{h(z)}, \qquad \frac{d}{dz}\frac{z}{h(z)} = \frac{1+z[z+h(z)]}{h(z)} > 0,$$

where $1+z[z+h(z)] > 0$ is immediate for $z \geq 0$ and follows for $z < 0$ from $|z|\,[z+h(z)] < h(z)[z+h(z)] < 1$. So ξ falls as x rises, Λ is concave in $\ln x$, and (ii) applies. The optimum satisfies $\Lambda(x^*) = \ell$, that is $h(z^*) = s/(1-\ell)$, and substituting into $\beta = \xi/(\xi+\ell)$ gives (A.8). Directly, $\beta = 1/(1+R)$ with $R = [h(z^*)-s]/[z^*+h(z^*)]$. As ℓ rises, $h(z^*) = s/(1-\ell)$ rises and z^* falls; the numerator $h(z^*)-s = s\ell/(1-\ell)$ rises, and the positive denominator falls, because $d[z+h(z)]/dz = 1-h(z)[z+h(z)] \in (0,1)$; so R rises and β falls. At $\ell = 0$, $h(z^*) = s$ and $\beta = 1$; as $\ell \to 1$, $z^* \to -\infty$, $z^*+h(z^*) \to 0$ and $h(z^*)-s \to \infty$, so $\beta \to 0$. The script checks/verify_tier_model.py in the replication package and checks/sim_check.py verify (A.7) and (A.8) against the numerical optimum. □

For $s = 1$ the elasticity is 0.63 at a cost share of one quarter, 0.30 at one half and 0.05 at four fifths; for $s = 0.7$ it is 0.76, 0.44 and 0.09. An increment whose price mostly covers inference cost is therefore priced almost uniformly by a profit-maximising vendor even absent any other friction, and an increment with a thin cost share is priced closer to proportionally to income. Proposition A.2 concerns increments; what it implies for the price level of the frontier tier depends on how the menu is priced, which Appendix A.3 takes up.

A.3 The menu: separability, pass-through and the frontier price level

The vendor earns $-c_F$ on a free user, $p_{L,c} - c_L$ on an entry subscriber and $p_{H,c} - c_H$ on a frontier subscriber, so in the nested regime its profit in storefront c is

$$\Pi_c = M_c \kappa_c \Big[-c_F \Phi(z_{D,c}) + (p_{L,c} - \Delta c_L)\,\Phi(z_{L,c}) + (\Delta p_{H,c} - \Delta c_H)\,\Phi(z_{H,c}) \Big], \tag{A.9}$$

with $\Delta p_{H,c} = p_{H,c} - p_{L,c}$, $\Delta c_L = c_L - c_F$ and $\Delta c_H = c_H - c_L$. Because $z_{D,c}$ depends on neither price, $z_{L,c}$ only on $p_{L,c}$ and $z_{H,c}$ only on $\Delta p_{H,c}$, (A.9) is the sum of two increment problems of the form of Proposition A.2.

Proposition A.3 (Menu prices and pass-through)**.** *Maintain the assumptions of Proposition A.2(i)–(ii). Suppose that marginal cost is convex in service, $\Delta c_L/(q_L - q_F) < \Delta c_H/(q_H - q_L)$, and that τ is small enough for the free-tier threshold to lie below the optimal entry threshold.*

(i) The optimal menu is nested in every market and consists of the two optimal increments, $p^*_{L,c} = (q_L - q_F)\, y_c\, x^*(\hat{c}_{L,c})$ and $\Delta p^*_{H,c} = (q_H - q_L)\, y_c\, x^*(\hat{c}_{H,c})$, where $\hat{c}_{k,c} = \Delta c_k/(\Delta q_k\, y_c)$ and $x^*(\cdot)$ is the normalised optimum of Proposition A.2. The frontier price is their sum, $p^*_{H,c} = p^*_{L,c} + \Delta p^*_{H,c}$.

(ii) For any entry price in the nested region, the frontier price that maximises Π_c is that entry price plus $\Delta p^*_{H,c}$: a dollar off the entry price is a dollar off the optimal frontier price. The income elasticity of the optimal frontier price level is the share-weighted average

$$\beta_H = \frac{p^*_{L,c}}{p^*_{H,c}}\,\beta_L + \frac{\Delta p^*_{H,c}}{p^*_{H,c}}\,\beta_{\Delta H}, \tag{A.10}$$

where β_L and $\beta_{\Delta H}$ are the elasticities (A.7) *of the two increments, with cost shares $\ell_L = \Delta c_L/p^*_{L,c}$ and $\ell_H = \Delta c_H/\Delta p^*_{H,c}$.*

(iii) $\beta_L > \beta_{\Delta H} > 0$. At the optimum the entry price, the dollar increment and the frontier price all rise with income, and the within-storefront ratio $p^*_{H,c}/p^*_{L,c} = 1 + \Delta p^*_{H,c}/p^*_{L,c}$ falls with income.

(iv) If the listed frontier price does not vary with income ($\beta_H = 0$) while the entry price does ($\beta_L > 0$), the dollar increment falls with income, $\beta_\Delta = -p_{L,c}\beta_L/(p_{H,c} - p_{L,c}) < 0$, which (iii) excludes at the vendor's optimum. A frontier price held at the platform default while the entry tier is localised therefore reveals a binding constraint on the frontier price.

(v) If the frontier price is held at a level $\bar{p}_H \geq p^*_{H,c}$, the optimal entry price given $\bar{p}_H$ is at least $p^*_{L,c}$: holding the frontier price up restrains, and never deepens, the localisation of the entry tier.

Proof. Write (A.9) as $M_c\kappa_c[-c_F\Phi(z_{D,c}) + A_c(p_{L,c}) + B_c(\Delta p_{H,c})]$, where the increment profits A_c and B_c have the form of Proposition A.2 and are strictly quasi-concave, with maxima at $p^*_{L,c}$ and $\Delta p^*_{H,c}$. (i) Both increments face the same distribution of $\tilde{v}$, so $T^*_{L,c} = y_c x^*(\hat{c}_{L,c})$ and $T^*_{H,c} = y_c x^*(\hat{c}_{H,c})$; since x^* is strictly increasing ($\rho > 0$), $T^*_{L,c} < T^*_{H,c}$ if and only if $\hat{c}_{L,c} < \hat{c}_{H,c}$, which is the convexity condition and does not involve y_c. Menus outside the nested region do no better. If the entry tier is not chosen, the menu is $\{F,H\}$ with threshold $T = p_{H,c}/(q_H - q_F)$ and margin $p_{H,c} - c_H + c_F = [T(q_L - q_F) - \Delta c_L] + [T(q_H - q_L) - \Delta c_H]$, so its profit is $A_c(T(q_L - q_F)) + B_c(T(q_H - q_L)) \leq \max A_c + \max B_c$; if no one buys the frontier tier, profit is $A_c(p_{L,c}) < \max A_c + \max B_c$, because $\max B_c > 0$. (ii) Given $p_{L,c}$, Π_c depends on $p_{H,c}$ only through $B_c(p_{H,c} - p_{L,c})$. Differentiating $\ln(p^*_{L,c} + \Delta p^*_{H,c})$ with respect to $\ln y_c$ gives (A.10). (iii) By Proposition A.2(ii), β is the same strictly decreasing function of $\hat{c}$ for both increments, and $\hat{c}_{L,c} < \hat{c}_{H,c}$, so $\beta_L > \beta_{\Delta H}$; $\beta_{\Delta H} > 0$ by Proposition A.2(i). Hence $d\ln(\Delta p^*_{H,c}/p^*_{L,c})/d\ln y_c = \beta_{\Delta H} - \beta_L < 0$. (iv) As in the proof of Proposition A.1. (v) Let $p_L < p^*_{L,c}$. Then $A_c(p_L) < A_c(p^*_{L,c})$, and $\bar{p}_H - p_L > \bar{p}_H - p^*_{L,c} \geq \Delta p^*_{H,c}$, so $B_c(\bar{p}_H - p_L) < B_c(\bar{p}_H - p^*_{L,c})$, because B_c falls beyond its maximum. Every entry price below $p^*_{L,c}$ therefore earns less than $p^*_{L,c}$. □

In a lognormal example calibrated so that the optimum in the reference market is \$8 for the entry tier and \$20 for the frontier tier, with $s = 1$ and cost shares of one quarter for the entry price and one half for the frontier increment, the optimal frontier price in a market with a fifth of the reference income is \$12.7, 37% lower: \$4.1 of the reduction is the lower entry price passed through and \$3.2 the lower optimal increment. The local income elasticity there is 0.28 for the entry price, 0.12 for the increment and 0.17 for the frontier price level (checks/sim_check.py). The unconstrained vendor localises the frontier tier, through the entry price, although inference cost makes its own increment nearly uniform.

A.4 The platform default, arbitrage and the binding constraint

Proposition A.3(iv) shows that a localised entry tier combined with a frontier tier at the platform default is not the vendor's unconstrained optimum in the maintained model: something must hold the frontier price up. The model contains two candidates.

The override cost. The listed price is the default unless the vendor overrides it, at a fixed cost K per plan and storefront. Let $G_{L,c} = \max_{p_L} \Pi_c(p_L, P_H) - \Pi_c(P_L, P_H)$ be the gain from localising the entry tier with the frontier tier left at the default, and $G_{H,c} = \max_{p_L,p_H} \Pi_c(p_L, p_H) - \max_{p_L} \Pi_c(p_L, P_H)$ the further gain from also re-pricing the frontier tier.

Arbitrage. A consumer whose home storefront is c' can buy any plan in storefront c at a friction χ, the cost of holding an account and a payment method in that foreign storefront; χ may differ across pairs of storefronts but not across the plans bought. Prices in this paragraph and in Proposition A.4(ii)–(iii) are the tax-inclusive dollar prices that she compares.

Proposition A.4 (Override and arbitrage)**.** *(i) $G_{L,c}$ and $G_{H,c}$ are proportional to the eligible population $M_c\kappa_c$; per eligible person they depend on the storefront only through its income, and they are zero at the reference income if the U.S. prices are optimal there. The vendor localises the entry tier and leaves the frontier tier at the default in storefront c if $G_{L,c} > K \geq G_{H,c}$ and localising the entry tier alone is worth more than localising the frontier tier alone.*

(ii) A consumer who buys tier k at home buys the same tier in c if and only if $p_{k,c'} - p_{k,c} > \chi$. A consumer who buys the frontier tier at home gains at most $p_{L,c'} - p_{L,c} - \chi$ from buying the entry tier in c, and every other cross-tier purchase gains at most the same-tier gap of the tier bought, less χ. Prices are therefore free of arbitrage if and only if $p_{k,c} \geq p_{k,c'} - \chi$ for both paid tiers and every home storefront c': arbitrage limits the dollar gap between two storefronts, by the same amount for every tier.

(iii) If prices are free of arbitrage and the entry tier is listed g_L dollars cheaper in c than in c', the friction between the two storefronts is at least g_L, and the arbitrage floor of any tier k in c, against consumers from c', lies at least g_L dollars, a proportion $g_L/p_{k,c'}$, below its price in c'. A tier whose price in c is less than that proportion below its price in c' is not held there by arbitrage from c'.

Proof. (i) By (A.9), $\Pi_c/(M_c\kappa_c)$ depends on the storefront only through y_c, and at the reference income the default prices are the optimum. The vendor compares four options (no override, entry tier only, frontier tier only, both), and the stated inequalities make the entry tier alone the best. (ii) The first statement is the definition of arbitrage for an identical product. A consumer who buys the frontier tier at home has $v \geq (p_{H,c'} - p_{L,c'})/(q_H - q_L)$, so the capped tier is worth $v(q_H - q_L) \geq p_{H,c'} - p_{L,c'}$ less to her, and her gain from the foreign entry tier is $p_{H,c'} - p_{L,c} - v(q_H - q_L) - \chi \leq p_{L,c'} - p_{L,c} - \chi$. In the

same way, the nested thresholds at home bound the gain of an entry subscriber from the foreign frontier tier, and of a free user or non-user from either foreign tier, by $p_{k,c'} - p_{k,c} - \chi$ for the tier k bought. (iii) Immediate from (ii). □

Proposition A.4 assigns the two candidates different roles. Arbitrage cannot produce the pattern on its own: a binding floor cuts every tier by the same number of dollars, a large proportional discount on a cheap capped tier and a small one on a dear frontier tier, but it does not leave the frontier tier at the default. And because ChatGPT Go is listed \$3–4 below its U.S. price in the poorest storefronts, part (iii) implies that arbitrage from the United States could hold a \$20 tier there no closer than 15–20% to the U.S. price, against 0.6–4% for the \$100–500 tiers: arbitrage is a candidate for keeping the high-usage tiers at the default, not the \$20 tiers.[4] The override cost is the candidate for the \$20 tiers, and this is where the cost share of the frontier increment matters. In the example of Appendix A.3, the further gain $G_{H,c}$ from re-pricing the frontier tier, once the entry tier is localised, is 8% of the gain $G_{L,c}$ from localising the entry tier at a fifth of the reference income and 3% at a tenth; at a fifth of the reference income the ratio is 2.6, 0.49, 0.08 and 0.007 for frontier-increment cost shares of 0.3, 0.4, 0.5 and 0.6 (checks/sim_check.py). A fixed override cost therefore binds on the frontier tier and not on the entry tier when inference cost is a large share of the frontier increment. Inference cost does not make the frontier price uniform at the vendor's optimum, which follows the entry price down; it makes the profit at stake in re-pricing the frontier tier too small to justify departing from the platform's default. Because gains scale with the eligible population, part (i) also predicts that, at a given income, the entry tier departs from the default more often in larger storefronts.

Remark 1 (Frontier subscribers who do not trade down). The pass-through of Proposition A.3(ii) rests on the marginal frontier subscriber's alternative being the entry tier. If instead the capped tier is of no use for the tasks of frontier subscribers ($q_L = q_F$ for them) while the mass market does not value the frontier tier's additional capability, the two tiers serve separate segments and each is priced against the free tier. The frontier price is then a level chosen as in Proposition A.2, with $\Delta p_H = p_{H,c}$ and cost share $(c_H - c_F)/p_{H,c}$; it does not respond to the entry price, and its income elasticity $\sigma_H \xi_H/(\xi_H + \ell_H)$ is close to zero if the cost share is high or if frontier subscribers' valuations do not scale with local income (σ_H near zero, Proposition A.2(iii)). Listed prices cannot separate this case from a held frontier price; the sign test of Proposition A.3(iv) excludes only the unconstrained optimum of the nested menu.

A.5 What the model predicts and what listed prices can test

Table A.1 collects the predictions of the model by the constraint that binds on the frontier price. Three of them can be confronted with cross-sectional listed prices. A frontier price at the default (P1) predicts a deviation slope of zero with bunching at the default, a sharp implication of a fixed override cost. The entry tier's deviation (P2) should rise with income, with a slope between zero and one. The within-storefront dollar increment (P3) should fall with income if the frontier price is held at the default and rise with income at the unconstrained optimum, and the observed increment of ChatGPT Plus over Go, relative to the default increment, has an income slope of -0.047 (s.e. 0.005; 161 storefronts) and is 25%

[4] Under the default, storefronts with higher taxes than the United States list every tier above its U.S. price, by up to about a quarter. If their consumers could buy elsewhere at a friction common to plans, the persistence of these gaps, tens of dollars on the \$100–500 tiers, would bound that friction from below by as much, and arbitrage from those storefronts could not hold any tier near the default either.

wider in the median low- and lower-middle-income storefront; in dollars relative to the U.S. increment the slope is -0.010 (s.e. 0.007), which under the unconstrained optimum would have to exceed the schedule's own tax slope of 0.033, so both normalisations reject that benchmark; that of Character.AI+ over Character.AI Lite has a slope of -0.059 (s.e. 0.012). The within-storefront price ratio rises as income falls under every regime, so it confirms P1 and P2 free of currency and tax effects but does not discriminate between regimes. Other statements are consistent with the data rather than tested by it: the cost share of the frontier increment is not observed, and listed prices cannot separate a high cost share from frontier subscribers whose valuations do not scale with local income (Proposition A.2(iii) and Remark 1); the implication of the override cost that, at a given income, the entry tier departs from the default more often in larger storefronts is not tested; the arbitrage friction is bounded from below by the entry tier's discount but not measured; the within-firm contrasts (P4) hold the dispersion and income scaling of valuations fixed across products; the composition of the menus (P5) cannot separate a usage cap from a weaker model, because service is one-dimensional in the model; and the demand-side gradients of Proposition A.1 are related to usage indicators that do not observe paid adoption.

Remark 2 (What a PPP denominator measures). For a service priced in dollars, the burden relative to income at market exchange rates is the relevant cash-cost quantity. Dividing the dollar price by income at purchasing-power parity instead yields $12p/y_c^{PPP} = 12\,(p \cdot PLR_c)/y_c$, where $PLR_c = y_c/y_c^{PPP}$, the ratio of Atlas to PPP income, is a proxy for the price level relative to the United States.[5] The same algebra equals the Atlas-income burden under a constructed price $p_c = p \cdot PLR_c$, that is, the burden that would obtain if the vendor indexed its price to local price levels. Because price levels rise with income (Balassa, 1964; Samuelson, 1964), and PPP conversion factors are designed for broad consumption baskets rather than internationally priced digital services (Deaton and Heston, 2010), the PPP-based ratio is best read as this counterfactual pricing scenario rather than as a robustness check on the same quantity.

[5] Because the Atlas method averages exchange rates over three years, the ratio differs from the same-year GDP price-level ratio by a median of 3.1% (90th percentile 8.3%) across 197 economies. Its income elasticity is 0.228, against 0.224 for the GDP price-level ratio.

Table 2: Data sources

Variable	Source and definition	Coverage
Income, groups	World Development Indicators (World Bank, 2026b): GNI per capita, Atlas method (current US$) and PPP, household consumption; FY2027 income groups (World Bank, 2026a)	201 economies; 2025 for 182
Distribution	WDI income/consumption shares by quintile (latest survey 2010–2025)	163 economies
Local prices, wave 1	Apple App Store product pages of ChatGPT, Claude and Gemini: in-app purchase lists and storefront currency, 27 Sep 2026; FX from open.er-api.com (cross-checked)	218 codes; 170 with prices; 162–165 with income by app (95–165 by plan), including the U.S.
Local prices, wave 2	The same pages for 34 apps (the three assistants, six other AI assistants, 21 general subscription apps, four apps with embedded AI features), 16 Oct 2026; page archive with SHA-256 fingerprints	218 codes; 173 with prices; 7,412 queries; 30,817 item prices
Platform schedule	30 small-developer subscription apps in the same storefronts, 16 Oct 2026: modal local price per U.S. price point and storefront multiplier	167 storefronts; 164 with a multiplier; 161 with income
Cross-channel prices	Google Play product pages: lowest and highest in-app price for 12 apps in 30 countries; Microsoft web store: Microsoft 365 Personal, Family and Premium in 57 countries; both 16 Oct 2026 (Appendix C)	360 app-country pages; 63 locale pages
Exchange-rate history	fawazahmed0 currency-api snapshots, six dates from March 2024 to October 2026 (vintage test, Section 5.3)	37 currencies in the price sample
Any-use diffusion	Microsoft AI User Share, H1 2025–Q2 2026 (Misra et al., 2025; Microsoft AI Economy Institute, 2026b)	147 economies; 111 country-specific
Claude activity	Anthropic AI Usage Index (AUI), Aug 2025–May 2026 (Anthropic, 2026a)	113–120 economies above the publication floor; 165–169 geolocated
Gemini usage rank	Google ATLAS v1.0 per-capita adoption quintile, Apr 2026 (Iscenko et al., 2026)	195 economies
Entry-tier rollout	First availability of ChatGPT Go by economy, from OpenAI's release notes and press reports (Appendix F)	27 economies dated individually; 71 undisclosed economies dated 14 Oct 2025
Complements	Internet use, electricity, tertiary enrolment, urbanisation (WDI); Global Findex 2025 (Klapper et al., 2025); ITU ICT price baskets (International Telecommunication Union, 2025b)	142–217 economies

Notes: Income groups follow the World Bank classification for fiscal year 2027 (effective 1 July 2026), based on 2025 GNI per capita: low $\leq$ \$1,175; lower-middle \$1,176–4,635; upper-middle \$4,636–14,375; high $>$ \$14,375. "LMIC" in the tables below denotes low- and lower-middle-income storefronts together.

Table 3: Income gradients of relative prices within firm and storefront, 16 October 2026

		Slope of $\ln(p_A/p_B)$ on ln GNI p.c.				LMIC	Share	Share
Pair (A vs B)	N	Slope	(HC1)	[currency]	RI p	median	< 0.9 (%)	at 1 (%)
ChatGPT Go vs Plus	163	0.104	(0.006)	[0.007]	<0.0005	0.62	80	12
ChatGPT Pro 100 vs Plus	163	−0.021	(0.004)	[0.011]	<0.0005	1.00	20	48
ChatGPT Pro 200 vs Plus	163	−0.007	(0.003)	[0.001]	0.0035	1.00	1	87
ChatGPT Pro 500 vs Plus	163	−0.022	(0.004)	[0.012]	<0.0005	1.00	21	45
YouTube Premium vs Google AI Pro	119	0.204	(0.042)	[0.086]	<0.0005	0.43	92	1
YouTube Premium Lite vs Google AI Pro	73	0.340	(0.039)	[0.078]	<0.0005	0.27	89	0
YouTube Music vs Google AI Pro	103	0.327	(0.034)	[0.071]	<0.0005	0.27	88	1
Google AI Plus vs AI Pro	159	−0.012	(0.003)	[0.007]	<0.0005	1.00	6	62
Google AI Ultra vs AI Pro	97	0.000	(0.001)	[0.001]	0.9770	1.00	1	89
X Premium Basic vs SuperGrok	151	0.151	(0.011)	[0.024]	<0.0005	0.50	94	2
X Premium vs SuperGrok	153	0.158	(0.011)	[0.024]	<0.0005	0.45	96	2
SuperGrok Lite vs SuperGrok	153	−0.009	(0.005)	[0.014]	0.0675	1.00	19	54
X Premium Plus vs SuperGrok	140	0.011	(0.005)	[0.002]	0.0130	1.00	11	47
SuperGrok Plus vs SuperGrok	154	−0.008	(0.005)	[0.013]	0.1115	1.00	29	54
SuperGrok Heavy vs SuperGrok	128	0.000	(0.001)	[0.001]	0.9285	1.00	1	95
X Premium vs X Premium Plus	150	0.150	(0.009)	[0.023]	<0.0005	0.45	95	2
X Premium Basic vs SuperGrok Lite	150	0.159	(0.011)	[0.036]	<0.0005	0.50	83	15
Microsoft 365 Personal vs Premium	160	0.011	(0.006)	[0.009]	0.0780	0.85	61	11
Character.AI Lite vs Character.AI+	151	0.081	(0.016)	[0.033]	<0.0005	0.60	77	5
Claude Max 5× vs Pro	161	0.009	(0.003)	[0.007]	0.0015	1.00	2	47
Claude Max 20× vs Pro	161	0.007	(0.003)	[0.008]	0.0275	1.00	4	50

Notes: Monthly App Store prices in storefronts other than the United States with income data. The outcome is $d_c = \ln(p_{A,c}/p_{B,c}) - \ln(P_A^{US}/P_B^{US})$, the price of A relative to B in storefront c net of the U.S. ratio; a positive slope means that A is localised more than B. Pairs: each plan of Table 4, and each of Anthropic's, as A, relative to its firm's standard frontier tier as B (SuperGrok for X Corp/xAI, Character.AI+ for Character.AI), whether A is a lower or a higher tier, plus two X Corp pairs (footnote in the text). Standard errors: HC1 in parentheses, clustered by storefront currency (18–41 clusters, one of them the U.S.-dollar bloc) in brackets. RI p: two-sided randomisation-inference p-value from 2,000 permutations of income across storefronts; "<0.0005" means that no permutation reached the observed slope. LMIC median: median of e^{d_c} in low- and lower-middle-income storefronts. Share < 0.9: storefronts with $e^{d_c} < 0.9$; share at 1: storefronts with $|d_c| < 0.005$. The Character.AI pair excludes the storefronts in which only an annual Character.AI+ listing was captured (Section 4). X Premium includes increased Grok limits and X Premium Plus includes SuperGrok (X Corp, 2026); Microsoft 365 Premium includes the full Copilot allowance and Personal a small one; Google AI Plus, SuperGrok Lite and Character.AI Lite are capped versions of the products above them.

Table 4: Relative prices within firm and storefront: stacked regression by type of product

		γ_g (s.e.), storefronts			
Group g	Items	All	U.S.-dollar	Local currency	RI p
Capped AI tier	ChatGPT Go, Google AI Plus, SuperGrok Lite, Character.AI Lite	0.036 (0.031)	0.032 (0.028)	0.020 (0.043)	0.4145
No AI allowance	YouTube Premium, YouTube Premium Lite, YouTube Music, X Premium Basic	0.204 (0.044)	0.103 (0.030)	0.380 (0.090)	0.0005
Small AI allowance	X Premium, Microsoft 365 Personal	0.116 (0.051)	0.069 (0.037)	0.169 (0.064)	0.0001

Notes: OLS of $r_{ic} = \ln(p_{ic}/P_i^{US})$ on firm-by-storefront fixed effects, item fixed effects and ln(GNI per capita) interacted with the group of item i, for the 22 items of five firms (OpenAI, Google, X Corp/xAI, Microsoft, Character.AI) in 169 storefronts (3,253 observations; 104 dollar storefronts, 65 local-currency storefronts); the inclusion rule is in the footnote of Section 5.1. The omitted group is the firm's AI tiers with large allowances (ChatGPT Plus, Pro 100, Pro 200 and Pro 500; Google AI Pro and AI Ultra; X Premium Plus, SuperGrok, SuperGrok Plus and SuperGrok Heavy; Microsoft 365 Premium; Character.AI+). Standard errors two-way clustered by storefront and item. RI p: share of the 252,000 relabellings that permute the group labels among the items within each firm, holding the number of items per group and firm fixed, in which $|\gamma_g|$ is at least the observed value.

Table 5: AI plans in U.S. dollars and relative to the platform's schedule, 16 October 2026

	U.S.		USD price	Deviation from the platform's schedule				Exact
Plan	price	N	slope (s.e.)	slope (s.e.)	within ±3% (%)	>10% below (%)	LMIC median (%)	ref. (%)
ChatGPT Go	8.00	161	0.143 (0.006)	0.110 (0.006)	14	75	−38	0
ChatGPT Plus	19.99	161	0.039 (0.005)	0.004 (0.003)	87	7	0	76
ChatGPT Pro 100	100.00	161	0.018 (0.004)	−0.015 (0.004)	60	10	0	0
ChatGPT Pro 200	200.00	161	0.032 (0.006)	0.000 (0.003)	73	2	0	0
ChatGPT Pro 500	500.00	161	0.016 (0.004)	−0.016 (0.004)	60	11	0	0
Claude Pro	20.00	159	0.026 (0.006)	−0.007 (0.004)	60	1	0	0
Claude Max 5×	124.99	159	0.035 (0.006)	0.002 (0.005)	56	1	0	0
Claude Max 20×	249.99	159	0.033 (0.006)	0.000 (0.005)	57	1	0	0
Google AI Plus	4.99	154	0.028 (0.004)	−0.003 (0.008)	59	29	0	25
Google AI Pro	19.99	159	0.038 (0.005)	0.002 (0.005)	60	16	0	76
Google AI Ultra	199.99	94	0.032 (0.006)	0.001 (0.004)	72	11	0	0
Perplexity Pro	20.00	161	0.026 (0.006)	−0.007 (0.003)	61	0	0	0
SuperGrok Lite	10.00	152	0.018 (0.005)	−0.016 (0.004)	66	13	0	0
SuperGrok	30.00	151	0.029 (0.008)	−0.005 (0.004)	70	1	0	0
SuperGrok Heavy	300.00	127	0.030 (0.008)	−0.006 (0.005)	68	1	0	0
Microsoft 365 Personal	9.99	159	0.047 (0.009)	0.021 (0.007)	23	62	−15	31
Microsoft 365 Premium	19.99	157	0.038 (0.005)	0.002 (0.004)	55	11	0	78

Notes: Monthly App Store prices on 16 October 2026 in storefronts other than the United States with income data and an estimated storefront multiplier (161 at most; Nigeria and Türkiye have no multiplier), so N and the U.S.-dollar slopes differ slightly from Appendix Table E.9, which uses every storefront with income data. USD price slope: OLS of ln(local price in dollars at market rates) on ln(GNI per capita), HC1 standard errors; the platform's schedule itself has a slope of 0.033 (s.e. 0.006, $N = 161$). Deviation: ln(local price / schedule price), where the schedule price is the panel's modal price for the plan's U.S. price point in that storefront when one exists and the storefront multiplier times the U.S. price otherwise (Section 5.3); >10% below: log deviation below −0.10; LMIC median: median deviation in low- and lower-middle-income storefronts as a percentage, $100(e^d - 1)$. Exact ref.: share of the plan's storefronts whose reference is an observed modal price rather than an extrapolation.

Table 6: Validation of the estimated platform schedule

Check	Result
Multipliers and ladder rebuilt from the panel items	164 multipliers, largest $\lvert\Delta\ln\rvert$ 0; 2,414 ladder entries, 0 differ
Income slope of the schedule, main rule (modal share ≥ 1/2)	0.0327 (0.0060); $N = 161$
Strict majority rule (modal share > 1/2)	0.0272 (0.0052); $N = 147$
Items of type subscription only	0.0331 (0.0061); $N = 160$
Items of type consumable only	0 storefronts with a multiplier (20 needed for a slope)
Leave one app out (29 refits): income slope	0.0291 to 0.0346; $N \geq 159$
Leave one app out: largest change of a storefront's multiplier over the refits, $\lvert\Delta\ln\rvert$	median over storefronts 0.000; 90th percentile 0.068; maximum 0.701
Split half (300 random splits of the apps): multipliers of the two halves	median $\lvert\Delta\ln\rvert$ 0.00001; 67.2% within 3%; 87.9% within 10%; 110 storefronts in common (median)
Split half: income slope of the half samples	0.021 to 0.041 (5th to 95th percentile)
Frontier plans on an exact ladder point: listed at the panel's modal price	257 of 375 (68.5%)
ChatGPT Plus; Google AI Pro; Microsoft 365 Premium	117 of 126; 73 of 124; 67 of 125
in the 33 storefronts where the schedule departs from FX parity by more than 20%	24 of 69 (34.8%)
Ladder entries with a unique mode (modal share ≥ 1/2)	1,551 of 2,414 (64.3%); median modal share 0.545; 27.1% of entries have a share ≥ 0.75

Notes: Panel of 30 small-developer apps (28 with a priced subscription item; one lists only a consumable and one no priced item), 16 October 2026. Income slope: OLS of ln(multiplier / market exchange rate) on ln(GNI per capita), HC1 standard errors. Split-half: 300 random partitions of the apps (seed 20261016); multipliers of the two halves compared on their common storefronts. Exact coincidence: frontier plans whose U.S. price point has a modal panel price in the storefront, listed within 0.5% of it. The modal-share and weak-storefront detail is in Appendix Table E.14.

Table 7: The within-storefront dollar increment between a higher and a lower tier of the same firm

	Relative to the schedule increment			In dollars, relative to the U.S.	
Pair	N	Slope (s.e.)	Median ratio: HIC / UMIC / LMIC / LIC	N	Slope (s.e.)
ChatGPT Plus − Go	161	−0.047 (0.005)	1.04 / 1.17 / 1.25 / 1.25	163	−0.010 (0.007)
Character.AI+ − Lite	143	−0.059 (0.012)	1.09 / 1.40 / 1.40 / 1.40	150	−0.043 (0.011)
Google AI Pro − AI Plus	154	0.001 (0.005)	1.00 / 1.00 / 1.00 / 1.00	159	0.041 (0.005)
ChatGPT Pro 200 − Plus	161	−0.001 (0.003)	1.00 / 1.00 / 1.00 / 1.00	163	0.031 (0.006)
Microsoft 365 Premium − Personal	157	−0.014 (0.008)	1.05 / 1.15 / 1.15 / 1.15	160	0.032 (0.008)

Notes: Left: OLS of $\ln[(p_{H,c}-p_{L,c})/(P^D_{H,c}-P^D_{L,c})]$ on ln(GNI per capita), with P^D the plans' reference prices (Section 5.3), HC1 standard errors, storefronts with income data and a reference for both plans; medians of the ratio by income group. Right: OLS of $\ln[(p_{H,c}-p_{L,c})/(P^{US}_H-P^{US}_L)]$ with both prices in dollars at market rates. A negative slope means that the dollar gap is wider where incomes are lower.

Table 8: Affordability of the $20 tier and of the localised entry tier by income group (medians)

		$20 plan, annual price as % of						Entry tier	$20 plan, 2% benchmark	
Income group	N	GNI p.c.	Plus local	PPP	Consumption	Poorest 20%	Richest 20%	local price	Above (%)	Subsidy (%)
High income	75	0.65	0.75	0.40	1.18	1.64	0.32	0.27	0	0
Upper-middle income	58	3.09	3.27	1.20	4.53	9.17	1.35	0.97	78	35
Lower-middle income	47	8.73	8.89	3.38	11.85	32.83	4.41	2.58	100	77
Low income	21	28.92	28.90	10.67	35.57	73.20	13.05	7.21	100	93

Notes: Medians across the economies of each group of $1200p/y \times 100$. Unless stated, the $20 columns apply a uniform $20 reference price, before tax, to every economy with income data (N in the second column, including the 31 economies with income data but without a storefront of their own). GNI p.c.: Atlas GNI per capita (2025 for 182 economies, 2022–2024 for 19). Plus local: the tax-inclusive local first-wave price of ChatGPT Plus in the economies with a listing (62, 51, 36 and 15, including the United States). PPP: GNI per capita at purchasing-power parity, which equals the burden if the price were indexed to local price levels (Remark 2). Consumption: household final consumption per capita. Quintile columns: mean income of the poorest and richest fifth, approximated as GNI per capita times the survey share relative to 20% (163 economies with survey data). Entry tier: ChatGPT Go at its tax-inclusive local first-wave price, in the economies with a local listing (62, 51, 36 and 15, including the United States). Above: share of economies whose $20 burden exceeds 2%. Subsidy: the share of the $20 price that would bring the median economy to the benchmark, $100\max(0, 1-y/12{,}000)$. Sensitivity to the benchmark, the income vintage and the price schedule is in Appendix Tables E.6 and E.7.

Table A.1: Predictions of the tier model by the constraint that binds on the frontier price

	Unconstrained nested optimum (Prop. A.3)	Frontier held at the default by the override cost (Prop. A.4(i))	Arbitrage floor binding, common friction (Prop. A.4(ii))	Segmented demand (Remark 1)
Frontier deviation from the default	falls as income falls; elasticity (A.10), between $\beta_{\Delta H}$ and β_L	zero, with bunching at the default	a discount of χ dollars where the desired gap exceeds χ	elasticity $\sigma_H\xi_H/(\xi_H+\ell_H)$; zero if overridden nowhere
Entry deviation	slope $\beta_L \in (0,1)$	positive slope where $G_{L,c} > K$; restrained by Prop. A.3(v)	a discount of at most χ dollars	slope $\sigma_L\xi_L/(\xi_L+\ell_L)$
Dollar increment $p_{H,c}-p_{L,c}$	rises with income	falls with income	unchanged where both floors bind	falls with income if the frontier price is nearly uniform
Ratio $p_{H,c}/p_{L,c}$	rises as income falls	rises as income falls largely; not the overrides to the U.S. price in taxed dollar storefronts (Table E.13)	rises as income falls possibly the $100–500 tiers; Appendix A.4 suggests frictions too large to bind	rises as income falls
Consistent with the audit	no (increment)			yes

Notes: Deviations are $\ln(p_{k,c}/P^D_{k,c})$, increments and ratios are measured within a storefront, net of the platform's conversion. Audit values: deviation slopes 0.004 for ChatGPT Plus and 0.110 for ChatGPT Go (Section 5.3); increment slope −0.047 (s.e. 0.005) for Plus over Go.

B Data construction and replication

Replication package. The package contains the raw extracts (World Bank API pulls, the Microsoft dataset at commit 507c316, Anthropic Economic Index files, Google ATLAS data, ITU price baskets, both App Store audit waves, the default-panel records, the cross-channel extracts, historical exchange-rate snapshots and current exchange rates), the processing scripts, the analysis, and the figure and table generators, together with a README that follows the template of the Social Science Data Editors. Every number in the text can be reproduced from it: most are written to output/results.json and output/results _wave2.json by the analysis scripts, the sensitivity analyses are written by the scripts in extensions/ and checks/, and the rest are simple calculations from those values or are read from the output tables, the processed data or the scripts. Re-running the pipeline from the shipped raw extracts reproduces every table body and figure byte for byte, and independent implementations of the key regressions match the reported estimates.

App Store audit, wave 1. Requests were made on 27 September 2026 (UTC) to https://apps.apple.com/<cc>/app/id<app> for ChatGPT (6448311069), Claude (6473753684) and Google Gemini (6477489729), using the two-letter codes of all 217 World Bank economies plus Taiwan. Of 654 requests, 500 returned local pages, 135 were redirected to the U.S. store and 19 returned "not found"; the 135 redirects correspond to 45 codes without a storefront of their own (Table E.4 reports them as "No store"), and the 19 "not found" pages to apps that are not offered in a storefront. Prices were parsed from the displayed strings, including locale-specific separators and abbreviations (e.g. Indonesian "ribu" and "juta"). Four storefront-plan cells in which the observed page listed only an annual price were dropped. In three cells two different candidate prices are listed for the same plan: Google AI Pro in India and Türkiye, where one of five listings carries a lower price, and in Benin, where a 10 TB variant is listed alongside the standard plan; the modal price is used, and taking the lowest listed price instead changes the Google AI Pro slope by 0.0001. Re-implementing the selection rules identifies a single surviving candidate price in every ChatGPT Go, ChatGPT Plus and Claude Pro storefront-plan cell, and annual-price exclusion cutoffs of three, five and eight times the lowest listed price select the same listings. The first wave stored the parsed listing records, not the pages. The first-wave regressions include the U.S. storefront, so their samples (162–165 storefronts by app) differ by one from the second-wave samples, which exclude it.

App Store audit, wave 2, and the page archive. On 16 October 2026 the same request was made for 34 apps in all 218 codes (7,412 requests; requests that returned HTTP 429 were repeated until a page or a definitive status was obtained, and the last definitive record per storefront and app is used). Apple's product page had changed from an embedded JSON record to an HTML list between the waves; both formats are parsed, and the full pages of the three assistants and the in-app purchase block of every other page were stored compressed, each with the SHA-256 fingerprint of the page as received, so that the listing behind every parsed record can be inspected and the extraction replayed. The fingerprints travel with the parsed records in the package; the page archive is deposited as `appstore_wave2_page_archive_2026-10-16.zip` in the Zenodo record of version 3.1: 12,860 files covering the 12,225 pages with HTTP 200, 109 MB, and the hashes of all 635 full pages match the parsed records. Plans that OpenAI renamed between the waves (Pro 5× and 20× to Pro 100 and Pro

200) are matched by their position in the menu, and the new Pro 500 tier is recorded but has no first-wave counterpart. Items are classified as consumables when their name denotes credits, gems, boosts, packs or similar one-off purchases, and as subscriptions otherwise. The monthly price of a subscription is the modal price among its listed variants after excluding variants priced at five or more times the lowest, the same rule as in the first wave; because the product page lists at most ten purchases, the monthly variant can be missing from a full list while an annual variant remains, so a price recorded as monthly is dropped when it exceeds three times the platform's reference for the item's U.S. monthly price (the storefront multiplier of the default panel times the U.S. price, or three times the U.S. price itself for single-variant listings in storefronts without a multiplier). The guard flags 151 of 21,938 item-storefront cells with a U.S. price and income data (157 before the income restriction), in 14 items, 129 of them in Super Duolingo, Headspace Plus, Gemini's 100 GB plan, Character.AI+ and Evernote Plus (Table E.11); a storefront whose monthly variant is censored is thereby missing rather than mispriced. Items that the same firm lists under two apps or two spellings are counted once in the item-level statistics (five of 152 items), keeping the listing with more storefronts.

Estimated platform schedule. For each of the 30 small-developer apps and each storefront the monthly price of every subscription item is recorded, and items are keyed by their U.S. price point. For storefront c and U.S. price point P, the schedule price is the modal local price among the items at that point when at least three items are observed and the mode is unique and accounts for at least half of them; 1,551 of 2,414 storefront-point cells satisfy this, and 22 cells are two-way ties at exactly one half, which yield no schedule price. The storefront multiplier is the median of the schedule price divided by P over points with $P \geq \$2.99$, computed for storefronts with at least three such points (164); its spread, the range of the point-wise ratios divided by the median, is below 0.2 in 111 storefronts, which form the clean subsample (108 with income data). A plan's reference price is the exact schedule price for its U.S. price point when one exists, and otherwise the multiplier times its U.S. price; Table 5 reports the share of each plan's storefronts with an exact reference. Resolving ties by file order, by the lower price or by the higher price instead changes the schedule's slope within 0.0325–0.0329 and the ChatGPT Go deviation slope within 0.110–0.111; discarding every cell whose mode accounts for exactly half of the items changes them to 0.027 and 0.110 (Table 6). The multiplier divided by the market exchange rate is the tax factor that the platform applies, which is one in the median U.S.-dollar storefront and above one in every European storefront priced in its own currency (1.19–1.36 in 31 of the 34 European and Central Asian storefronts outside the dollar, and 1.05–1.12 in Romania, Cyprus and Montenegro), in line with their value-added tax rates and Apple's rounding to local price conventions. The vintage test of Section 5.3 uses the range of the log exchange rate of each storefront currency against the dollar over six dates between March 2024 and October 2026 (data/raw/fx_hist/).

Microsoft region-imputed economies. Six pools with identical four-period series: West Africa (Benin, Burkina Faso, Ghana, Guinea, Guinea-Bissau, Liberia, Mali, Mauritania, Niger, Nigeria, Sierra Leone, Togo); East Africa (Burundi, Eritrea, Ethiopia, Somalia, South Sudan, Sudan, Tanzania, Uganda); Central Africa (Cameroon, Central African Republic, Chad, Congo, Congo DR); Southern Africa (Angola, Madagascar, Malawi, Mozambique); the Guianas (French Guiana, Guyana, Suriname) and Venezuela; and Afghanistan, Tajikistan and Turkmenistan. These 36 economies coincide with those marked as region-imputed in Misra et al. (2025, Appendix Table 2). French Guiana and Taiwan are not

among the 217 World Bank WDI economies (the App Store audit adds Taiwan to those 217), so 145 of the 147 Microsoft economies are matched and 142 have income data; of the 111 country-specific economies, 109 have income data and enter the cross-sectional regressions of Appendix D, and 33 of the 36 region-imputed economies have income data.

Anthropic windows and samples. Table B.1 lists the windows, sampling and product coverage of the releases used, from their data documentation. The index pools free and paid accounts in every release. Published indices are used for April and May 2026. For August and November 2025 and February 2026 the index is recomputed as usage share over working-age population share among geolocated countries, because the August 2025 published values include un-geolocated traffic in the usage denominator (scaling all values by a common factor of 0.843) and later releases publish counts rather than indices. The recomputed index exists for 165–169 geolocated economies (157–161 observed in consecutive windows), and the cross-sectional regressions apply Anthropic's 200-conversation publication floor, which leaves 114–115 economies in the 2025 and February 2026 windows and 113–120 published economies in April and May 2026; the rollout comparison of Appendix F uses the recomputed index without the floor, which is why its comparison groups count 141–145 economies (109 for February–May 2026, whose May value is the published index).

Table B.1: Anthropic Economic Index releases used

Release	Window	Sample	Accounts and products	Publication rule
15 Sep 2025	4–11 Aug 2025	about 1 million conversations	Claude.ai Free and Pro	countries with at least 200 sampled conversations
15 Jan 2026	13–20 Nov 2025	about 1 million conversations	Free, Pro and Max (the raw file's label still reads "Free and Pro")	same
24 Mar 2026	5–12 Feb 2026	about 1 million conversations	Free, Pro and Max	same
26 Jun 2026	calendar months April and May 2026	fixed number of conversations per hour, 10 Apr–10 Jun 2026	Claude chat and Cowork; Free, Pro and Max (web and desktop)	geography sample floor per period; published index used

Notes: From the data documentation of each release (data/raw/anthropic/DATA_DICTIONARY.md) (Handa et al., 2025; Appel et al., 2025, 2026; Massenkoff et al., 2026a; Anthropic, 2026a). The August 2025, November 2025 and February 2026 indices are recomputed from the sampled counts as described above; the April and May 2026 values are the published indices.

C Prices in other channels: Google Play and Microsoft's web store

The audit observes one channel. If the App Store prices of the AI plans were specific to that channel, for instance because an in-app price list adds the channel's own tax or fee, the same plan would be priced differently on the web or on Google Play; if they are the prices the vendor sets for the country, the channels would agree. Two extracts collected on the day of the second wave, 16 October 2026, allow the comparison for the plans whose prices other channels display (Section 4); Table C.1 summarises it, and Table C.2 lists the Google Play comparison by country.

Microsoft's web store. The page https://www.microsoft.com/<locale>/microsoft-365/buy/compare-all-microsoft-365-products lists the monthly and annual prices of Microsoft 365 Personal, Family and Premium. Of 107 locale paths tried, 63 returned prices, covering 57 countries (several countries appear under two locales with identical prices); 42 returned no page, among them Nigeria, Kenya, Pakistan, Egypt and Bangladesh, where the product pages exist but show no price, so the web store does not sell the consumer plans there; Russia lists the products without prices; and the Chinese page offers Personal and Family only. The prices follow the locale path rather than the requesting address, since every page's currency matched its country, and the pages do not state whether prices include tax. Within a country the three plans share a currency and a tax treatment, so the ratio of the Personal to the Premium price, relative to the same ratio in the United States, is a default-free measure in the same spirit as Section 5.1: its income slope across the 54 countries with both plans and income data is 0.082 (HC1 s.e. 0.018), the ratio is 31% below the U.S. ratio in India, the only low- or lower-middle-income country with a web price, and 9% below it in the median high-income country, which reproduces on the web the moderate localisation of Personal relative to Premium that the App Store shows (Table 3). Where the web currency equals the storefront currency, the web price is within 5% of the App Store price in all 49 countries for both plans (98 comparisons; median log difference 0.000; identical to the cent in 37 of 49 for each plan and within 0.5% in 45; the widest gap, Sweden, is 4.3% for Personal and 2.4% for Premium), so Microsoft sets essentially one price per country across the two channels.

Google Play. The Play web page of an app, requested with the country parameter `gl`, shows the range of its in-app purchase prices, from the lowest to the highest item, but not the items themselves. For ChatGPT the lowest item in the United States is $8.00, the Go price, and for Claude $20.00, the Pro price, so the lowest item is compared with the App Store price of that plan; Gemini's page shows no range in any of the 30 countries because its plans are sold through Google One, and the ranges of the other apps cover items that cannot be assigned to plans. Where the Play page and the App Store storefront share a currency (25 of the 30 countries; Argentina, Ukraine and Kenya have dollar-priced storefronts and Bangladesh has no storefront of its own), the lowest ChatGPT item lies within 5% of the App Store Go price in 22 of the 25 countries and is identical to it, within the 0.5% band of the stability comparison, in 20 (19 to the cent; Brazil R$39.99 against R$39.90; median log difference 0.000), and in dollars at the exchange rates of 15 October 2026 it is 0.63 times the U.S. price in the median lower-income country with a storefront (0.64 including Bangladesh), the same discount that the App Store shows. For Claude the lowest item is within 5% of the App Store Pro price in 12 of 25 countries and differs in both directions elsewhere, by -0.45 to $+0.13$ log points, so Claude's lowest Play item is not reliably its monthly Pro price and the comparison is weaker; in dollars it is 1.05 times the U.S. price in

the median lower-income country, with no sign of a lower price where incomes are lower. The Microsoft Copilot listing on Play is the former Office hub app, whose range mixes Microsoft 365 and Copilot items and is not used.

What the check does and does not show. Of the two localised AI tiers, ChatGPT Go and Character.AI Lite, only ChatGPT is among the apps collected on Google Play, and for Go the localisation is the vendor's country price and not an App Store artefact; for Microsoft the two channels agree within 5%, usually to the cent; for the frontier tiers of OpenAI and Google the check is silent, because Play shows only the lowest item and the vendors' own web pages localise by the requesting address and could not be collected for many countries at zero cost. No official per-country price table is published by OpenAI or Anthropic; their help pages give the U.S. price and state that prices are localised in some markets, and country prices appear only in press reports. The raw pages of both extracts are deposited as `cross_channel_page_archive_2026-10-16.zip` (487 files, 77 MB) in the same record; each row of the two extracts carries the archive path and the SHA-256 of the page it was parsed from.

Table C.1: Cross-channel comparison of listed prices, 16 October 2026

Comparison	Result
Microsoft 365 web store: income slope of the Personal / Premium price ratio relative to the U.S.	0.082 (0.018); $N = 54$
median of the same ratio, low and lower-middle income countries	-31% ($N = 1$: India)
median of the same ratio, high-income countries	-9% ($N = 39$)
Web store against App Store, Microsoft 365 Premium, same currency	49 comparisons; median log difference 0.000; 100% within 5%; identical to the cent in 37, within 0.5% in 45; widest gap 2.4% (Sweden)
Web store against App Store, Microsoft 365 Personal, same currency	49 comparisons; median log difference 0.000; 100% within 5%; identical to the cent in 37, within 0.5% in 45; widest gap 4.3% (Sweden)
Google Play lowest in-app item against the App Store price of ChatGPT Go (US minimum $8.00)	25 of 30 countries comparable; median log difference 0.000; 88% within 5%
Google Play lowest in-app item against the App Store price of Claude Pro (US minimum $20.00)	25 of 30 countries comparable; median log difference 0.000; 48% within 5%
ChatGPT on Google Play: lowest in-app item in low and lower-middle income countries, US$ price relative to the U.S. (median)	0.63 ($N = 5$)
Claude on Google Play: the same	1.05 ($N = 5$)
Gemini on Google Play	no in-app price range shown in any of the 30 countries

Notes: Microsoft rows: monthly web prices from 63 locale pages (57 countries); the ratio is $\ln(\text{Personal}_c/\text{Premium}_c) - \ln(\text{Personal}_{US}/\text{Premium}_{US})$ in countries with both plans and income data, HC1 standard error; medians are reported as $100(e^m - 1)$ percent. Web-against-App-Store rows compare the same product in the same currency. Google Play rows: the lowest in-app item against the App Store price of ChatGPT Go or Claude Pro where the two share a currency; the U.S. is the anchor and not compared; the dollar ratio uses the exchange rates of 15 October 2026 and the median over the low- and lower-middle-income countries with a storefront of their own.

Table C.2: Google Play's lowest in-app price and the App Store price, ChatGPT and Claude, 16 October 2026

Country	Income group	Currency	Play lowest item	US$	App Store	ln difference
United States	High income	USD	8.00	8.00	8.00	–
Australia	High income	AUD	13.00	9.04	13.00	0.000
Germany	High income	EUR	7.99	8.97	7.99	0.000
Canada	High income	CAD	11.00	7.75	12.99	−0.166
United Kingdom	High income	GBP	6.99	9.26	6.99	0.000
United Arab Emirates	High income	AED	29.99	8.17	29.99	0.000
Japan	High income	JPY	1,400	8.88	1,400	0.000
Korea, Rep.	High income	KRW	13,000	9.64	13,000	0.000
Saudi Arabia	High income	SAR	35.00	9.33	35.00	0.000
Poland	High income	PLN	34.99	8.99	34.99	0.000
Chile	High income	CLP	7,990	8.11	7,990	0.000
Turkiye	Upper-middle income	TRY	249.99	5.06	249.99	0.000
Argentina	Upper-middle income	ARS	8,955.05	5.92	US$5.99	–
Mexico	Upper-middle income	MXN	129.00	6.98	129.00	0.000
Malaysia	Upper-middle income	MYR	23.99	5.88	23.99	0.000
Brazil	Upper-middle income	BRL	39.99	7.92	39.90	0.002
Peru	Upper-middle income	PEN	19.90	5.78	19.90	0.000
Colombia	Upper-middle income	COP	24,900	7.80	24,900	0.000
Thailand	Upper-middle income	THB	259.00	7.72	259.00	0.000
South Africa	Upper-middle income	ZAR	149.00	8.99	149.99	−0.007
Ukraine	Upper-middle income	UAH	299.99	6.66	US$6.99	–
Indonesia	Upper-middle income	IDR	75,000	4.18	75,000	0.000
Viet Nam	Upper-middle income	VND	120,000	4.63	132,000	−0.095
Philippines	Upper-middle income	PHP	300.00	4.76	300.00	0.000
Egypt, Arab Rep.	Lower-middle income	EGP	220.00	4.14	249.99	−0.128
Bangladesh	Lower-middle income	BDT	699.00	5.66	no store	–
India	Lower-middle income	INR	399.00	4.20	399.00	0.000
Kenya	Lower-middle income	KES	745.00	5.74	US$5.99	–
Pakistan	Lower-middle income	PKR	1,400.00	5.05	1,400.00	0.000
Nigeria	Lower-middle income	NGN	7,000.00	5.26	6,900.00	0.014
United States	High income	USD	20.00	20.00	20.00	–
Australia	High income	AUD	33.99	23.63	30.00	0.125
Germany	High income	EUR	22.00	24.70	22.00	0.000
Canada	High income	CAD	28.00	19.72	28.00	0.000
United Kingdom	High income	GBP	18.00	23.85	20.00	−0.105
United Arab Emirates	High income	AED	79.99	21.78	79.99	0.000
Japan	High income	JPY	3,400	21.56	3,000	0.125
Korea, Rep.	High income	KRW	30,000	22.25	33,000	−0.095
Saudi Arabia	High income	SAR	89.99	24.00	89.99	0.000
Poland	High income	PLN	89.99	23.12	99.99	−0.105
Chile	High income	CLP	19,990	20.28	22,990	−0.140
Turkiye	Upper-middle income	TRY	999.99	20.22	999.99	0.000
Argentina	Upper-middle income	ARS	29,900.00	19.76	US$20.00	–
Mexico	Upper-middle income	MXN	399.00	21.59	399.00	0.000
Malaysia	Upper-middle income	MYR	99.90	24.48	99.90	0.000
Brazil	Upper-middle income	BRL	110.00	21.79	129.90	−0.166
Peru	Upper-middle income	PEN	79.90	23.23	89.90	−0.118
Colombia	Upper-middle income	COP	63,490	19.89	99,900	−0.453
Thailand	Upper-middle income	THB	799.00	23.82	699.00	0.134
South Africa	Upper-middle income	ZAR	399.99	24.15	399.99	0.000
Ukraine	Upper-middle income	UAH	979.99	21.77	US$25.00	–
Indonesia	Upper-middle income	IDR	349,000	19.45	349,000	0.000
Viet Nam	Upper-middle income	VND	499,000	19.27	599,000	−0.183
Philippines	Upper-middle income	PHP	1,300.00	20.61	1,290.00	0.008
Egypt, Arab Rep.	Lower-middle income	EGP	1,099.99	20.71	999.99	0.095
Bangladesh	Lower-middle income	BDT	2,700.00	21.87	no store	–
India	Lower-middle income	INR	1,999.00	21.05	1,999.00	0.000
Kenya	Lower-middle income	KES	3,000.00	23.11	US$25.00	–
Pakistan	Lower-middle income	PKR	4,900.00	17.69	4,900.00	0.000
Nigeria	Lower-middle income	NGN	33,500.00	25.18	29,900.00	0.114

Notes: Upper block: ChatGPT, against the App Store price of ChatGPT Go; lower block: Claude, against Claude Pro; countries in descending order of GNI per capita. Play lowest item: the lower end of the in-app purchase range on the Play web page, in the currency of the page. US$: that price at the exchange rates of 15 October 2026. App Store: the monthly price of the plan in the storefront of the same country, in the storefront's currency (a dollar price marks a storefront priced in U.S. dollars, which is not compared). ln difference: ln(Play lowest item / App Store price) where the currencies coincide; "–" where they do not, where the country has no storefront (Bangladesh), or for the United States, which anchors the comparison.

D The public usage indicators: tables and figures

This appendix holds the estimates summarised in Section 6: the income gradient of any use (Table D.1, Figure D.1, with controls in Table D.2), the comparison of any use with per-capita Claude activity (Figure D.3, Tables D.4, D.5 and D.6), and the accounting split of the any-use gradient into an internet-use component and the ratio of AI users to internet users (Table D.3, Figure D.2).

Any use. On country-specific estimates, a one-log-point increase in income is associated with 8.3 percentage points higher use, and the elasticity is 0.370 (s.e. 0.021; $R^2 = 0.71$). Including pooled regional values lowers the elasticity to 0.336; collapsing each pool to one observation restores it (0.367). The elasticity is unchanged by controls for internet use, tertiary enrolment, urbanisation, age structure, digital payments and region fixed effects (0.355–0.367), and conditional on income none of the controls is individually significant except urbanisation in columns (3)–(4) of Table D.2 ($p \approx 0.04$), which loses significance once region fixed effects are added. Between the first half of 2025 and the second quarter of 2026 the median user share rose by 5.5 points in the high-income group, 3.7 in the upper-middle-, 2.0 in the lower-middle- and 0.9 in the low-income group (Figure D.1b): proportional growth was similar across the three higher income groups (21–26%) and slightly faster in lagging economies (-0.026 per log point of the initial share, s.e. 0.011), while in percentage points each log point of income is associated with a 1.37-point larger gain (s.e. 0.13), which reconciles reports of faster proportional growth in lower-income countries (Chatterji et al., 2025) with reports of a widening divide (Microsoft AI Economy Institute, 2026a; Liu et al., 2025).

The accounting split. Define the ratio $R_c = D_c/I_c$, where D_c is Microsoft's working-age AI-use share and I_c is the all-age internet-use share, so that $\ln D_c = \ln I_c + \ln R_c$ exactly. Because Microsoft's estimator scales device telemetry by internet penetration, R_c backs that scaling out and is best read as the telemetry signal per online user rather than as an observed probability of adopting AI conditional on being online; the denominator also has a different age base (all ages, against 15–64 for D_c) and often a different year, and in lower-income economies with young populations the all-age rate understates the working-age rate, which raises R_c there relative to older, richer economies. In the 109-economy estimation sample no ratio exceeds one. The slopes on log income are 0.182 for internet use and 0.189 for the ratio, summing to 0.371 against the directly estimated slope of 0.370 (a rounding difference), so the internet-use component accounts for 49% of the gradient as an accounting matter; holding income fixed, the elasticity of AI use with respect to internet use is 0.05 (s.e. 0.20). Across groups (Table D.3, Figure D.2) the connectivity share of the gap to high-income economies rises from 21% for upper-middle-income to 49% for lower-middle-income and 67% for the three low-income economies with country-specific data. The "AI / internet" column of Table D.3 reports the arithmetic mean of the country-level ratios within each group (37.5% in high-income economies and 22.7%, 23.6% and 23.0% from the upper-middle- to the low-income group), whereas the log-gap columns are differences of group means of logs; the ratio's income slope across non-high-income economies is -0.03 (s.e. 0.06). The low-income row contains only three economies.

Per-capita Claude activity. Anthropic's own report documents an income gradient of Claude activity near 0.7 using GDP per working-age capita (Appel et al., 2025); the contribution here is the comparison

with a differently measured indicator on a common sample of 96 economies (42 high-, 33 upper-middle-, 20 lower-middle- and one low-income economy, Rwanda), so the comparison describes the gradient from lower-middle to high income and says little about low-income economies. The difference of 0.340 between the two elasticities (clustered s.e. 0.034; bootstrap 95% interval 0.27–0.40) persists when the user share is expressed in log-odds (0.50 versus 0.71), when high-income economies are excluded (0.33 versus 0.69), conditional on internet use, tertiary enrolment and urbanisation (0.31 versus 0.55), in earlier measurement windows (0.38 versus 0.62) and with pooled regional values included (0.36 versus 0.73); it is 0.266 with region fixed effects and 0.232 with controls, and leaving out each economy in turn gives 0.329–0.351 (Table D.5). On the 111 economies observed in all five Anthropic windows the elasticity rose from 0.676 in August 2025 to 0.741 in February 2026 (a change of 0.064, clustered s.e. 0.018) and was 0.734 in May 2026 (Table D.6).

Table D.1: Income gradient of generative-AI use (Microsoft AI User Share, Q2 2026)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Level (pp)	Log user share					
ln GNI per capita	8.326	0.370	0.336	0.367	0.360	0.357	0.367
	(0.655)	(0.021)	(0.015)	(0.019)	(0.024)	(0.062)	(0.067)
Controls	No	No	No	No	No	Yes	Yes
Region fixed effects	No	No	No	No	No	No	Yes
Sample	CS	CS	All	Coll.	CS†	CS†	CS†
Observations	109	109	142	115	97	97	97
R^2	0.67	0.71	0.72	0.73	0.69	0.73	0.74

Notes: OLS with HC3 standard errors in parentheses. CS: country-specific estimates only; All: including region-imputed values; Coll.: each region-imputed pool collapsed to one observation with population-weighted log income. †Common sample with non-missing controls (internet use, tertiary enrolment, urban share, population 65+, share making digital online merchant payments). Coefficients on controls in Table D.2.

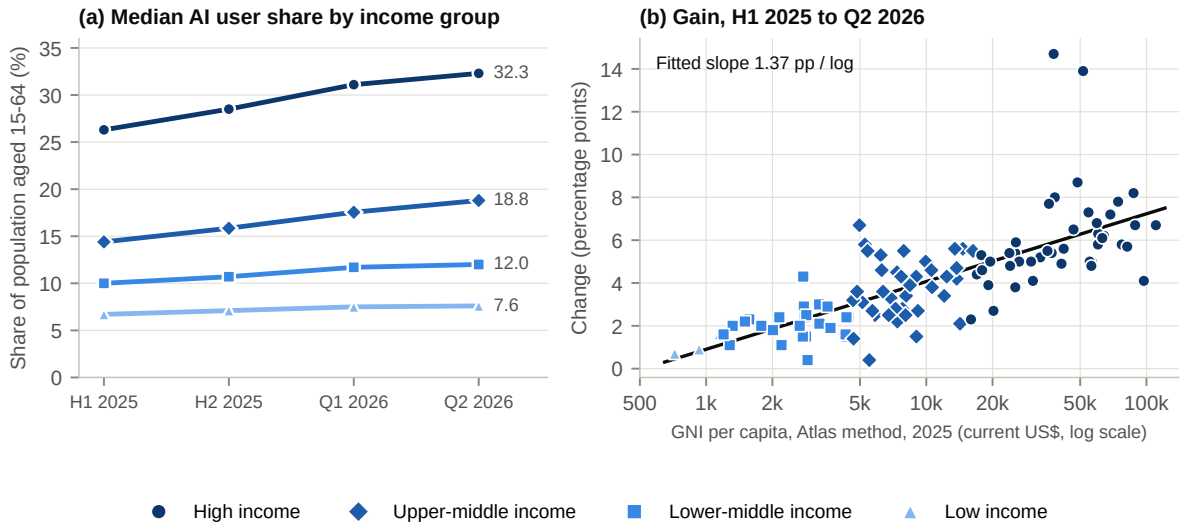


Figure D.1: Diffusion dynamics. (a) Median AI user share by income group, country-specific Microsoft estimates (low-income: three economies). (b) Change in the AI user share between H1 2025 and Q2 2026 against GNI per capita, with the fitted log-linear slope.

Table D.2: Income gradient of AI use with controls (log user share, country-specific, common sample)

	(1)	(2)	(3)	(4)	(5)
ln GNI per capita	0.3595	0.3554	0.3659	0.3566	0.3675
	(0.024)	(0.047)	(0.063)	(0.062)	(0.067)
Internet users (%)		0.0004	−0.0006	0.0002	0.0016
		(0.004)	(0.004)	(0.004)	(0.005)
Tertiary enrolment (% gross)			−0.0015	−0.0011	−0.0006
			(0.002)	(0.002)	(0.002)
Urban population (%)			0.0064	0.0063	0.0040
			(0.003)	(0.003)	(0.004)
Population 65+ (%)			−0.0077	−0.0083	−0.0011
			(0.008)	(0.008)	(0.011)
Paid online merchant digitally (%)				−0.0022	−0.0024
				(0.002)	(0.002)
Region fixed effects	No	No	No	No	Yes
Observations	97	97	97	97	97
R^2	0.69	0.69	0.72	0.73	0.74

Notes: OLS with HC3 standard errors. Controls measured in percent; Findex 2025 for digital payments.

Table D.3: Accounting split of the AI-use gap into internet use and the AI-to-internet-user ratio (country-specific estimates, Q2 2026)

		Group means (%)			Log gap to high income			Connectivity
Income group	N	AI users	Internet	AI / internet	Total	Connect.	Ratio	share (%)
High income	43	34.9	92.6	37.5	0.00	0.00	0.00	–
Upper-middle income	38	18.5	81.8	22.7	−0.63	−0.13	−0.50	21
Lower-middle income	25	12.2	59.6	23.6	−1.05	−0.51	−0.54	49
Low income	3	8.8	38.3	23.0	−1.34	−0.90	−0.44	67

Notes: $\ln D = \ln I + \ln(D/I)$ with D the AI user share (population 15–64) and I internet users (% of population, latest year). Group means are arithmetic means of country values, including the mean of the country-level ratios D/I. Log gaps are differences between group means of log values (log ratios of geometric means), so they differ from log ratios of the arithmetic means shown. The low-income row rests on three economies and is indicative only. The ratio D/I mixes age bases and backs out the internet-penetration scaling of Microsoft's estimator; no cap binds in the estimation sample.

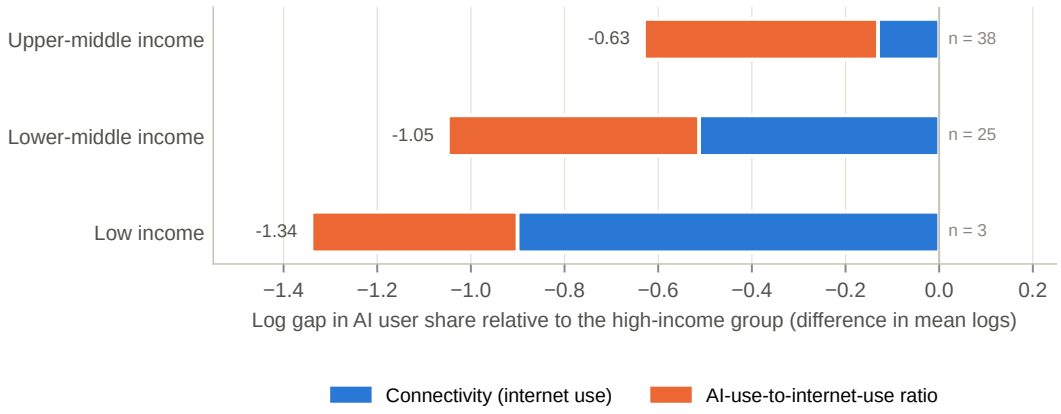


Figure D.2: Accounting split of the log gap in AI use relative to the high-income group (difference in group means of log shares) into internet use and the AI-use-to-internet-use ratio. Country-specific Microsoft estimates, Q2 2026; the low-income bar rests on three economies.

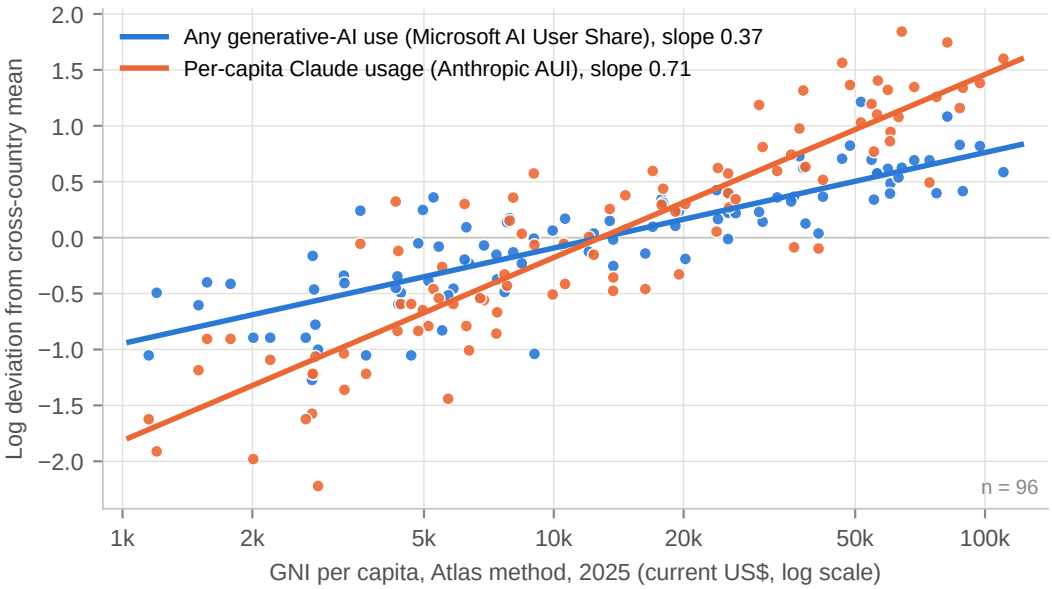


Figure D.3: Income gradients in two differently measured AI-use indicators. Log deviation from the cross-country mean of the Microsoft AI User Share (any generative-AI use, Q2 2026, country-specific estimates) and of the Anthropic AI Usage Index (per-capita Claude use, May 2026) against GNI per capita, for the 96 economies covered by both. Lines are OLS fits; slopes are income elasticities.

Table D.4: Income elasticities of any use and of per-capita Claude activity

Specification	Any use (MS)	Claude activity (AUI)	N
Baseline: country-specific MS, Q2 2026 vs AUI, May 2026	0.371 (0.026)	0.711 (0.036)	96
MS in log-odds (ceiling check)	0.496 (0.037)	0.711 (0.036)	96
Excluding high-income economies	0.334 (0.060)	0.688 (0.089)	54
Conditional on internet use, tertiary enrolment, urbanisation	0.315 (0.050)	0.547 (0.064)	92
Earlier windows: MS H2 2025 vs AUI Aug 2025	0.380 (0.028)	0.619 (0.047)	93
Including region-imputed MS values	0.358 (0.020)	0.733 (0.030)	108
Difference in baseline elasticities (AUI − MS)	0.340 (0.034); bootstrap 95% CI [0.27, 0.40]		96

Notes: Cells are OLS slopes on log GNI per capita in common samples; outcomes are in logs except the explicitly labelled Microsoft log-odds row, which is not the same elasticity. Parentheses give HC3 errors except for the stacked difference row. MS: Microsoft AI User Share; AUI: Anthropic AI Usage Index. The difference test stacks both outcomes with an outcome-specific intercept and slope, clustering by economy. The bootstrap resamples economies 2,000 times. The main sample comprises 42 high-, 33 upper-middle-, 20 lower-middle- and one low-income economy. Excluding high-income economies leaves 54.

Table D.5: Sensitivity of the difference between the two usage-indicator income elasticities

Specification	Difference	HC3 SE	95% interval	N
Baseline	0.340	0.034	[0.273, 0.407]	96
2025 GNI only	0.356	0.034	[0.289, 0.423]	92
Non-high-income	0.354	0.085	[0.187, 0.521]	54
Region fixed effects	0.266	0.050	[0.169, 0.363]	96
Internet, education, urbanisation	0.232	0.068	[0.099, 0.365]	92
Population weighted	0.476	0.086	[0.308, 0.644]	96

Notes: Dependent variable is $\ln(AUI/D)$, whose slope on log income equals the difference of the two elasticities on a common sample; baseline uses 96 common economies. Intervals use the normal approximation. Population weighting changes the estimand.

Table D.6: Income elasticity of the Anthropic AI Usage Index over time

	All qualifying economies				Common sample ($N = 111$)	
Window	Elasticity	(s.e.)	N	R^2	Elasticity	(s.e.)
Aug 2025	0.676	(0.041)	114	0.71	0.676	(0.043)
Nov 2025	0.690	(0.036)	115	0.76	0.685	(0.040)
Feb 2026	0.750	(0.035)	115	0.80	0.741	(0.037)
Apr 2026	0.767	(0.033)	113	0.84	0.769	(0.033)
May 2026	0.741	(0.028)	120	0.85	0.734	(0.031)

Notes: OLS of log AUI on log GNI per capita (2025) with HC3 standard errors; economies with at least 200 sampled conversations (Aug 2025–Feb 2026) or with published values (Apr–May 2026). Common sample: economies qualifying in all five windows; change Aug 2025 to Feb 2026: 0.064 (clustered s.e. 0.018); Feb to May 2026: -0.006 (0.016). Anthropic's releases changed over this period, adding Cowork sessions and monthly windows, so the series describes measured Claude use rather than a strictly like-for-like quantity; the steepening is in line with Anthropic's own finding that low-adoption countries fell slightly further behind (Massenkoff et al., 2026a).

E Additional tables

Tables E.1 to E.4 report the first-wave estimates and their sensitivity; Tables E.5 to E.7 the affordability sensitivities; Tables E.8 to E.12 and Figure E.1 the item-level results of the second wave, including the product-class summary that Section 5.1 replaces by within-firm comparisons; and Tables E.13 to E.15 the checks on the estimated platform schedule referred to in Section 5.3.

Table E.1: Local prices and localisation slopes by plan, first wave (27 September 2026)

	U.S.	Median local price (US$)				Income slope		
Plan	price	High	Upper-mid.	Lower-mid.	Low	$\hat{\beta}$	(s.e.)	N
ChatGPT Go	8.00	9.05	5.99	5.34	4.99	0.143	(0.006)	164
Google AI Plus	4.99	5.58	4.99	4.99	4.99	0.026	(0.004)	160
ChatGPT Plus	19.99	22.99	19.99	19.99	19.99	0.039	(0.006)	164
Claude Pro	20.00	23.94	20.00	20.00	20.00	0.026	(0.006)	162
Google AI Pro	19.99	22.77	19.99	19.99	19.99	0.037	(0.005)	165
ChatGPT Pro 5×	100.00	114.53	100.00	100.00	100.00	0.019	(0.004)	164
Claude Max 5×	124.99	149.99	124.99	129.99	124.99	0.035	(0.006)	162
ChatGPT Pro 20×	200.00	234.06	200.00	200.00	200.00	0.033	(0.006)	164
Google AI Ultra	199.99	205.60	199.99	199.99	199.99	0.030	(0.006)	95

Notes: Monthly App Store prices, 27 September 2026, tax-inclusive except in the U.S. and Canada; storefronts with income data, including the United States. $\hat{\beta}$: OLS slope of ln(local dollar price) on ln(GNI per capita), heteroskedasticity-robust (HC3) standard errors. For reference, the slope on log income of the log price-level proxy (Atlas over PPP GNI) is 0.228 (s.e. 0.014) across 197 economies and 0.243 on this sample. Proportional pricing would imply a slope of 1. Restricting to economies with 2025 income data gives 0.149 for Go and 0.046 for Plus.

Table E.2: First-wave price slopes under alternative standard-error assumptions

Plan or ratio	Slope	HC3 SE	Currency SE	Cluster 95% CI	N
ChatGPT Go	0.143	0.006	0.014	[0.114, 0.173]	164
ChatGPT Plus	0.039	0.006	0.020	[−0.0002, 0.0790]	164
Claude Pro	0.026	0.006	0.014	[−0.002, 0.055]	162
Google AI Pro	0.037	0.005	0.016	[0.004, 0.070]	165
Go / Plus ratio	0.104	0.006	0.007	[0.089, 0.119]	164

Notes: All outcomes are in logs. Currency-clustered CR1 errors with t_{G-1} intervals; 39 clusters except Google AI Pro (40). Currency clusters are highly unbalanced (102 USD, 25 EUR, 37 singletons). The ratio uses the two plans as listed in the same storefront. For ChatGPT Plus the clustered interval includes zero, so the small positive standard-tier slope is not distinguishable from zero under currency dependence, which the decomposition of Section 5.3 attributes to the platform schedule; the Go slope's clustered interval excludes zero.

Robustness of the deviation slopes (Section 5.3). Replacing the exchange rates with an independent source changes the dollar slopes by at most 0.0003 (Plus 0.040, Go 0.143). The modal rule admits four treatments of ties, which give schedule slopes of 0.0325–0.0329 and Go deviation slopes of 0.110–0.111; a strict-majority rule that discards every price point whose mode accounts for exactly half of the panel's items lowers the schedule slope to 0.027 on 147 storefronts and leaves the Go slope at 0.110 and the Plus slope below 0.01. Restricted to the 60 storefronts not priced in U.S. dollars, the deviation slope is 0.109 (0.025) for Go and −0.000 (0.024) for Plus; weighted by working-age population it is 0.150 (0.030) and 0.000 (0.007); HC3 standard errors differ from the HC1 standard errors of Table 5 by less than 0.0002.

Table E.3: The Go-to-Plus price ratio by currency sample

Sample	Slope	HC3 SE	HC3 95% interval	N
All matched storefronts	0.104	0.006	[0.093, 0.115]	164
USD only	0.113	0.008	[0.098, 0.128]	102
Exclude USD	0.112	0.020	[0.073, 0.151]	62
Exclude USD and EUR	0.125	0.023	[0.079, 0.171]	37

Notes: OLS of the log Go-to-Plus listing ratio on log Atlas GNI per capita, first wave. Intervals use HC3 standard errors with a normal reference. Each restriction changes the sample; no low-income economy remains in the non-USD samples.

Table E.4: Storefront availability, currency and the cheapest locally listed paid tier, first wave

		ChatGPT storefront (% of econ.)			USD	Cheapest paid tier		Plus
Income group	N	Local	No store	Not offered	priced (%)	US$	Burden	burden
High income	86	73	23	3	35	5.65	0.17	0.75
Upper-middle income	59	86	10	3	69	4.99	0.81	3.27
Lower-middle income	47	77	21	2	86	4.99	2.15	8.89
Low income	25	64	36	0	100	4.99	7.21	28.90

Notes: All 217 World Bank economies. "No store": the request was redirected to the U.S. store, a platform rule (Section 2.2). "Not offered": the request returned no app page. These are audit retrieval outcomes, not complete service-availability determinations. Cheapest paid tier: the lower of ChatGPT Go and Google AI Plus in local storefronts. Burdens are medians of $12p/y \times 100$ at tax-inclusive local prices among economies with a local listing and income data.

Table E.5: Burden of a $20 plan by income quintile, and comparison with mobile broadband (medians)

		$20 plan, % of quintile mean income			Mobile broadband	AI plan /
Income group	N	Poorest 20%	Middle 20%	Richest 20%	2 GB, % GNI p.c.	broadband
High income	47	1.6	0.7	0.3	0.43	1.7
Upper-middle income	53	9.2	3.9	1.4	1.16	2.6
Lower-middle income	44	32.8	13.1	4.4	2.31	5.1
Low income	19	73.2	36.1	13.0	8.22	6.4

Notes: Quintile shares $s_{cq} \in (0,1)$ from the latest household survey in WDI (median survey year 2021–2023); quintile mean income is approximated as $y_c s_{cq}/0.2$. Mobile broadband: ITU data-only basket with at least 2 GB per month, the selected 2024 basket, as a share of monthly GNI per capita (International Telecommunication Union, 2025b). The last column is the median of $20 divided by the 2024 broadband basket price in current dollars. Broadband and survey columns use their available samples, which need not coincide.

Table E.6: Share of economies and population above alternative benchmarks, $20 plan

Cutoff	Economies above	Share of economies	Share of covered population
1%	140	69.7%	85.6%
2%	113	56.2%	60.1%
5%	71	35.3%	44.6%
10%	42	20.9%	16.7%

Notes: 201 economies. Population is ages 15–64; the covered economies hold 98.6% of the world's. The 2% row is the broadband benchmark used in the text. Restricting to the 182 economies with 2025 GNI gives 60.0% at 2%, with median burdens of 0.67%, 3.09%, 8.87% and 28.9% across the four groups.

Table E.7: Burden of the $20 tier under alternative price schedules, and the subsidy required under uniform pricing

		Median burden, % of GNI per capita					Subsidy to the 2% benchmark (medians)		
Income group	N	Uniform	Entry slope	Entry deviation	Price level	Proportional	Rate (%)	US$ a year	Cost, % GNI
High income	75	0.65	0.57	0.59	0.40	0.27	0	0	0.00
Upper-middle income	58	3.09	2.18	2.36	1.20	0.27	35	84	0.08
Lower-middle income	47	8.73	5.30	5.95	3.38	0.27	77	185	0.43
Low income	21	28.92	14.79	17.27	10.67	0.27	93	223	1.55

Notes: Burden $= 1200p/y$ with y GNI per capita (Atlas). Schedules anchored at the U.S. price ($20) and income: entry slope $p = 20(y/y_{US})^{0.143}$, the U.S.-dollar income slope of ChatGPT Go in the first wave (0.143 in the second as well), which includes the 0.033 of the platform schedule; entry deviation $p = 20(y/y_{US})^{0.110}$, Go's deviation from the schedule, that is, the vendor's own localisation; price-level index $p = 20\,y/y^{PPP}$ (Remark 2); proportional $p = 20\,y/y_{US}$. Subsidy under uniform pricing: rate $100\max(0, 1 - y/12{,}000)$ percent; dollars $\max(0, 240 - 0.02y)$ per subscriber; cost of that subsidy for one tenth of the population aged 15–64 as a share of GNI. Medians of economy-level values; the uniform column repeats the $20 column of Table 8. Under uniform pricing 113 economies need a subsidy; the entry-slope and entry-deviation schedules each leave 57% and the price-level schedule 41% of the covered working-age population above the benchmark, against 60% under uniform pricing.

Table E.8: Income slopes of local prices by product class, 16 October 2026

Product class	Items	Median slope	Interquartile range	Slope > 0.10 (%)
Non-AI subscriptions	70	0.039	0.022–0.118	29
AI-feature apps' subscriptions	15	0.046	0.019–0.060	20
AI subscriptions	28	0.030	0.023–0.034	7
AI consumables (credits)	17	0.026	0.015–0.028	0

Notes: 130 of the 147 items listed with a U.S. price in at least 60 storefronts after de-duplication (the four Microsoft 365 plans appear in Table E.9; the 13 consumables of non-AI and AI-feature apps are omitted). Slope: OLS of ln(local dollar price / U.S. price) on ln(GNI per capita) in storefronts other than the United States with income data, HC1 standard errors. AI subscriptions: the plans of ChatGPT, Claude, Gemini, Perplexity, Grok, Mistral, Poe and Character.AI; AI-feature apps: Canva, CapCut, Notion and Grammarly; non-AI: the 21 general subscription apps (including one AI tier sold through the Google One app, Google AI Plus 2 TB, slope 0.026). The class shares do not identify a difference between AI and non-AI apps: at the app level, 8 of the 21 non-AI apps and 2 of the 8 AI apps have at least one subscription item with a slope above 0.10 (Fisher exact $p = 0.67$). The identifying comparison is within firm (Section 5.1).

Table E.9: Within-firm contrasts: income slopes of every monthly subscription of the five firms

Firm	Product	Group	U.S. price	N	USD slope (s.e.)	Deviation slope (s.e.)	LMIC price / U.S. price
Google	YouTube Premium	No AI allowance	20.99	122	0.244 (0.044)	0.196 (0.041)	0.43
	YouTube Premium Lite	No AI allowance	11.99	74	0.396 (0.041)	0.338 (0.033)	0.27
	YouTube Music	No AI allowance	15.99	105	0.372 (0.035)	0.325 (0.029)	0.27
	YouTube Premium Family	Family variant	34.99	102	0.360 (0.039)	0.305 (0.033)	0.34
	YouTube Music Family	Family variant	24.99	98	0.375 (0.037)	0.324 (0.031)	0.30
	Google One 100 GB	Storage (no AI)	1.99	93	0.033 (0.007)	−0.008 (0.008)	1.00
	Google AI Plus 2 TB (Google One app)	Capped AI (duplicate)	9.99	128	0.026 (0.005)	0.000 (0.008)	1.00
	Google AI Plus	Capped AI	4.99	159	0.025 (0.004)	−0.003 (0.008)	1.00
	Google AI Pro	Frontier AI	19.99	164	0.036 (0.005)	0.002 (0.005)	1.00
	Google AI Ultra	Frontier AI	199.99	97	0.030 (0.006)	0.001 (0.004)	1.00
X Corp	X creator subscription	Creator product	4.00	146	0.032 (0.009)	−0.010 (0.004)	1.00
	X Premium Basic	No AI allowance	4.00	164	0.170 (0.011)	0.142 (0.010)	0.50
	X Premium	Small AI allowance	11.00	166	0.180 (0.011)	0.151 (0.010)	0.45
	X Premium Plus	Frontier AI	50.00	150	0.035 (0.005)	0.005 (0.004)	1.00
	SuperGrok Lite	Capped AI	10.00	155	0.017 (0.005)	−0.016 (0.004)	1.00
	SuperGrok	Frontier AI	30.00	154	0.026 (0.008)	−0.005 (0.004)	1.00
	SuperGrok Plus	Frontier AI	100.00	156	0.018 (0.005)	−0.014 (0.004)	1.00
	SuperGrok Heavy	Frontier AI	300.00	130	0.027 (0.008)	−0.006 (0.005)	1.00
Microsoft	Microsoft 365 Basic	Storage (no AI)	1.99	132	0.029 (0.011)	−0.008 (0.014)	1.00
	Microsoft 365 Personal	Small AI allowance	9.99	164	0.046 (0.009)	0.021 (0.007)	0.85
	Microsoft 365 Family	Family variant	12.99	164	0.056 (0.010)	0.023 (0.008)	0.81
	Microsoft 365 Premium	Frontier AI	19.99	160	0.039 (0.006)	0.002 (0.004)	1.00
Character.AI	Character.AI Lite	Capped AI	4.99	167	0.116 (0.015)	0.091 (0.012)	0.60
	Character.AI+	Frontier AI	9.99	151	0.034 (0.009)	0.002 (0.004)	1.00
OpenAI	ChatGPT Go	Capped AI	8.00	163	0.142 (0.006)	0.110 (0.006)	0.62
	ChatGPT Plus	Frontier AI	19.99	163	0.038 (0.005)	0.004 (0.003)	1.00
	ChatGPT Pro 100	Frontier AI	100.00	163	0.017 (0.004)	−0.015 (0.004)	1.00
	ChatGPT Pro 200	Frontier AI	200.00	163	0.031 (0.006)	0.000 (0.003)	1.00
	ChatGPT Pro 500	Frontier AI	500.00	163	0.016 (0.004)	−0.016 (0.004)	1.00

Notes: Every monthly subscription item of the five firms listed with a U.S. price in at least 60 storefronts, except annual variants and X's post-promotion product; 16 October 2026, in every storefront other than the United States with income data (up to 167; 170 storefronts have income data in the second wave). Group: the classification of the stacked regression (Table 4), which includes every item except the storage plans (Google One 100 GB, Microsoft 365 Basic), the Google One app's second listing of Google AI Plus (2 TB), the family variants (YouTube Premium Family, YouTube Music Family, Microsoft 365 Family) and X's creator subscription, whose price is set by the creator; the storage plans, the second AI Plus listing and the creator subscription are not localised either, and each family variant moves with the plan it extends. USD slope as in Table E.8. Deviation slope: slope on log income of ln(local price / estimated platform schedule), on the plan's storefronts with a schedule reference (a smaller sample than N); "–" if none. Last column: median ratio of the local dollar price to the U.S. price in low- and lower-middle-income storefronts. X Premium carries a small Grok allowance and X Premium Basic none; X Premium Plus includes SuperGrok; Microsoft 365 Premium includes the full Copilot allowance, Personal and Family a small one; Character.AI Lite is a capped version of Character.AI+. The within-storefront relative prices of the plans in the stacked regression are in Table 3.

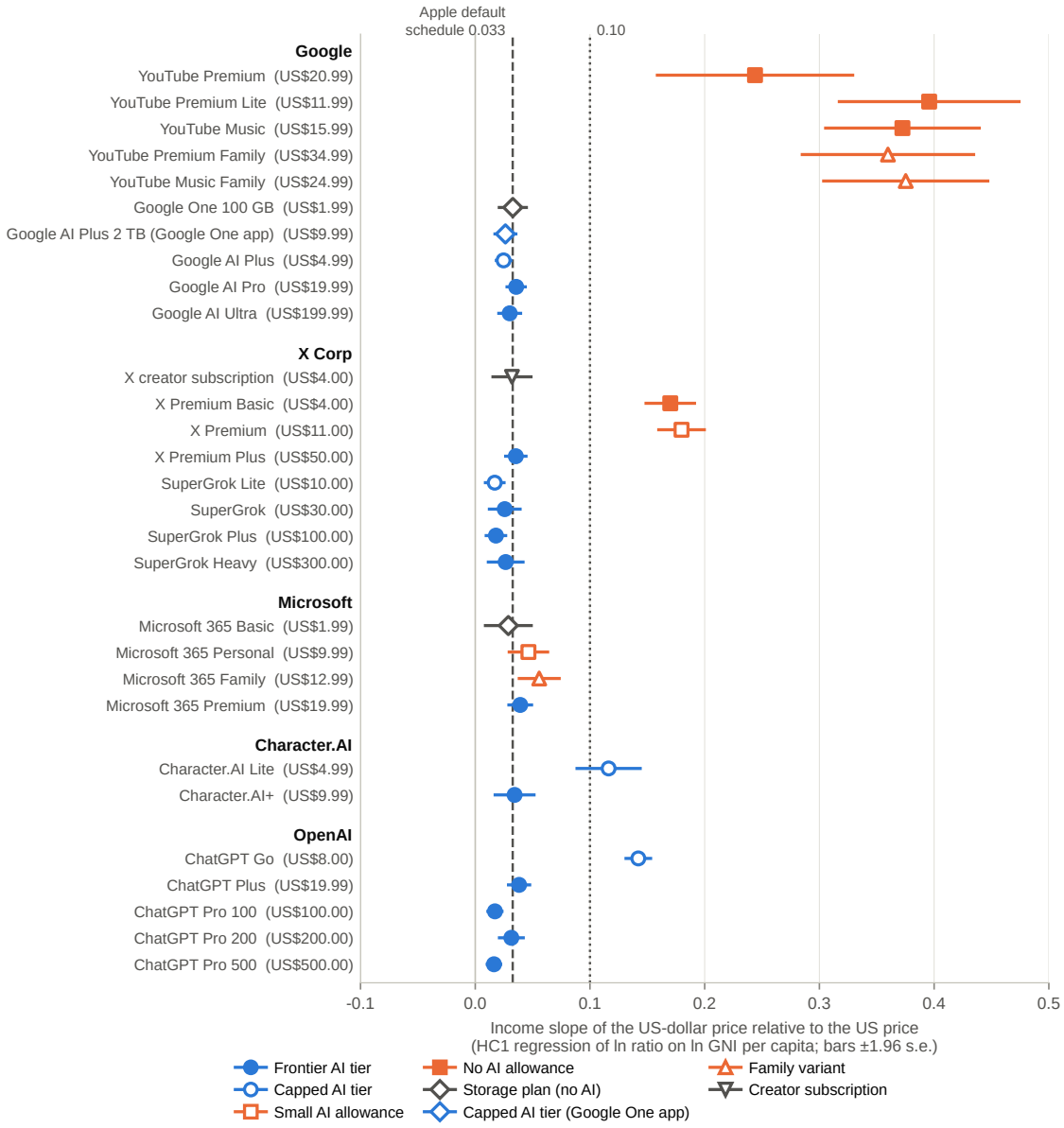


Figure E.1: Income slopes of local dollar prices for AI and other products of the same firms, 16 October 2026. Bars are ±1.96 HC1 standard errors. The dashed line is the slope of the estimated platform schedule (0.033); the dotted line marks 0.10.

Table E.10: Relative prices against a common frontier reference, 16 October 2026

			Relative to ChatGPT Plus		Relative to frontier median		Deviation
Plan	US$	N	Slope (s.e.)	LMIC (%)	Slope (s.e.)	LMIC (%)	slope (s.e.)
ChatGPT Go	8.00	163	0.104 (0.006)	−38	0.106 (0.005)	−38	0.110 (0.006)
ChatGPT Plus	19.99	163	–	–	0.002 (0.002)	0	0.004 (0.003)
ChatGPT Pro 100	100.00	163	−0.021 (0.004)	0	−0.019 (0.004)	0	−0.015 (0.004)
ChatGPT Pro 200	200.00	163	−0.007 (0.003)	0	−0.005 (0.004)	0	0.000 (0.003)
ChatGPT Pro 500	500.00	163	−0.022 (0.004)	0	−0.020 (0.004)	0	−0.016 (0.004)
Claude Pro	20.00	161	−0.014 (0.004)	0	−0.012 (0.004)	0	−0.007 (0.004)
Claude Max 5×	124.99	161	−0.004 (0.004)	0	−0.003 (0.004)	0	0.002 (0.005)
Claude Max 20×	249.99	161	−0.007 (0.004)	0	−0.005 (0.005)	0	0.000 (0.005)
Google AI Plus	4.99	156/157	−0.015 (0.004)	0	−0.013 (0.003)	0	−0.003 (0.008)
Google AI Pro	19.99	161/162	−0.002 (0.003)	0	0.000 (0.002)	0	0.002 (0.005)
Google AI Ultra	199.99	94/95	0.001 (0.004)	0	0.000 (0.003)	0	0.001 (0.004)
Google One 100 GB	1.99	92	−0.016 (0.008)	0	−0.015 (0.006)	0	−0.008 (0.008)
Google AI Plus (2 TB)	9.99	123/124	−0.016 (0.006)	0	−0.015 (0.005)	0	0.000 (0.008)
YouTube Premium	20.99	116/117	0.203 (0.044)	−58	0.205 (0.043)	−58	0.196 (0.041)
YouTube Premium Lite	11.99	72	0.349 (0.039)	−73	0.352 (0.039)	−73	0.338 (0.033)
YouTube Music	15.99	101/102	0.334 (0.034)	−73	0.334 (0.033)	−73	0.325 (0.029)
YouTube Premium Family	34.99	99/100	0.315 (0.038)	−66	0.318 (0.037)	−66	0.305 (0.033)
YouTube Music Family	24.99	94/95	0.337 (0.036)	−71	0.337 (0.035)	−71	0.324 (0.031)
X creator subscription	4.00	140/141	−0.007 (0.009)	0	−0.006 (0.009)	0	−0.010 (0.004)
X Premium Basic	4.00	158/159	0.134 (0.009)	−50	0.136 (0.009)	−50	0.142 (0.010)
X Premium	11.00	160/161	0.143 (0.009)	−55	0.145 (0.009)	−55	0.151 (0.010)
X Premium Plus	50.00	145/146	−0.002 (0.002)	0	−0.001 (0.003)	0	0.005 (0.004)
SuperGrok Lite	10.00	154/155	−0.023 (0.005)	0	−0.020 (0.004)	0	−0.016 (0.004)
SuperGrok	30.00	153/154	−0.012 (0.005)	0	−0.011 (0.005)	0	−0.005 (0.004)
SuperGrok Plus	100.00	155/156	−0.021 (0.005)	0	−0.018 (0.004)	0	−0.014 (0.004)
SuperGrok Heavy	300.00	129/130	−0.012 (0.005)	0	−0.011 (0.006)	0	−0.006 (0.005)
Microsoft 365 Basic	1.99	129/130	−0.015 (0.013)	0	−0.012 (0.011)	0	−0.008 (0.014)
Microsoft 365 Personal	9.99	161/162	0.011 (0.008)	−15	0.013 (0.008)	−15	0.021 (0.007)
Microsoft 365 Family	12.99	161/162	0.020 (0.008)	−19	0.022 (0.008)	−19	0.023 (0.008)
Microsoft 365 Premium	19.99	159/160	0.002 (0.005)	0	0.003 (0.004)	0	0.002 (0.004)
Character.AI Lite	4.99	162/163	0.081 (0.012)	−40	0.083 (0.012)	−40	0.091 (0.012)
Character.AI+	9.99	146/147	−0.004 (0.010)	0	−0.001 (0.009)	0	0.002 (0.004)
Perplexity Pro	20.00	163/164	−0.013 (0.003)	0	−0.012 (0.004)	0	−0.007 (0.003)
Super Duolingo	12.99	125/126	−0.131 (0.046)	+132	−0.111 (0.049)	+141	−0.118 (0.046)
Telegram Premium	4.99	161/162	0.003 (0.006)	0	0.005 (0.006)	0	0.011 (0.007)
Snapchat+	4.99	158/159	0.292 (0.034)	−80	0.297 (0.033)	−80	0.293 (0.033)

Notes: For each plan and storefront, ln of the local price relative to the local price of ChatGPT Plus, or to the median local price of the three $19.99 frontier plans (ChatGPT Plus, Google AI Pro, Microsoft 365 Premium), each net of the same ratio in the United States; slopes on log GNI per capita with HC1 standard errors, and the median in low- and lower-middle-income storefronts as $100(e^m - 1)$ percent. N: storefronts in the two comparisons where they differ. Rows: the plans of Table 4, Anthropic's plans, the same firms' storage, family and creator products, Perplexity Pro and three other apps. The last column repeats the deviation slope from the estimated platform schedule (Tables 5 and E.9). Unlike Table 3, the reference plan is sold by another firm, so the comparison holds the storefront fixed but not the firm. Super Duolingo's relative price falls with income because its monthly listing is censored in many higher-income storefronts (Table E.11); the row is shown for completeness.

Table E.11: Item-storefront cells dropped by the annual-as-monthly guard, by item

Item	Class	Storefronts	Cells dropped	Slope before (s.e.)	Slope after (s.e.)
Super Duolingo	Non-AI	167	36	0.045 (0.056)	−0.076 (0.048)
Gemini 100 GB	AI	129	36	−0.150 (0.069)	0.033 (0.007)
Headspace Plus	Non-AI	117	35	−0.104 (0.086)	0.202 (0.055)
Character.AI+	AI	167	16	−0.092 (0.038)	0.034 (0.009)
Evernote Plus	Non-AI	106	6	0.046 (0.034)	0.032 (0.014)

Notes: The five items with the most cells dropped, of 14 items and 151 cells in all (of 21,938 item-storefront cells with a U.S. price and income data). A cell is dropped when the price recorded as monthly exceeds three times the platform's reference for the item's U.S. monthly price (Appendix B). Slopes: OLS of ln(local dollar price / U.S. price) on ln(GNI per capita), HC1 standard errors, before and after the guard. The Super Duolingo slope remains negative after the guard, which drops the flagged cells but cannot recover the censored monthly listings; the item enters Table E.8 as it stands.

Table E.12: Local prices of the first-wave plans, 27 September and 16 October 2026

Plan	Storefronts compared	Identical	Share (%)
ChatGPT Go	167	167	100
ChatGPT Plus	167	167	100
ChatGPT Pro 100 (was 5×)	167	167	100
ChatGPT Pro 200 (was 20×)	167	167	100
Claude Pro	164	164	100
Claude Max 5×	164	164	100
Claude Max 20×	164	164	100
Google AI Plus	163	163	100
Google AI Pro	168	168	100
Google AI Ultra	95	95	100
All	1586	1586	100

Notes: A local price is identical when the two listings, in the storefront's own currency, differ by less than 0.5%. ChatGPT Pro 5× and 20× were renamed Pro 100 and Pro 200 between the waves. Google AI Ultra in the Philippines was listed in the first wave only, so 1,586 of 1,587 first-wave listings are compared.

Table E.13: Frontier tiers in U.S.-dollar storefronts whose schedule embeds a tax of more than 5%

	All 16 storefronts			Low and lower-middle income	
Plan	Storefronts	At U.S. price	Within 3% of schedule	Storefronts	At U.S. price
ChatGPT Plus	16	5	11	10	5
Google AI Pro	16	15	1	10	10
Google AI Ultra	8	8	0	3	3
Microsoft 365 Premium	16	10	3	10	8
Claude Pro	16	0	0	10	0
Perplexity Pro	16	0	0	10	0
SuperGrok	15	0	0	9	0

Notes: The 16 storefronts priced in U.S. dollars whose tax factor (multiplier divided by the exchange rate) exceeds 1.05, so that the platform's schedule price for a plan is above its U.S. price. "At the U.S. price": the plan is listed at its U.S. price to within 0.6%, so the vendor has overridden the schedule and absorbs the tax; "within 3% of the schedule": the plan is listed at the tax-inclusive reference. Google AI Ultra is listed in eight of the 16 storefronts and SuperGrok in 15. Ten of the 16 storefronts are low- or lower-middle-income economies; the last two columns repeat the counts for them. In all 16 storefronts the two $20.00 plans of different vendors, Claude Pro and Perplexity Pro, are listed at the same price, $25.00, which points to the platform's conversion of a round-number price point rather than to two vendors' decisions (Section 5.3).

Table E.14: Storefronts in which the estimated schedule is weak at the $19.99 price point

Storefront	Ladder points	With a unique mode	Median modal share	Multiplier spread	Frontier plan at an exact point
India	17	8	0.43	0.87	no
Brazil	17	8	0.44	0.80	no
Egypt, Arab Rep.	16	6	0.37	1.70	no
Uganda	14	7	0.50	1.19	no
Nigeria	17	3	0.40	–	no
Turkiye	17	1	0.33	–	no
Indonesia	17	4	0.40	0.07	no
Pakistan	16	6	0.41	0.13	no
Philippines	17	7	0.40	0.58	no
Viet Nam	17	4	0.38	0.40	no
Mexico	17	11	0.50	0.16	yes
South Africa	17	8	0.46	0.06	yes

Notes: Upper block: the six lower-income storefronts in which no price point of $19.99 has a unique mode among the panel's items, so the frontier plans' references there are extrapolated with the storefront multiplier or missing (Nigeria and Türkiye have no multiplier); lower block: six other emerging-market storefronts for comparison. Ladder points: U.S. price points with at least three panel items in the storefront; with a unique mode: points whose modal price accounts for at least half of the items; multiplier spread: range of the point-wise ratios divided by their median (below 0.2 defines the clean subsample); last column: whether any frontier plan's U.S. price point has an exact schedule price in the storefront.

Table E.15: Vintage test: deviations from the estimated schedule and exchange-rate movements since March 2024

		Slope on ln GNI (s.e.)		Coefficient on FX range (s.e.)		Median	Within 3% (%):
Sample	*N*	alone	with FX range	in the deviation	in |deviation|	deviation	US$ / stable / moved
Frontier plans, pooled	1,489	−0.0014 (0.0013)	−0.0074 (0.0025)	0.403 (0.115)	0.542 (0.082)	0.000	81 / 41 / 22
ChatGPT Go	161	0.1103 (0.0056)	0.1080 (0.0097)	0.156 (0.416)	−0.453 (0.229)	−0.289	17 / 14 / –
ChatGPT Plus	161	0.0037 (0.0035)	−0.0028 (0.0076)	0.437 (0.370)	0.543 (0.280)	0.000	92 / 89 / –
ChatGPT Pro 200	161	−0.0004 (0.0033)	−0.0070 (0.0072)	0.443 (0.352)	0.568 (0.267)	0.000	83 / 63 / –
Claude Pro	159	−0.0074 (0.0038)	−0.0146 (0.0071)	0.490 (0.320)	0.527 (0.276)	0.000	73 / 37 / –
Claude Max 20×	159	0.0003 (0.0049)	−0.0072 (0.0080)	0.508 (0.349)	0.688 (0.247)	0.000	84 / 9 / –
Google AI Pro	159	0.0020 (0.0046)	−0.0031 (0.0085)	0.341 (0.389)	0.679 (0.178)	0.000	82 / 23 / –
Google AI Ultra	94	0.0011 (0.0043)	−0.0039 (0.0052)	0.315 (0.198)	0.364 (0.132)	0.000	87 / 50 / –
Perplexity Pro	161	−0.0065 (0.0034)	−0.0133 (0.0070)	0.461 (0.332)	0.497 (0.284)	0.000	73 / 37 / –
SuperGrok	151	−0.0045 (0.0041)	−0.0096 (0.0081)	0.359 (0.384)	0.527 (0.277)	0.000	81 / 46 / –
SuperGrok Heavy	127	−0.0058 (0.0051)	−0.0116 (0.0083)	0.436 (0.368)	0.523 (0.294)	0.000	79 / 44 / –
Microsoft 365 Premium	157	0.0021 (0.0036)	0.0004 (0.0056)	0.117 (0.238)	0.375 (0.088)	0.000	74 / 18 / –

Notes: Deviation = ln(local price / schedule reference). FX range: the range of ln(units of the storefront currency per U.S. dollar) over six dates between 6 March 2024 and 15 October 2026 (zero for dollar storefronts). Columns 3–4: OLS of the deviation on log GNI per capita without and with the FX range; column 5: the coefficient on the FX range in that regression; column 6: OLS of the absolute deviation on the FX range; HC1 standard errors. The pooled row stacks the ten frontier plans (ChatGPT Plus and Pro 200, Claude Pro and Max 20×, Google AI Pro and Ultra, Perplexity Pro, SuperGrok and SuperGrok Heavy, Microsoft 365 Premium), ten rows per storefront, so its HC1 errors understate the uncertainty; the text quotes the pooled coefficients with standard errors clustered by storefront, −0.0014 (0.0028), −0.0074 (0.0066), 0.403 (0.330) and 0.542 (0.233); clustered by currency they are −0.0014 (0.0031), −0.0074 (0.0074), 0.403 (0.341) and 0.542 (0.240). Last column: share of storefront-plan rows within 3% of the schedule among dollar storefronts, local-currency storefronts whose currency moved by at most 10% (stable) and storefronts whose currency moved by more than 30% (moved); "–" for a class with fewer than ten rows.

F The ChatGPT Go rollout: a descriptive record

OpenAI's release notes and press reports date the arrival of ChatGPT Go in India (19 August 2025), Indonesia (22 September 2025), sixteen further Asian economies (8 October 2025: Afghanistan, Bangladesh, Bhutan, Brunei, Cambodia, Lao PDR, Malaysia, Maldives, Myanmar, Nepal, Pakistan, Philippines, Sri Lanka, Thailand, Timor-Leste and Viet Nam), 71 undisclosed economies (14 October 2025), Brazil (28 October) and eight European economies (30 October), after which further economies followed before the worldwide rollout of 16 January 2026 (OpenAI, 2026d, 2025; Mehta, 2025). India also received twelve months of the tier at no charge from 4 November 2025, and in the same months Indian subscribers of Airtel and Jio received Perplexity Pro (from July 2025) and Google AI Pro (from October 2025) at no charge; app-analytics firms recorded a ninefold rise in Perplexity's downloads in India while ChatGPT's and Claude's stayed flat (Singh, 2026). The rollout record, with sources, is data/raw/openai/chatgpt_go_rollout.csv.

Against these dates the public indicators offer four semi-annual or quarterly windows of any use (H1 2025 before any rollout; H2 2025, in which India, Indonesia and the sixteen Asian economies were exposed for 2.8–4.4 months and most others for at most 2.6; Q1 and Q2 2026 with the tier universal) and five windows of Claude activity (4–11 August 2025, before India's launch; 13–20 November 2025; February, April and May 2026). Table F.1 and Figure F.1 describe how the early economies' indicators moved relative to the later-exposed ones. For each window the change in the log indicator is regressed on log income and the initial log level in the economies that did not receive the tier early, every economy's residual change is computed from those coefficients, and the residual of India and the mean residual of the sixteen Asian economies are reported with their rank among all residuals and, for the sixteen, with the coefficient on a cohort indicator. Because all sixteen economies, and India and Indonesia, lie in South and South-East Asia, the specification with World Bank region fixed effects is the one reported first; the specification without them was written into the analysis script before the outcomes were computed and was not registered externally, and the region-fixed-effects specification was added afterwards.

Table F.1: Changes in use around the first rollouts of ChatGPT Go

	Later-exposed comparison econ.		India		Sixteen Asian economies	
Indicator and window	N	Mean Δln	Δln	Residual (p)	Contrast with region FE (s.e.)	Contrast without region FE (s.e.) [p]
Any use, H1 2025–H2 2025	96	0.067	0.100	0.048 (0.08)	0.021 (0.020)	0.024 (0.010) [0.0005]
Any use, H2 2025–Q1 2026	96	0.094	0.114	0.033 (0.21)	0.016 (0.024)	0.010 (0.009) [0.2560]
Any use, Q1 2026–Q2 2026	96	0.040	0.050	0.017 (0.30)	0.011 (0.015)	0.008 (0.007) [0.0955]
Claude activity, Aug 2025–Nov 2025	141	0.033	−0.176	−0.166 (0.39)	0.030 (0.102)	−0.057 (0.058) [0.3270]
Claude activity, Nov 2025–Feb 2026	145	0.035	0.077	0.113 (0.58)	−0.140 (0.126)	−0.180 (0.063) [0.0025]
Claude activity, Feb 2026–May 2026	109	0.063	0.044	−0.052 (0.84)	−0.181 (0.201)	−0.032 (0.077) [0.6765]

Notes: Δln is the change in the log indicator between the two windows; the table reports log differences, and the percentage changes quoted in the text are $e^{\Delta} - 1$. Comparison economies: country-specific Microsoft estimates (any use) or the recomputed Anthropic index without the publication floor (Claude activity; the February–May 2026 row uses the published May 2026 index) for economies outside the two early cohorts; they include the 71 economies that received the tier on 14 October 2025 and the nine of 28–30 October, which is why the design is a comparison of exposure length rather than of treatment against no treatment. Residuals are from an OLS of Δln on log GNI per capita and the initial log level fitted on the comparison economies; p in parentheses is the share of all economies with an absolute residual at least as large. The contrast is the coefficient on an indicator for the sixteen Asian economies in the pooled regression (India and Indonesia excluded), with HC3 standard errors, with World Bank region indicators (lead specification) and without them; p in brackets is the share of 2,000 random groups of the same size drawn from the comparison economies with a mean residual at least as large in absolute value in the specification without region indicators. The sixteen contribute 11 economies to the any-use rows and 14, 14 and 9 to the Claude rows. Indonesia's residuals are 0.017, 0.006 and 0.010 for any use and 0.307, −0.592 and 0.205 for Claude activity.

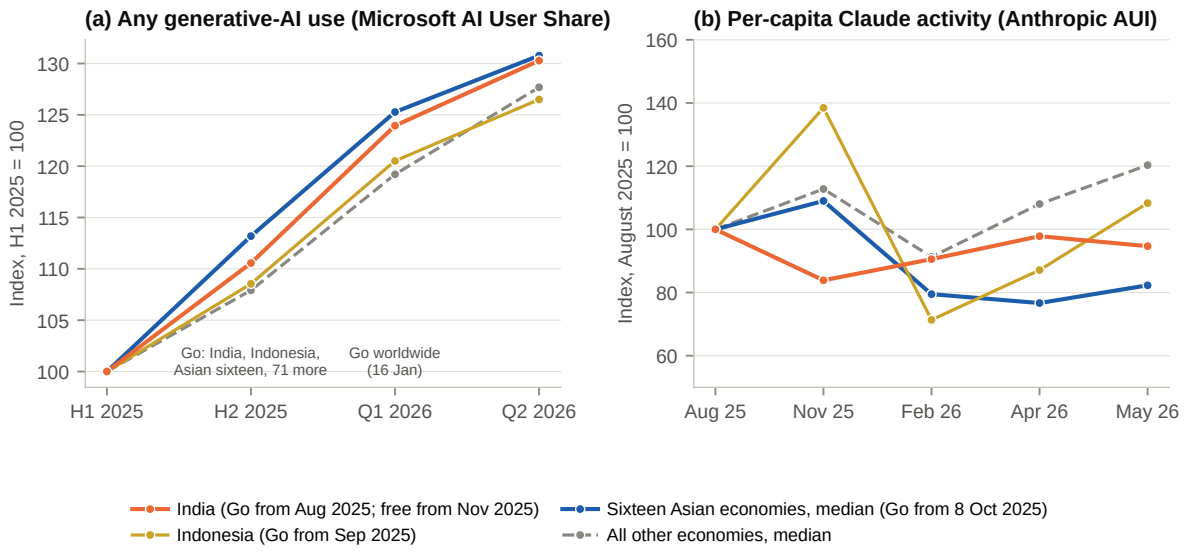


Figure F.1: Usage indicators around the first rollouts of ChatGPT Go. (a) Microsoft AI User Share indexed to H1 2025 = 100 for India, Indonesia, the median of the eleven of the sixteen Asian economies with country-specific estimates and the median of all other country-specific economies. (b) Anthropic AI Usage Index indexed to August 2025 = 100 for the economies published in all five windows. The tier reached India on 19 August 2025 (free from 4 November), Indonesia on 22 September, the sixteen Asian economies on 8 October, 71 further economies on 14 October 2025, and everywhere on 16 January 2026.

The record shows the following. With region fixed effects, in the half-year in which the sixteen Asian economies received the tier a week before 71 other economies and two to three months before the rest, their share of adults using any generative-AI product grew 0.021 log points faster than income, starting level and region predict (s.e. 0.020), and their Claude activity changed by -0.140 log points relative to others between November 2025 and February 2026 (s.e. 0.126); neither contrast is distinguishable from zero, and the later windows are smaller still. Without region fixed effects the same contrasts are 0.024 (s.e. 0.010; 2.5% faster growth), a difference exceeded by one of 2,000 random groups of comparison economies, with ten of the eleven economies showing positive residuals, and -0.180 (s.e. 0.063; a 16.5% relative fall), with eleven of fourteen residuals negative. India grew 4.9% faster than predicted in any use (0.048 log points), a residual exceeded in absolute value by 8% of economies, and its Claude activity fell 15.3% relative to others between August and November 2025 (-0.166 log points), a residual that 39% of economies exceed; Indonesia's residuals are small.

This record does not support a causal interpretation, for reasons the table makes visible. The comparison group received the tier within six days to three weeks of the sixteen, so the H2 2025 contrast is a difference in exposure of a few weeks within a six-month window. The early cohorts lie in one region, and the region fixed effects that this geography requires double the standard errors and leave no contrast distinguishable from zero. The rollout order was chosen by the vendor, launches came with marketing, India's free year coincided with two telecom bundles that gave other frontier tiers away, and the only pre-period is a single half-year, so parallel trends cannot be checked. The episode is recorded here as an account of what the public indicators show around the dates, and as the kind of variation that a design with higher-frequency outcomes, registered specifications and estimators for staggered adoption (Callaway and Sant'Anna, 2021) could exploit.

G A protocol for recording access conditions

A recurring fee does not determine the cost of completing useful work: an unchanged fee can correspond to different entitlements by account cohort and date, and a new tier can add eligibility rights rather than replace a price, as the tariff change of 29 September 2026 illustrated (Section 2). Table G.1 lists the fields that a reproducible observation of access conditions should record, as a proposed extension of the present audit rather than a newly collected dataset; the replication package supplies the field dictionary (checks/access_measurement_fields.csv). Missing terms should remain missing rather than be assigned a provider-wide constant, and a prospective evaluation of what a task costs should report total expenditure and success rates alongside cost per successful task, since excluding failed attempts would understate costs.

Table G.1: Fields of a record of access conditions

Group	Fields
Observation	country or storefront; date and time (UTC); billing channel (app store, web, carrier, institution); page fingerprint
Product	provider; plan name; billing period; subscriber cohort (new, existing, grandfathered)
Price	listed price; currency; tax convention (inclusive, exclusive, not collected); checkout status; introductory or promotional terms
Entitlements	included allowances and reset rules; eligible models and service modes; tool and feature eligibility; overage terms
Access	payment methods accepted; identity or residence checks; availability of the storefront to residents

Notes: The present audit observes the first three groups for the App Store channel (the page fingerprint from the second wave) and, for the plans of Appendix C, for Google Play and Microsoft's web store; the entitlement and access groups are documented from vendors' pages for the United States only.